MEAT key-rate engine (meat-core)

Statements digest, version 1.32.0

Every definition, fact, assumption, result and requirement of the contract, with the contract's numbers; remarks, examples, technical steps and all proofs are left out. Each heading opens its statement in the contract.

Mathematics

1 The declaration and the protocol channel

Definition 1.1 (The protocol declaration as the MEAT core reads it)**.** In this module a protocol declaration is a tuple

$$\mathfrak{D} = \Big(A,\ \tau_A;\ Q,\ \mathbb{S};\ \{M_x\}_{x\in\mathcal{X}};\ \{F_v\}_{v\in\mathcal{V}};\ \mathfrak{P};\ a_A,\ a_B;\ \phi;\ r,\ \mathsf{h};\ \mathfrak{L}\Big)$$

of Definition "Protocol declaration" (protocol-compiler/def:declaration), in its basic format or in its general format, (DX1)–(DX6) there, whose components satisfy the well-formedness conditions of Definition 1.2 and the two conditions on its raw-key map below. The two values of its marginal slot, $\tau_A \in S_=(A)$ and $\tau_A = \mathsf{none}$, are the two source modes, trusted-source mode and untrusted-source mode. Write $\mathcal{K}$ for the announcement-class alphabet of that definition, $\mathcal{C}_A \times \mathcal{C}_B$ in the basic format and the product of the message alphabets of (DX5) in the general one, $a(x,v) \in \mathcal{K}$ for the joint announcement it assigns to an outcome pair, and κ for a class. *Sifting is announcement-determined*: there is $\varsigma : \mathcal{K} \to \{\text{pass}, \text{fail}\}$ with $r(x,v) = \perp \iff \varsigma(a(x,v)) = \text{fail}$. *The raw key is party-computable*: for the declared key holder $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$ there is $r_{\mathsf{h}} : \mathcal{O}_{\mathsf{h}} \times \mathcal{K} \to \mathcal{Z}_\perp$ with

$$r(x,v) = r_{\mathsf{h}}\big(\pi_{\mathsf{h}}(x,v),\ a(x,v)\big) \qquad \text{for all } (x,v) \in \mathcal{X}\times\mathcal{V} \tag{1.1}$$

(Definition 1.2 (viii)). We write

$$\widehat{\mathcal{C}}_{\text{key}} \coloneqq \big\{\hat{c} \in \widehat{\mathcal{C}} :\ \exists\, \kappa \in \phi^{-1}(\hat{c}) \text{ with } \varsigma(\kappa) = \text{pass}\big\} \tag{1.2}$$

for the set of *key-live acceptance value*s, ς being the predicate of the sifting condition above. That condition fixes ς on every class some outcome pair realises, and on a class $\kappa \notin a(\mathcal{X}\times\mathcal{V})$, which no outcome pair realises, we take $\varsigma(\kappa) = \text{fail}$; so $\widehat{\mathcal{C}}_{\text{key}}$ is the set of acceptance values to which some realised class passing sifting maps, and it is fixed by the declaration.

What this module reads of $\mathfrak{P}$. The party partition $\mathfrak{P}$ of that definition consists of (a) its two readout maps π_{A} and π_{B}, naming what each party physically observes in a round, and (b) *system metadata*: the party label for

A and, for each $s \in \mathbb{S}$, the declared tensor factorization $\mathcal{H}(Q_s) = \bigotimes_i \mathcal{H}(Q_{s,i})$ with a party label per factor (a comma-indexed *tensor* factor $Q_{s,i}$, never the parenthesized superscript $X^{(s)}$, which this module reserves for *sector blocks* of an operator). The *default partition* is that of Definition "Target conditions of a well-formed declaration" (protocol-compiler/def:target): $\pi_{\mathsf{A}}(x, v) = x$, $\pi_{\mathsf{B}}(x, v) = v$ with A labelled A and every Q_s labelled B. Part (a) is consumed by the derivations, through Definition 1.2 (viii). Part (b) is consumed by no derivation; it carries an obligation of the compiler: the declaration's POVMs must be realizable by each party measuring the factors it is labelled with, so that π_{A} and π_{B} are what the parties can in fact read. Part (b) is read by the premises that place the parties' laboratories: Assumption 13.8 takes all its labels as naming the parties' laboratories, and A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) reads the label of A; so a run attests that obligation there, and the claims that assume those premises rest on it through them. This module never verifies the obligation.

The declared summation order. The declaration fixes, as part of the declared alphabet $\mathcal{V}$ of outcomes, a declared order $v_1 \prec v_2 \prec \cdots \prec v_{|\mathcal{V}|}$ on it. For each sector s the completeness sum is the sum in the declared order of the blocks $F_v^{(s)}$, the sequential sum

$$\Sigma_0^{(s)} := 0, \qquad \Sigma_i^{(s)} := \Sigma_{i-1}^{(s)} + F_{v_i}^{(s)} \quad (1 \le i \le |\mathcal{V}|), \qquad \Sigma^{(s)} := \Sigma_{|\mathcal{V}|}^{(s)}, \tag{1.3}$$

formed left to right, entrywise, with no re-ordering and no re-association. In exact arithmetic (1.3) is the unordered sum and the order is immaterial; in binary64 it is not (Definition "Sums in a declared order and sums that round nowhere" (foundations/def:declared-order)), so the order is part of the datum and this module reads it off the declared alphabet rather than choosing one.

Exact admissibility. A declaration reaches this module with its operator data exactly admissible: every F_v exactly Hermitian and exactly positive semidefinite per sector block, every $\hat{O}_k$ exactly Hermitian per sector block (positivity is asked of no constraint operator anywhere in this module), every M_x and the declared τ_A exactly Hermitian and exactly positive semidefinite, and the completeness sum (1.3) of each sector exactly the identity with every addition rounding nowhere. These are the conditions the compiler decides on the numbers it emits (Definition "Exact admissibility of declared operator data" (protocol-compiler/def:exact)).

The general format. The general format is that of Definition "Protocol declaration" (protocol-compiler/def:declaration), (DX1)–(DX6) there, and the basic format is its case of one Alice sector, one measured factor, POVM form, two messages neither of which reads the other, and no party-local entry. This contract refers to five of that definition's displays under its own labels, each read as it is defined there: the product sector map of (DX1), over the ordered set $\mathbb{F}$ of measured factors, factor f having the sectors $s_f \in \mathbb{S}^f$

with projectors $\Pi^f_{s_f}$ and block dimensions d_{f,s_f},

$$\begin{gathered} Q=\bigotimes_{f\in\mathbb{F}} Q_f, \qquad \mathbb{S}=\prod_{f\in\mathbb{F}} \mathbb{S}^f, \qquad \mathcal{H}(Q_s)=\bigotimes_{f} \mathcal{H}(Q_{f,s_f}), \\ \Pi_s=\bigotimes_{f} \Pi^f_{s_f}, \qquad d_s=\prod_{f} d_{f,s_f}; \end{gathered} \tag{1.4}$$

the product family of (DX3),

$$F_v \coloneqq \bigotimes_{f\in\mathbb{F}} F^f_{v_f}, \qquad F^{(s)}_v=\bigotimes_{f\in\mathbb{F}} F^{f,(s_f)}_{v_f} \qquad \big(v=(v_f)_f,\ s=(s_f)_f\big); \tag{1.5}$$

the two conditions on the declared $\{N_x\}$ of the ensemble form of (DX4), whose blocks lie along the index set $\mathbb{T}$ of Alice's sectors, $\mathcal{H}(A)=\bigoplus_{t\in\mathbb{T}}\mathcal{H}(A_t)$, and the family of the POVM-form declaration an ensemble-form declaration denotes, with P_t the projector onto the support of $\tau^{(t)}_A$ and the inverse square root taken on that support and zero off it,

$$\sum_{x\in\mathcal{X}} N^{(t)}_x=\tau^{(t)}_A \quad \text{and} \quad \tau^{(t)}_A\neq 0 \qquad \text{for every } t\in\mathbb{T}, \tag{1.6}$$

$$M^{(t)}_x \coloneqq \big(\tau^{(t)}_A\big)^{-1/2} N^{(t)}_x \big(\tau^{(t)}_A\big)^{-1/2} + \frac{\operatorname{tr} N^{(t)}_x}{\operatorname{tr}\tau^{(t)}_A}\big(\mathbb{1}_{A_t}-P_t\big); \tag{1.7}$$

and the joint announcement function a transcript protocol of (DX5) derives by composition from its L messages, message m having the sender π_m, whose readout map is π_{π_m}, the alphabet $\mathcal{C}_m$ and the table a_m,

$$\begin{gathered} \kappa_m(x,v) \coloneqq a_m\big(\pi_{\pi_m}(x,v),\ \kappa_1(x,v),\dots,\kappa_{m-1}(x,v)\big), \\ a(x,v) \coloneqq \big(\kappa_m(x,v)\big)_{m=1}^{L} \in \mathcal{K} \coloneqq \prod_{m=1}^{L} \mathcal{C}_m. \end{gathered} \tag{1.8}$$

Exact admissibility in the general format. The exact-admissibility paragraph above holds with these readings: the conditions on $\{F_v\}$ are decided per measured factor, on the family $\{F^f_{v_f}\}$, its blocks and its declared order, and on no product; the conditions on τ_A, on every M_x and on the right-hand side Θ_k of a general entry are decided block by block along $\mathbb{T}$, the trace of τ_A accumulated over the blocks in their declared order and rounding nowhere; and in ensemble form the conditions on $\{M_x\}$ are replaced by those of (DX4) on $\{N_x\}$, the denoted $\{M_x\}$ being a POVM by Lemma 1.3 and decided by nothing.

Definition 1.2 (Admissible declaration)**.** The tuple $\mathfrak{D}$ of Definition 1.1 is an *admissible declaration* when it satisfies: (i) all systems finite-dimensional,

all alphabets finite, all maps total; (ii) the marginal slot holds $\tau_A \in S_=(A)$ (trusted-source mode) or the value none (untrusted-source mode); in particular $\operatorname{tr}\tau_A = 1$ whenever a marginal is declared, so $\tau_A = 0$ is inadmissible; (iii) $\{M_x\}$ and $\{F_v\}$ are POVMs and every F_v and every $\hat{O}_k$ is sector-block-diagonal; (iv) r maps into $\mathcal{Z} \sqcup \{\perp\}$; (v) sifting is announcement-determined via ς; (vi) $\bowtie_k \in \{\preceq, =, \succeq\}$ and each entry carries one of the two declared forms of Definition 2.2, so that Θ_k is Hermitian on A, in trusted-source mode, respectively $\bowtie_k \in \{\leq, =, \geq\}$ and $\Theta_k = c_k \in \mathbb{R}$ in untrusted-source mode; (vii) $\mathcal{W} \neq \varnothing$; (viii) *party computability:* the declaration names a party partition $\mathfrak{P}$ (Definition 1.1) and

- each party announces a function of its own readout: a_A is defined on $\mathcal{O}_\mathsf{A}$ and a_B on $\mathcal{O}_\mathsf{B}$, with $a(x,v) = (a_A(\pi_\mathsf{A}(x,v)), a_B(\pi_\mathsf{B}(x,v)))$;
- the declared key holder $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$ computes the raw key from *its own readout* and the announcements, $r(x,v) = r_\mathsf{h}(\pi_\mathsf{h}(x,v), a(x,v))$ with r_h a declared total function, Eq. (1.1).

Under the default partition ($\pi_\mathsf{A}(x,v) = x$, $\pi_\mathsf{B}(x,v) = v$) these conditions read: $a_A : \mathcal{X} \to \mathcal{C}_A$, $a_B : \mathcal{V} \to \mathcal{C}_B$, and $r(x,v) = r_\mathsf{A}(x, a(x,v))$ resp. $r_\mathsf{B}(v, a(x,v))$.

In the general format of Definition 1.1 (DX1)–(DX6) the same eight conditions read as follows, and at one Alice sector, one measured factor, POVM form and two messages they are the conditions above. (iii): every M_x, the declared τ_A and the right-hand side of every general entry are block diagonal along Alice's sectors $\mathbb{T}$, *by their type as block lists against a declared factor tagged* A, and $\{M_x\}$ is a POVM; in ensemble form the declared $\{N_x\}$ satisfy (1.6) and $\{M_x\}$ is the family (1.7), a POVM block diagonal along $\mathbb{T}$ by Lemma 1.3 (i); $\{F_v\}$ is the product family (1.5) of per-factor POVMs, each block diagonal for its factor's sector map, and every $\hat{O}_k$ is block diagonal for the sector map of its scope. (vi): an entry may in addition carry the party-local form of Definition 2.2, in either mode, with $\bowtie_k \in \{\preceq, =, \succeq\}$ and $c_k \in \mathbb{R}$. (viii): each message of the transcript is a function of its sender's own readout and of the earlier messages, a is the composition (1.8), and (1.1) holds with that a.

Lemma 1.3 (The measurement an ensemble-form declaration denotes)**.** Let $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$, and for every t let $\tau_A^{(t)} \in \operatorname{Pos}(A_t)$ be non-zero and $N_x^{(t)} \in \operatorname{Pos}(A_t)$, $x \in \mathcal{X}$, with $\sum_x N_x^{(t)} = \tau_A^{(t)}$. Let P_t be the projector onto the support of $\tau_A^{(t)}$, let $(\tau_A^{(t)})^{-1/2}$ be the inverse square root on that support, zero off it, and let $M_x^{(t)}$ be the operator of (1.7), $M_x \coloneqq \bigoplus_t M_x^{(t)}$. Then:

(i) $M_x^{(t)} \succeq 0$ and $\sum_x M_x^{(t)} = \mathbb{1}_{A_t}$ for every t, so $\{M_x\}_{x\in\mathcal{X}}$ is a POVM on A and every element is block diagonal along $\mathbb{T}$;

(ii) $(\tau_A^{(t)})^{1/2} M_x^{(t)} (\tau_A^{(t)})^{1/2} = N_x^{(t)}$, and in particular $\operatorname{tr}[M_x \tau_A] = \sum_t \operatorname{tr} N_x^{(t)}$;

(iii) $M_x^{(t)} = 0$ if and only if $N_x^{(t)} = 0$;

(iv) conversely, if $\tau_A \in \mathrm{Pos}(A)$ and $\{M_x\}$ is a POVM on A, both block diagonal along $\mathbb{T}$, then $N_x := \tau_A^{1/2} M_x \tau_A^{1/2}$ is positive semidefinite and block diagonal along $\mathbb{T}$ and $\sum_x N_x = \tau_A$.

Definition 1.4 (Generated registers, isometries and maps)**.** The classical registers are X (alphabet $\mathcal{X}$), Y (alphabet $\mathcal{V}$), the announcement register $C_A C_B$ and I (alphabet $\mathcal{K}$ each), $\widehat{C}$ (alphabet $\widehat{\mathcal{C}}$), R and Z (alphabet $\mathcal{Z}_\perp$). The announcement register is one register, named after the two messages of the basic format; in the general format it holds the whole tuple $a(x, v)$ of messages. Define

$$V_{\mathrm{msm}} := \sum_{x \in \mathcal{X}} \sum_{v \in \mathcal{V}} |x, v\rangle_{XY} \otimes \left(\sqrt{M_x} \otimes \sqrt{F_v}\right), \quad \mathcal{H}(AQ) \to \mathcal{H}(XY\,AQ), \tag{1.9}$$

$$V_{\mathrm{ann}} := \sum_{x,v} |a(x,v)\rangle_{C_A C_B} \otimes |x,v\rangle\langle x,v|_{XY}\,, \qquad \text{controlled on } XY, \tag{1.10}$$

$$V_I := \sum_{\kappa \in \mathcal{K}} |\kappa\rangle_I \otimes |\kappa\rangle\langle\kappa|_{C_A C_B}\,, \qquad \text{controlled on } C_A C_B, \tag{1.11}$$

$$V_{\widehat{C}} := \sum_{\kappa \in \mathcal{K}} |\phi(\kappa)\rangle_{\widehat{C}} \otimes |\kappa\rangle\langle\kappa|_{C_A C_B}\,, \qquad \text{controlled on } C_A C_B, \tag{1.12}$$

$$V_R := \sum_{x,v} |r(x,v)\rangle_R \otimes |x,v\rangle\langle x,v|_{XY}\,, \qquad \text{controlled on } XY, \tag{1.13}$$

$$V_Z := \sum_{j \in \mathcal{Z}_\perp} |j\rangle_Z \otimes Z_j, \qquad Z_j := |j\rangle\langle j|_R\,, \qquad \text{controlled on } R, \tag{1.14}$$

each acting as the identity on all registers not named. The full protocol isometry and the *single-round protocol map* are

$$\begin{aligned} W &:= V_R\, V_{\widehat{C}}\, V_I\, V_{\mathrm{ann}}\, V_{\mathrm{msm}}, \\ \mathcal{M}(\omega) &:= \operatorname{tr}_{AQ\,XY\,C_A C_B\,R}\left[V_Z\, W\, \omega\, W^\dagger V_Z^\dagger\right], \qquad \mathcal{M} : AQ \to Z\widehat{C}I. \end{aligned} \tag{1.15}$$

The *key pinching* is $\mathcal{Z}(\tau) := \sum_{j \in \mathcal{Z}_\perp} Z_j\, \tau\, Z_j$ on R, the pinching of Definition "Operator conventions and pinching" (foundations/def:pinching) by the projectors Z_j (extended by the identity to any ambient system), and the *statistics map* is

$$\boldsymbol{\Phi}[\omega] := \sum_{\hat{c} \in \widehat{\mathcal{C}}} \left(\sum_{(x,v):\, \phi(a(x,v)) = \hat{c}} \operatorname{tr}\left[(M_x \otimes F_v)\,\omega\right] \right) \hat{e}_{\hat{c}}, \qquad \boldsymbol{\Phi} : \mathcal{L}(AQ) \to \mathbb{R}^{\widehat{\mathcal{C}}}, \tag{1.16}$$

with $\hat{e}_{\hat{c}}$ the unit vector at position $\hat{c}$.

Theorem 1.5 (The declaration generates a protocol channel)**.** Under Definition 1.2:

(i) Each of (1.9)–(1.14), and hence W, is an isometry, and $\mathcal{M}$ is a channel $AQ \to Z\widehat{C}I$ with $\widehat{C}$, I, Z classical.

(ii) For every system E and every $\omega \in S_{\leq}(AQE)$,

$$(\mathcal{M} \otimes \mathrm{id}_E)(\omega) = \sum_{x,v} |r(x,v)\rangle\langle r(x,v)|_Z \otimes |\phi(a(x,v))\rangle\langle \phi(a(x,v))|_{\widehat{C}} \otimes |a(x,v)\rangle\langle a(x,v)|_I \otimes \mathrm{tr}_{AQ}\big[(M_x \otimes F_v \otimes \mathbb{1}_E)\,\omega\big]. \tag{1.17}$$

(iii) The $\widehat{C}$-marginal of $(\mathcal{M} \otimes \mathrm{id}_E)(\omega)$ is the probability vector $\mathbf{\Phi}[\omega_{AQ}]$; $\mathbf{\Phi}$ is affine, and $\mathbf{\Phi}[\omega]$ is a probability distribution on $\widehat{\mathcal{C}}$ for every $\omega \in S_{=}(AQ)$.

Definition 1.6 (Branch maps)**.** For $\hat{c} \in \widehat{\mathcal{C}}$, a class $\kappa \in \phi^{-1}(\hat{c})$ with $\varsigma(\kappa) = \text{pass}$ is a *pass class* of $\hat{c}$, and one with $\varsigma(\kappa) = \text{fail}$ a *fail class*; let $\mathcal{K}_{\text{pass}}(\hat{c}) := \{\kappa \in \phi^{-1}(\hat{c}) : \varsigma(\kappa) = \text{pass}\}$ and $\mathcal{K}_{\text{fail}}(\hat{c}) := \{\kappa \in \phi^{-1}(\hat{c}) : \varsigma(\kappa) = \text{fail}\}$. For each class κ define the operator

$$K_\kappa := \sum_{(x,v)\in a^{-1}(\kappa)} |r(x,v)\rangle_R \otimes |x,v\rangle_{XY} \otimes |\kappa\rangle_{C_A C_B} \otimes \big(\sqrt{M_x} \otimes \sqrt{F_v}\big) \\ :\ \mathcal{H}(AQ) \to \mathcal{H}(R\,XY\,C_A C_B\,AQ), \tag{1.18}$$

and the CP *branch maps*, the pass map $\mathcal{G}_{\hat{c}}$ and the fail map $\Theta_{\hat{c}}$,

$$\mathcal{G}_{\hat{c}}(\omega) := \sum_{\kappa\in\mathcal{K}_{\text{pass}}(\hat{c})} K_\kappa\,\omega\,K_\kappa^\dagger, \qquad \Theta_{\hat{c}}(\omega) := \sum_{\kappa\in\mathcal{K}_{\text{fail}}(\hat{c})} K_\kappa\,\omega\,K_\kappa^\dagger. \tag{1.19}$$

For $s \in \mathbb{S}$, the sector versions $K_\kappa^{(s)}$, $\mathcal{G}_{\hat{c},s}$, $\Theta_{\hat{c},s}$ are (1.18)–(1.19) with $\sqrt{F_v}$ replaced by $\sqrt{F_v^{(s)}}$, acting $\mathcal{H}(AQ_s) \to \mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$.

Lemma 1.7 (Properties of the pass and fail branch maps)**.** Under Definition 1.2, for every $\hat{c} \in \widehat{\mathcal{C}}$:

(i) $\mathcal{G}_{\hat{c}}$ and $\Theta_{\hat{c}}$ are CP and trace-non-increasing, with

$$\mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega) + \mathrm{tr}\,\Theta_{\hat{c}}(\omega) = \mathbf{\Phi}[\omega](\hat{c}) \qquad \text{for all } \omega \in \mathrm{Pos}(AQ). \tag{1.20}$$

(ii) $\mathcal{G}_{\hat{c}}(\omega)$ is supported on $\mathrm{span}\{|j\rangle_R : j \in \mathcal{Z}\}$ and $\Theta_{\hat{c}}(\omega) = |\perp\rangle\langle\perp|_R \otimes \theta_{\hat{c}}(\omega)$ for a CP map $\theta_{\hat{c}}$; in particular their supports are orthogonal and $\mathcal{Z}(\Theta_{\hat{c}}(\omega)^p) = \Theta_{\hat{c}}(\omega)^p$ for every $p > 0$.

(iii) (Sector decomposition.) If $\omega = \bigoplus_{s\in\mathbb{S}} \omega_s$ is block-diagonal with respect to the decomposition $\mathcal{H}(AQ) = \bigoplus_s \mathcal{H}(AQ_s)$, then

$$\mathcal{G}_{\hat{c}}(\omega) = \bigoplus_{s\in\mathbb{S}} \mathcal{G}_{\hat{c},s}(\omega_s), \qquad \Theta_{\hat{c}}(\omega) = \bigoplus_{s\in\mathbb{S}} \Theta_{\hat{c},s}(\omega_s), \qquad \mathbf{\Phi}[\omega] = \sum_{s\in\mathbb{S}} \mathbf{\Phi}_s[\omega_s], \tag{1.21}$$

where the direct sums refer to the (mutually orthogonal) subspaces $\mathcal{H}(RXY\,C_AC_B) \otimes \mathcal{H}(AQ_s)$ and $\mathbf{\Phi}_s$ is (1.16) with $F_v^{(s)}$ in place of F_v.

(iv) If $\hat{c} \notin \widehat{\mathcal{C}}_{\text{key}}$ then $\mathcal{G}_{\hat{c}} = 0$.

The complete positivity of $\mathcal{G}_{\hat{c}}$ and $\Theta_{\hat{c}}$ in (i), and all of (iii), hold as well when $\{F_v\}$ is any family of sector-block-diagonal positive semidefinite operators, complete or not, the other components satisfying Definition 1.2.

2 Accumulation

Fact 2.1 (Marginal-constrained entropy accumulation, sequential form)**.** *([1, Thm. 4.2a, Eqs. (100)–(101)].)* For each $j \in \{1,\dots,n\}$ take a state $\sigma^{(j-1)} \in S_=(A_{j-1})$ and a channel $\mathcal{M}_j \in \mathrm{CPTP}(A_{j-1}E_{j-1},\, S_j\widehat{C}_jE_j)$ with $\widehat{C}_j$ classical. Let

$$\rho_{S_1^n\widehat{C}_1^nE_n} = \big(\mathcal{M}_n \circ \cdots \circ \mathcal{M}_1\big)\big[\omega^0\big], \qquad \omega^0 \in S_=(A_0^{n-1}E_0),$$

subject to the product marginal constraint $\omega^0_{A_0^{n-1}} = \sigma^{(0)} \otimes \cdots \otimes \sigma^{(n-1)}$; E_0 is unconstrained, and ω^0 ranges over *all* normalized states compatible with the marginal constraint (purity is not assumed). Suppose $\rho = p_\Omega\,\rho_{|\Omega} + (1-p_\Omega)\rho_{|\bar{\Omega}}$ with $p_\Omega \in (0,1]$ and $\rho_{|\Omega}, \rho_{|\bar{\Omega}}$ normalized, all $\widehat{C}_j$ isomorphic to one register $\widehat{C}$ with alphabet $\widehat{\mathcal{C}}$, and let S_Ω be a convex set of probability distributions on $\widehat{\mathcal{C}}$ such that $\mathrm{freq}_{\hat{c}_1^n} \in S_\Omega$ for every frequency vector with nonzero probability in $\rho_{|\Omega}$. Then, for every $\alpha \in (1,\infty)$ (locator note (vi): the source also covers the endpoint $\alpha = \infty$, which no claim of this module uses),

$$\widetilde{H}^{\uparrow}_{\alpha}\big(S_1^n\big|\widehat{C}_1^nE_n\big)_{\rho_{|\Omega}} \;\geq\; n\,h^{\uparrow}_{\alpha} - \frac{\alpha}{\alpha-1}\log_2\frac{1}{p_\Omega}, \tag{2.1}$$

where

$$h^{\uparrow}_{\alpha} = \inf_{\mathbf{q}\in S_\Omega}\ \inf_{\nu\in\Sigma}\left(\frac{\alpha}{\alpha-1}\,D\big(\mathbf{q}\,\big\|\,\boldsymbol{\nu}_{\widehat{C}}\big) + \sum_{\hat{c}\in\mathrm{supp}(\boldsymbol{\nu}_{\widehat{C}})} q(\hat{c})\,\widetilde{H}^{\uparrow}_{\alpha}(S|E\widetilde{E})_{\nu_{|\hat{c}}}\right), \tag{2.2}$$

with D the classical (base-2) KL divergence of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) and $\Sigma =$

$\Sigma_{S\widehat{C}E\widetilde{E}}$ the *marginal-constrained convex range* ([1, Def. 4.2, Eq. (98)]) of the channels $\mathcal{M}_j \otimes \mathrm{id}_{\widetilde{E}}$ and states $\sigma^{(j-1)}$:

$$\Sigma \coloneqq \mathrm{conv}\Big(\bigcup_{j=1}^{n} \Sigma_j\Big), \qquad \Sigma_j \coloneqq \big\{(\mathcal{M}_j \otimes \mathrm{id}_{\widetilde{E}})[\omega] : \omega \in S_{=}(A_{j-1}E_{j-1}\widetilde{E}),\, \omega_{A_{j-1}} = \sigma^{(j-1)}\big\}, \tag{2.3}$$

where every output is embedded in common registers $S\widehat{C}E\widetilde{E}$ (the results hold for any choice of embeddings; [1, footnote 12]) and $\widetilde{E}$ is a register of large enough dimension to serve as a purifying register for any of the $A_{j-1}E_{j-1}$ registers (enlargeable without loss of generality; [1, footnote 17]).

Locator notes. (i) The statement above is the source's Thm. 4.2a with its registers written in this module's fonts; its Eq. (100) is (2.1), its Eq. (101) is (2.2), and its Def. 4.2, Eq. (98) is (2.3). Thm. 4.2a is the special case of the source's Thm. 4.2b in which the per-round secret test registers (the unbarred C_j of the source) are trivial, which is the case Theorem 1.5 generates. The source states its entropies in base 2 (bits). (ii) The divergence in the penalty term is the *KL* divergence D: the source obtains the rate objective as the convex conjugate of a log-sum-exponential ([1, Lemma 4.11, Eqs. (113)–(114)]) and evaluates it in its Eq. (118), stating there explicitly that the first term "is the KL divergence". (iii) An alternative import is the tensor-product specialization [1, Cor. 4.2, Eq. (137)], in which the round channels carry no E_j output and Eve's entire action is the choice of one global input state. The source presents that corollary as a register substitution of the theorems above, and its Sec. 5 applies the resulting analysis when every public announcement is made only after *all* measurements are complete; for per-round ("on-the-fly") announcements — the model Theorem 1.5 generates, whose round channel announces I_j in every round — the source's own route (its Sec. 5, Protocols 1–2 and the reduction between them) is the sequential form quoted here, in which Eve carries a quantum memory E_{j-1} between rounds and may adapt each round's action to all earlier announcements. The present import therefore covers a strictly larger attack model than that specialization would. (iv) The theorem has no announcement register of its own: announcements enter the sequential model inside the E_j outputs (the source's Sec. 5: Eve "immediately incorporates" each I_j into her side information). (v) Indexing is off by one between this module's rounds and the source's registers ([1, Rem. 5.1]): round j here uses the source's $A_{j-1}E_{j-1}$. (vi) The source states the theorem for $\alpha \in (1, \infty]$, and its Def. 2.6 defines the entropies at the endpoint; Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) defines the sandwiched family on $(1, \infty)$ only, so the Fact is quoted, and Theorems 2.10 and 3.1 are stated, on $(1, \infty)$. No claim uses the endpoint: every key-length statement is confined to $(1, 2)$. (vii) *Trivial marginal registers are admissible* — the import check for untrusted-source mode, read off the source

at the locators named. The hypotheses that [1, Thm. 4.2a] places on the marginal registers are, in full, that one "take a state $\sigma^{(j-1)} \in S_=(A_{j-1})$, and a channel $M_j \in \mathrm{CPTP}(A_{j-1}E_{j-1}, S_j\widehat{C}_jE_j)$, such that $\widehat{C}_j$ are classical", together with the product condition $\omega_{A_0^{n-1}} = \sigma^{(0)}_{A_0} \otimes \cdots \otimes \sigma^{(n-1)}_{A_{n-1}}$ on the input state; and [1, Def. 4.2, Eq. (98)] likewise only "consider[s] channels $E_j \in \mathrm{CPTP}(A_{j-1}\bar{A}_{j-1}, Z_j)$ and states $\sigma^{(j-1)} \in S_=(A_{j-1})$" (in the source's symbols, where E_j is a channel and Z_j its output register, not this module's register E_j and raw key Z_j). *No dimension bound and no rank condition is imposed on A_{j-1} at either locator*, and the source's standing convention is explicit that none is imposed globally either: "we will assume that all systems are finite-dimensional, but we will not impose any bounds on the system dimensions unless otherwise specified" ([1, Sec. 2, the convention stated below its Table 1]). A one-dimensional A_{j-1} is therefore an admissible instantiation, and at it the product marginal condition is satisfied by every normalized ω, since $S_=(A_0^{n-1})$ is then a singleton — the hypothesis discharges vacuously rather than failing. Independently, the source reads the same equivalence into its own definition of channel conditional entropy: see the sentence following [1, Eq. (1)]. Nothing else in Fact 2.1 mentions A_{j-1} beyond the purifier requirement, which asks $\widetilde{E}$ to purify $A_{j-1}E_{j-1}$ and is only weakened by shrinking A_{j-1}.

Definition 2.2 (Licensed constraint list and its two classes)**.** The declared forms of an entry of $\mathfrak{L}$ — homogeneous, carrying a real scalar c_k, general, carrying a Hermitian Θ_k on A, and, in the general format, party-local — are those of Definition "Licensed constraint list" (protocol-compiler/def:homog), and the list is *licensed* when it is licensed for the declared source mode there. Its two classes for an untagged entry are the following; in the general format an untagged entry is read for its lifted operator, and the cited class also licenses a party-local entry in either mode (the paragraph on party-local entries).

(a) *trusted-source mode:* every entry of $\mathfrak{L}$ is declared in the homogeneous form, carrying $c_k \in \mathbb{R}$; the right-hand side it denotes is then *constructed* as $\Theta_k \coloneqq c_k\,\tau_A$ from the declared c_k and the declared τ_A, and such a list is a *τ_A-homogeneous list* by construction;

(b) *untrusted-source mode:* $\mathfrak{L}$ is a *scalar list*, i.e. every right-hand side is a real number $\Theta_k = c_k$ and the entry reads $\mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\right] \bowtie_k c_k$ — a comparison of real numbers, with $\bowtie_k \in \{\leq, =, \geq\}$.

Licensing is therefore a *type* check in both modes and compares no numbers. A general entry remains admissible input, every structural and numerical claim of this module being proved for it, but it is never licensed, so a list containing one carries no key-length claim, which is the safe direction. The

two classes agree at $d_A = 1$ (Proposition 5.2 (iii)), and (b) is what the licensing condition (2.8) transports in untrusted-source mode, through (2.14): *the operator form $\Theta_k = c_k \mathbb{1}_A$ is not licensed.*

Party-local entries. In the general format an entry may be declared in the party-local form (Definition "Licensed constraint list" (protocol-compiler/def:homog)); it is written $(\hat{O}_k, \bowtie_k, c_k; f)$, and its state-level reading, in either source mode, is

$$\mathrm{tr}_{Q_f}\Big[\big(\mathbb{1}\otimes(\hat{O}_k - c_k\,\mathbb{1}_{Q_f})\big)\,\omega\Big] \;\bowtie_k\; 0 \qquad \text{as an operator on the systems of } \omega \text{ other than } Q_f, \tag{2.4}$$

those systems being A and the measured factors $Q_{f'}$, $f' \neq f$.

Definition 2.3 (Round model and licensed attack sets)**.** Fix a declaration $\mathfrak{D}$ with single-round protocol channel $\mathcal{M} \in \mathrm{CPTP}(AQ \to Z\widehat{C}I)$ (Theorem 1.5). The round model of this contract is the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) with per-round system Q, Alice's system A and the declared source mode, the round index of Q and A dropped. An attack sequence is thus a family $(\mathcal{A}_j)_{j=1}^{n}$ of channels whose output carries the systems the adversary supplies in the declared source mode, the display being that definition's with its per-round system S taken to be Q:

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}(E_{j-1},\; Q\,E'_j), & \text{trusted-source mode,} \\ \mathrm{CPTP}(E_{j-1},\; A\,Q\,E'_j), & \text{untrusted-source mode,} \end{cases} \tag{2.5}$$

where E_0 is Eve's initial register and E'_j her outgoing memory. In trusted-source mode A is *not* an output of $\mathcal{A}_j$: it is supplied by Alice, inaccessible to the adversary, and carried into the round channel as a separate input — that, and nothing else, is what A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) licenses. The generated *round channel*s are

$$\mathcal{M}_j := \begin{cases} (\mathcal{M}\otimes \mathrm{id}_{E'_j}) \circ (\mathrm{id}_A \otimes \mathcal{A}_j) \;\in\; \mathrm{CPTP}(A\,E_{j-1},\; Z\,\widehat{C}\,E_j), & \text{trusted mode,} \\ (\mathcal{M}\otimes \mathrm{id}_{E'_j}) \circ \mathcal{A}_j \;\in\; \mathrm{CPTP}(E_{j-1},\; Z\,\widehat{C}\,E_j), & \text{untrusted mode,} \end{cases}$$
$$E_j := E'_j\, I_j : \tag{2.6}$$

the attack delivers the round input on Q — and, in untrusted-source mode, on A as well — together with Eve's new memory, the protocol channel produces $(Z_j, \widehat{C}_j, I_j)$, and the announcement I_j — classical and public — is placed in Eve's register. Content of E_{j-1} that Eve wishes to retain (earlier announcements included) she carries inside E'_j; every physical attack is represented by a sequence that retains *all* announcements, and Theorem 2.10 uses that representation, so that $E_n = I_1^n \widehat{E}$ with $\widehat{E}$ Eve's retained quantum

memory. The untrusted-mode round channel has no A input at all, which is what makes it an instance of Fact 2.1 with the marginal register A_{j-1} trivial (Fact 2.1, locator note (vii); Theorem 2.10).

For a licensed list $\mathfrak{L}$ (Definition 2.2) and finite-dimensional registers E', E'', the *licensed attack set* is the attack set of $\mathfrak{L}$ of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) at $S = Q$; its displays at that system are, in trusted-source mode,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, QE'') := \Big\{ \mathcal{A} \in \mathrm{CPTP}(E', QE'') :\ \mathcal{A}^{\dagger}\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'} \quad \forall k \Big\}, \tag{2.7}$$

and, with the constraint operator lifted to the larger supplied system,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, AQE'') := \Big\{ \mathcal{A} \in \mathrm{CPTP}(E', AQE'') :\ \mathcal{A}^{\dagger}\big(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'} \ \ \forall k \Big\} \tag{2.8}$$

in untrusted-source mode; the two are distinguished by the output system named in the argument. An attack sequence is $\mathfrak{L}$-admissible (an admissible attack sequence for $\mathfrak{L}$) if $\mathcal{A}_j$ lies in the set of (2.7), resp. (2.8), with $E' = E_{j-1}$ and $E'' = E'_j$, for every j. Where $\mathfrak{L}$ carries entries declared party-local (Definition 2.2), each such entry $(\hat{O}_k, \bowtie_k, c_k; f)$ contributes to the licensed attack set, in place of the scalar condition of (2.7) or (2.8), the condition of an entry tagged with the factor f in Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) (iii) at $S = Q$. With Q_{f^c} the product of the measured factors other than f, and R equal to A in untrusted-source mode and trivial in trusted-source mode, its display at that system is

$$\begin{aligned} &\big(\mathrm{id}_{R'} \otimes \mathcal{A}\big)^{\dagger}\Big(X \otimes \big(\hat{O}_k - c_k\, \mathbb{1}_{Q_f}\big) \otimes \mathbb{1}_{E''}\Big) \ \bowtie_k \ 0 \\ &\qquad \text{for every finite-dimensional } R' \text{ and every } X \in \mathrm{Pos}(R' R\, Q_{f^c}), \end{aligned} \tag{2.9}$$

as an operator on $R'E'$, the tensor factors of the argument written in the order $R' R\, Q_{f^c}$, Q_f, E''; untagged entries, those not declared party-local, keep their conditions, read for their lifted operators (Definition "Protocol declaration" (protocol-compiler/def:declaration) (DX6)). The reference system R' is what carries Alice's retained system in trusted-source mode, where A is no output of $\mathcal{A}$. With R' trivial and $X = \mathbb{1}$ the condition is the scalar one for the lifted operator, so the set with tagged entries is contained in the set with the same entries untagged. For $\mathfrak{L} = \varnothing$ the condition is empty: $\mathfrak{A}_{\varnothing} = \mathrm{CPTP}$, every attack sequence is admissible, and the model is unrestricted. The adjoint condition in (2.7) is exactly the published squashed-attack-set format of [3, Eq. (151)], and restricting attention to a declared subset of channels is the standard modelling step of [3, Def. 8.3, Rem. 8.4]. Whether a given physical protocol reduces to this model with a given $\mathfrak{A}_{\mathfrak{L}}$ is *not* decided here: it is what A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) certifies.

Definition 2.4 (Constrained target set and single-round rate)**.** For a register $\widetilde{E}'$ and a (possibly empty) declared list $\mathfrak{L}$, let

$$\Sigma^{\mathfrak{L}}(\widetilde{E}') \coloneqq \big\{(\mathcal{M}\otimes \mathrm{id}_{\widetilde{E}'})(\sigma):\ \sigma\in S_{=}(AQ\widetilde{E}'),\ \sigma\models\mathfrak{D},\ \sigma_{AQ}\models\mathfrak{L}\big\}, \tag{2.10}$$

where $\sigma\models\mathfrak{D}$ abbreviates the declaration's marginal condition,

$$\sigma\models\mathfrak{D} \quad :\Longleftrightarrow \quad \begin{cases} \sigma_A=\tau_A, & \text{trusted-source mode,}\\ \text{no condition beyond } \operatorname{tr}\sigma=1, & \text{untrusted-source mode,}\end{cases} \tag{2.11}$$

and $\sigma_{AQ}\models\mathfrak{L}$ abbreviates "σ_{AQ} satisfies every entry of $\mathfrak{L}$" in the form Definition 2.2 gives it for the declared mode, i.e. $\operatorname{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\,\sigma_{AQ}]\bowtie_k\Theta_k$ in trusted-source mode and $\operatorname{tr}\big[(\mathbb{1}_A\otimes\hat{O}_k)\,\sigma_{AQ}\big]\bowtie_k c_k$ in untrusted-source mode, for all k; an entry declared party-local is read, in either mode, in the form (2.4) of Definition 2.2, an affine semidefinite condition on σ_{AQ}. $\Sigma^{\mathfrak{L}}(\widetilde{E}')$ is convex in either mode: the state set is the intersection of the convex $S_{=}(AQ\widetilde{E}')$ with affine equalities and operator- or scalar-inequality conditions, each convex, and $\mathcal{M}\otimes\mathrm{id}_{\widetilde{E}'}$ is affine. Define the *constrained single-round rate*

$$h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}}\ \inf_{\substack{\sigma\in S_{=}(AQ\widetilde{E})\\ \sigma\models\mathfrak{D},\ \sigma_{AQ}\models\mathfrak{L}}} \left(\frac{\alpha}{\alpha-1}\,D(\mathbf{q}\,\|\,\mathbf{\Phi}[\sigma_{AQ}]) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c})\,\widetilde{H}_\alpha^{\uparrow}(Z|I\widetilde{E})_{\nu^\sigma_{|\hat{c}}}\right), \tag{2.12}$$

with $\nu^\sigma\coloneqq(\mathcal{M}\otimes\mathrm{id}_{\widetilde{E}})(\sigma)$ and $\dim\mathcal{H}(\widetilde{E})\geq\dim\mathcal{H}(AQ)$; by Lemma 2.7 the value does not depend on that choice. For $\mathfrak{L}=\varnothing$ we write $h_\alpha^{\uparrow,\mathrm{QKD}}\coloneqq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$, the unconstrained single-round rate, which Theorem 2.10 writes out as (2.17). Since a nonempty $\mathfrak{L}$ shrinks the inner feasible set, $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})\geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ always.

Definition 2.5 (Admissible acceptance set)**.** An *admissible acceptance set* is a closed convex subset $\mathcal{S}_{\mathrm{acc}}$ of the probability simplex on $\widehat{\mathcal{C}}$.

Lemma 2.7 (Independence of the purifier size)**.** Fix $\mathfrak{L}$ (possibly empty) and write $h(\widetilde{E}')$ for the value of (2.12) computed with a register $\widetilde{E}'$ of $\dim\mathcal{H}(\widetilde{E}')\geq\dim\mathcal{H}(AQ)$ in place of $\widetilde{E}$. Then $h(\widetilde{E}')$ is the same for every such $\widetilde{E}'$, and the inner infimum is unchanged if restricted to pure σ.

Lemma 2.8 (Constraint transport and range inclusion)**.** Let $\mathfrak{L}$ be licensed (Definition 2.2) or empty, and let $\mathcal{A}_j$ be $\mathfrak{L}$-admissible for the declared source mode (Definition 2.3).

(i) *(Transport.) Trusted-source mode:* for every $\omega \in S_{=}(A\,E_{j-1}\,\widetilde{E})$ with $\omega_A = \tau_A$, the state $\varsigma := (\mathrm{id}_A \otimes \mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega) \in S_{=}(AQE'_j\widetilde{E})$ satisfies $\varsigma_A = \tau_A$ and $\varsigma_{AQ} \models \mathfrak{L}$. *Untrusted-source mode:* for every $\omega \in S_{=}(E_{j-1}\,\widetilde{E})$, the state $\varsigma := (\mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega) \in S_{=}(AQE'_j\widetilde{E})$ satisfies $\varsigma \models \mathfrak{D}$ — which in that mode is only $\mathrm{tr}\,\varsigma = 1$ — and $\varsigma_{AQ} \models \mathfrak{L}$ in the scalar form of Definition 2.2 (b). *In either mode*, every entry of $\mathfrak{L}$ declared party-local holds of ς_{AQ} in the form (2.4), the attack being admissible for it in the sense of (2.9).

(ii) *(Range inclusion.)* With $\mathcal{M}_j$ the round channel (2.6) and Σ_j its output set (2.3), $\Sigma_j \subseteq \Sigma^{\mathfrak{L}}(E'_j\widetilde{E})$ up to the regrouping of the registers of the output, $(Z\widehat{C})(E'_jI_j)\widetilde{E} \mapsto (Z\widehat{C}I)(E'_j\widetilde{E})$; hence, embedding the finitely many $E'_j\widetilde{E}$ into a common register $\widetilde{E}^*$ with $\dim \mathcal{H}(\widetilde{E}^*) = \max_j \dim \mathcal{H}(E'_j\widetilde{E})$,

$$\Sigma = \mathrm{conv}\left(\bigcup_j \Sigma_j\right) \;\subseteq\; \Sigma^{\mathfrak{L}}(\widetilde{E}^*).$$

(iii) *(Antitonicity.)* If $\Sigma \subseteq \Sigma'$ then the value of (2.2) computed over Σ is at least its value computed over Σ'.

Lemma 2.9 (Product marginal of Alice's retained systems)**.** Let the declaration be in trusted-source mode with $d_A \geq 2$, and let its A, τ_A and $\{M_x\}$ be those of a source specification (Definition "Source specification" (transmitter/def:source-spec)) that carries a source-replacement certificate Ψ for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against its virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$ on A'', in any of the forms of Definition "Source-replacement certificate" (transmitter/def:srcert), an ensemble presentation being read as the block-form certificate it defines; let its source segment satisfy the source identity $\Xi(\rho_x^{\mathrm{v}}) = \rho_x$ for every $x \in \mathcal{X}$, with Ξ the composite channel of the segment and $\Xi = \mathrm{id}$ when the segment is empty (Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (S3)). Let A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), A-SEED (Assumption 13.3) and A-LAB (Assumption 13.8) at the party to which the party partition assigns A hold of the physical protocol, let the adversary's initial register E be finite-dimensional, and let σ_E be the state in which it starts. Put

$$\omega^0 := \left(\bigotimes_{j=1}^{n} \Psi_{A_jA''_j}\right) \otimes \sigma_E \;\in\; S_{=}(A_1^n E_0), \qquad E_0 := (A'')_1^n\, E. \tag{2.15}$$

Then:

(i) *(source replacement over the n rounds)* against every attack, the physical protocol and the protocol that starts from ω^0, applies Ξ to A''_j in round j in the place of the source's emission, and applies $\mathcal{N}_{\mathsf{A}}$, Alice's measurement as the channel of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii), to A_j at the point of round j at which the declared protocol's measurements are made, produce the same state after every round, on every register that both then hold; in the second protocol A_j is an input of that measurement and of no other operation, which is how the round model of Definition 2.3 takes Alice's system in trusted-source mode;

(ii) $\omega^0_{A_1^n} = \tau_A^{\otimes n}$;

(iii) for every attack sequence of Definition 2.3 and every j, the state on which the round channel $\mathcal{M}_j$ acts, in a run of the round model from ω^0, has the marginal τ_A on A_j.

At $d_A = 1$, where $\tau_A = [1]$ and no certificate is declared, (ii) and (iii) hold of every state on $A_1^n E_0$.

Theorem 2.10 (Accumulated entropy bound under adaptive attacks). Let $\mathfrak{D}$ satisfy Definition 1.2, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let Assumption A-RHOM (Assumption "Round homogeneity of the declaration" (protocol-compiler/ass:rhom)) hold; in trusted-source mode let A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), 13.3 (A-SEED) and 13.8 (A-LAB) at the party to which the party partition assigns A hold as well, and, at $d_A \geq 2$, let the declaration's A, τ_A and $\{M_x\}$ be those of a source specification carrying a source-replacement certificate against its virtual ensemble, with a source segment satisfying the source identity, as Lemma 2.9 states them; together with these, the adversary's initial register being finite-dimensional as every register of the round model is, the five assumptions are the hypotheses of Lemma 2.9, so that by that lemma the protocol's n-round input in that mode has the marginal below (in untrusted-source mode the lemma is not invoked and none of the hypotheses named for that mode is used here). Let $(\mathcal{A}_j)_{j=1}^n$ be an *arbitrary* attack sequence (Definition 2.3; no admissibility restriction; its registers E_0 and E'_j finite-dimensional, of any dimensions, as every register of that model is), let

$$\rho_{Z_1^n \widehat{C}_1^n I_1^n \widehat{E}} = \big(\mathcal{M}_n \circ \cdots \circ \mathcal{M}_1\big)\big[\omega^0\big], \qquad \omega^0 \in S_=(A_1^n E_0) \text{ with } \omega^0_{A_1^n} = \tau_A^{\otimes n} \text{ (trusted mode)},$$

respectively $\omega^0 \in S_=(E_0)$ arbitrary (untrusted mode, where the round channels take no A input), with $\mathcal{M}_j$ the round channels (2.6) and Eve's final

register written $E_n = I_1^n \widehat{E}$ (Definition 2.3), and let Ω_{AT}, the acceptance event, be the event that the frequency vector $\mathrm{freq}_{\hat{c}_1^n}$ of the acceptance values of the n rounds lies in $\mathcal{S}_{\mathrm{acc}}$, with $\Pr[\Omega_{\mathrm{AT}}] > 0$. Then for every $\alpha \in (1, \infty)$,

$$\widetilde{H}_\alpha^\uparrow\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\geq\; n\, h_\alpha^{\uparrow,\mathrm{QKD}} - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]}, \tag{2.16}$$

with $h_\alpha^{\uparrow,\mathrm{QKD}} = h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ the single-round rate of Definition 2.4 at the empty list, which reads

$$h_\alpha^{\uparrow,\mathrm{QKD}} = \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\substack{\omega\in S_=(AQ\widetilde{E}) \\ \omega\models\mathfrak{D}}} \left(\frac{\alpha}{\alpha-1}\, D(\mathbf{q} \,\|\, \mathbf{\Phi}[\omega_{AQ}]) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c})\, \widetilde{H}_\alpha^\uparrow(Z|I\widetilde{E})_{\nu^\omega_{|\hat{c}}} \right),$$
$$\nu^\omega := (\mathcal{M} \otimes \mathrm{id}_{\widetilde{E}})(\omega), \tag{2.17}$$

with $\omega \models \mathfrak{D}$ as in (2.11) — so the inner infimum is over states with $\omega_A = \tau_A$ in trusted-source mode and over *all* normalized states on $AQ\widetilde{E}$ in untrusted-source mode. It ranges over all (not necessarily pure) such states, with $\dim \mathcal{H}(\widetilde{E}) \geq \dim \mathcal{H}(A) \cdot \dim \mathcal{H}(Q)$ (Lemma 2.7: the value is independent of that choice).

3 Constrained accumulation

Theorem 3.1 (Constrained accumulation under the attack-set licence)**.** Let the hypotheses of Theorem 2.10 hold in either source mode, let $\mathfrak{L}$ be licensed for that mode (Definition 2.2), let A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) hold, and let the attack sequence be $\mathfrak{L}$-admissible (Definition 2.3). Then for every $\alpha \in (1, \infty)$,

$$\widetilde{H}_\alpha^\uparrow\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\geq\; n\, h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]}, \tag{3.1}$$

with $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ the constrained single-round rate of Definition 2.4, and $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing) = h_\alpha^{\uparrow,\mathrm{QKD}}$.

4 Branch entropy and sectors

Theorem 4.1 (Branch entropy from the protocol marginal)**.** For $\alpha \in (1, \infty)$, $\omega \in \mathrm{Pos}(AQ)$ and $\hat{c} \in \widehat{\mathcal{C}}$ define

$$P_{\hat{c}}(\omega) := \mathbf{\Phi}[\omega](\hat{c}), \qquad T_{\hat{c}}(\omega) := \mathrm{tr}\Big[\Big(\mathcal{Z}\big(\mathcal{G}_{\hat{c}}(\omega)^{1/\alpha}\big)\Big)^\alpha\Big], \qquad K_{\hat{c}}(\omega) := T_{\hat{c}}(\omega) + P_{\hat{c}}(\omega) - \mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega), \tag{4.1}$$

and, for block-diagonal $\omega = \bigoplus_s \omega_s$, the sector deficits

$$f_{\hat{c},s}(\omega_s) := \mathrm{tr}\Big[\Big(\mathcal{Z}\big(\mathcal{G}_{\hat{c},s}(\omega_s)^{1/\alpha}\big)\Big)^\alpha\Big] - \mathrm{tr}\,\mathcal{G}_{\hat{c},s}(\omega_s). \tag{4.2}$$

Let $\alpha \in (1,\infty)$, let $\omega \in S_=(AQ\widetilde{E})$ be *pure*, let $\nu^\omega = (\mathcal{M}\otimes \mathrm{id}_{\widetilde{E}})(\omega)$, and let $\hat{c} \in \widehat{\mathcal{C}}$ with $P_{\hat{c}} \coloneqq P_{\hat{c}}(\omega_{AQ}) > 0$. Then:

(i) *(the identity)*

$$\widetilde{H}^{\downarrow}_{\alpha}(Z|I\widetilde{E})_{\nu^\omega_{|\hat{c}}} = \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}(\omega_{AQ})}{P_{\hat{c}}}; \tag{4.3}$$

(ii) *(the bounds)* $0 \le K_{\hat{c}}(\omega_{AQ}) \le P_{\hat{c}}$, so the right-hand side of (4.3) is non-negative;

(iii) *(the sector form, on the whole cone)* for every $\hat{c} \in \widehat{\mathcal{C}}$ and *every* block-diagonal $\varrho = \bigoplus_s \varrho_s \in \mathrm{Pos}(AQ)$, with none of the hypotheses above on ω and $P_{\hat{c}}$ (no purity, no normalization and no hypothesis $P_{\hat{c}}(\varrho) > 0$),

$$K_{\hat{c}}(\varrho) = P_{\hat{c}}(\varrho) + \sum_{s\in\mathbb{S}} f_{\hat{c},s}(\varrho_s). \tag{4.4}$$

The two bounds of (ii) hold on the whole cone, as a statement separate from the identity (i): for every $\alpha \in (1,\infty)$, every $\hat{c} \in \widehat{\mathcal{C}}$ and *every* $\varrho \in \mathrm{Pos}(AQ)$, with no purity, no normalization, no block-diagonality and no hypothesis $P_{\hat{c}}(\varrho) > 0$, one has $0 \le K_{\hat{c}}(\varrho) \le P_{\hat{c}}(\varrho)$. We call this the theorem's *cone clause*.

Corollary 4.2 (Concavity of the branch quantities on subnormalised states)**.** For $\alpha \in (1,\infty)$, the map $X \mapsto \mathrm{tr}\big[(\mathcal{Z}(X^{1/\alpha}))^\alpha\big] = Q_{1/\alpha}(X)$ is concave on $S_{\le}()$ of its space; consequently, for every $\hat{c}$, s, the deficit $f_{\hat{c},s}$ of (4.2) is concave on $\{\sigma \in \mathrm{Pos}(AQ_s) : \mathrm{tr}\,\sigma \le 1\}$, and $K_{\hat{c}}$ is concave on $\mathcal{W}$.

Lemma 4.3 (Concavity on the positive cone)**.** Let $\alpha \in (1,\infty)$, put $p \coloneqq 1/\alpha \in (0,1)$, and let $\mathcal{Z}$ be a pinching built from a complete family of mutually orthogonal projectors. On the cone $\mathrm{Pos}()$ of the space $\mathcal{Z}$ acts on, consider the functional

$$T(X) \coloneqq \mathrm{tr}\big[\big(\mathcal{Z}(X^p)\big)^\alpha\big]. \tag{4.7}$$

(a) T is positively 1-homogeneous: $T(tX) = t\,T(X)$ for every $t \ge 0$.

(b) T is concave on all of $\mathrm{Pos}()$.

(c) Consider a declaration whose components other than $\{F_v\}$ satisfy Definition 1.2 and whose $\{F_v\}$ is any family of sector-block-diagonal positive semidefinite operators, complete or not. Then each deficit $f_{\hat{c},s}$ of (4.2) is concave on all of $\mathrm{Pos}(AQ_s)$, and $K_{\hat{c}}$ of (4.1) is concave on the sector-block-diagonal part of $\mathrm{Pos}(AQ)$, in particular on $\mathcal{W}$.

Corollary 4.4 (Non-positivity of the sector deficits)**.** For every $\hat{c}$, s and *every* $\sigma \in \mathrm{Pos}(AQ_s)$, with no trace hypothesis (in particular not $\operatorname{tr}\sigma \leq 1$), $f_{\hat{c},s}(\sigma) \leq 0$, and $f_{\hat{c},s}(\sigma) = 0$ if and only if $\mathcal{G}_{\hat{c},s}(\sigma)$ is block-diagonal for the key pinching $\mathcal{Z}$.

Definition 4.5 (Key-inert pair)**.** For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}$ define the *effective outcome set* of a pass class $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ on sector s as

$$E_\kappa^{(s)} \coloneqq \big\{(x,v) \in a^{-1}(\kappa) :\ M_x \neq 0,\ F_v^{(s)} \neq 0\big\}.$$

The pair $(\hat{c}, s)$ is *key-inert* if for every $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ the raw-key map r is constant on $E_\kappa^{(s)}$. This is a finite, mechanically checkable predicate on the declaration.

Proposition 4.6 (Exact cancellation of key-inert sectors)**.** If $(\hat{c}, s)$ is key-inert, then for *every* $\sigma \in \mathrm{Pos}(AQ_s)$ the operator $\mathcal{G}_{\hat{c},s}(\sigma)$ is block-diagonal for the key pinching $\mathcal{Z}$, and consequently $f_{\hat{c},s}(\sigma) = 0$. In particular such sectors contribute nothing to $K_{\hat{c}}$ in (4.4), for every input; they remain optimization variables, entering through $\mathbf{\Phi}$, $P_{\hat{c}}$, and the marginal constraint.

Definition 4.7 (Key-active index set)**.** The *key-active index set* of $\hat{c}$ is $\mathbb{S}_K(\hat{c}) \coloneqq \{s \in \mathbb{S} : (\hat{c}, s) \text{ is not key-inert}\}$. By Proposition 4.6, $K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_{s \in \mathbb{S}_K(\hat{c})} f_{\hat{c},s}(\omega_s)$ for block-diagonal ω; the objective computed with the sum restricted to $\mathbb{S}_K(\hat{c})$ is *exactly equal* to the full sum, not a bound.

Definition 4.8 (Joint-sector deficits and the joint key-active index set)**.** Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1, with Alice sectors $t \in \mathbb{T}$ and joint sectors $(t,s) \in \mathbb{T} \times \mathbb{S}$. Regard $\mathrm{Pos}(A_tQ_s)$ as the subcone of $\mathrm{Pos}(AQ_s)$ of operators supported on $\mathcal{H}(A_t) \otimes \mathcal{H}(Q_s)$. For $\hat{c} \in \widehat{\mathcal{C}}$ the *joint-sector* maps, statistics and deficit are the restrictions of the sector objects of Definition 1.6 and of (4.2) to that subcone,

$$\mathcal{G}_{\hat{c},(t,s)} \coloneqq \mathcal{G}_{\hat{c},s}\big|_{\mathrm{Pos}(A_tQ_s)}, \qquad P_{\hat{c},(t,s)}(\sigma) \coloneqq P_{\hat{c}}(\sigma), \qquad f_{\hat{c},(t,s)}(\sigma) \coloneqq f_{\hat{c},s}(\sigma) \qquad (\sigma \in \mathrm{Pos}(A_tQ_s)), \tag{4.8}$$

so that on such a σ the Kraus operators (1.18) act through $\sqrt{M_x^{(t)}} \otimes \sqrt{F_v^{(s)}}$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and a pass class $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ the *effective outcome set on the joint sector* is

$$E_\kappa^{(t,s)} \coloneqq \big\{(x,v) \in a^{-1}(\kappa) :\ M_x^{(t)} \neq 0,\ F_v^{(s)} \neq 0\big\} \subseteq E_\kappa^{(s)}, \tag{4.9}$$

and the pair $(\hat{c}, (t,s))$ is *key-inert* if r is constant on $E_\kappa^{(t,s)}$ for every $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$. The *joint key-active index set* is $\mathbb{J}_K(\hat{c}) \coloneqq \{(t,s) : (\hat{c}, (t,s)) \text{ is not key-inert}\}$.

Key-inertness of a joint pair is decided on declared data. In POVM form $M_x^{(t)} \neq 0$ is a comparison of the declared block with the zero block. In ensemble form it is $N_x^{(t)} \neq 0$, by Lemma 1.3 (iii), again a comparison of a declared block with zero. For a factorised family $F_v^{(s)} = \bigotimes_f F_{v_f}^{f,(s_f)}$ is non-zero exactly when every factor block $F_{v_f}^{f,(s_f)}$ is, a tensor product of matrices vanishing exactly when one of them does; so the predicate is decided per factor and no product is formed. With $|\mathbb{T}| = 1$ one has $E_\kappa^{(t_0,s)} = E_\kappa^{(s)}$, and the pair $(\hat{c}, (t_0, s))$ is key-inert exactly when $(\hat{c}, s)$ is in the sense of Definition 4.5, with $\mathbb{J}_K(\hat{c}) = \{t_0\} \times \mathbb{S}_K(\hat{c})$.

Lemma 4.9 (The objective split over joint sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, let $\alpha \in (1, \infty)$ and $\hat{c} \in \widehat{\mathcal{C}}$.

(i) *(decomposition)* For $s \in \mathbb{S}$ and $\sigma = \bigoplus_t \sigma_t \in \mathrm{Pos}(AQ_s)$ with $\sigma_t \in \mathrm{Pos}(A_tQ_s)$,

$$\mathcal{G}_{\hat{c},s}(\sigma) = \bigoplus_{t\in\mathbb{T}} \mathcal{G}_{\hat{c},(t,s)}(\sigma_t), \qquad \Theta_{\hat{c},s}(\sigma) = \bigoplus_{t\in\mathbb{T}} \Theta_{\hat{c},s}(\sigma_t), \qquad \mathbf{\Phi}_s[\sigma] = \sum_t \mathbf{\Phi}_s[\sigma_t], \tag{4.10}$$

the direct sums along the mutually orthogonal subspaces $\mathcal{H}(RXYC_AC_B)\otimes \mathcal{H}(A_tQ_s)$.

(ii) *(additivity)* For such σ, $f_{\hat{c},s}(\sigma) = \sum_t f_{\hat{c},(t,s)}(\sigma_t)$. Consequently, for every $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$ block diagonal along the joint sectors,

$$P_{\hat{c}}(\omega) = \sum_{(t,s)} P_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big), \qquad K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_{(t,s)} f_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big), \tag{4.11}$$

and for every $\sigma \in \mathrm{Pos}(A_tQ_s)$: $f_{\hat{c},(t,s)}(\sigma) \leq 0$, $0 \leq P_{\hat{c},(t,s)}(\sigma) + f_{\hat{c},(t,s)}(\sigma)$, and $f_{\hat{c},(t,s)}$ is concave and positively 1-homogeneous on $\mathrm{Pos}(A_tQ_s)$.

(iii) *(key-inert joint pairs)* If $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $(\hat{c}, (t, s))$ is key-inert (Definition 4.8), then $f_{\hat{c},(t,s)}(\sigma) = 0$ for *every* $\sigma \in \mathrm{Pos}(A_tQ_s)$. Hence the second sum of (4.11) may be restricted to $(t, s) \in \mathbb{J}_K(\hat{c})$, as an identity and not as a bound.

(iv) *(relation to the sector notions)* If $(\hat{c}, s)$ is key-inert in the sense of Definition 4.5 then $(\hat{c}, (t, s))$ is key-inert for every t; the converse fails in general, and holds when $|\mathbb{T}| = 1$.

5 The program and its reduction

Definition 5.1 (Feasible set and computable objective)**.** For a declaration $\mathfrak{D}$, $\mathcal{W}$ is the feasible set of Definition "Feasible set" (protocol-compiler/def:

feasible). Read as a set of operators on AQ, a list of sector blocks being the block-diagonal operator it denotes, it is

$$\mathcal{W} \coloneqq \Big\{\omega \in \mathrm{Pos}(AQ) :\ (\mathbb{1}_A \otimes \Pi_s)\,\omega\,(\mathbb{1}_A \otimes \Pi_{s'}) = 0\ \forall s \neq s', \quad \operatorname{tr}\omega = 1, \quad \omega \models \mathfrak{D}, \quad \omega \models \mathfrak{L}\Big\}, \tag{5.1}$$

writing ω_s for the sector blocks, with $\omega \models \mathfrak{D}$ the marginal condition of (2.11) and $\omega \models \mathfrak{L}$ the constraint conditions in the form Definition 2.2 gives them for the declared mode. Written out, $\mathcal{W}$ is

$$\begin{aligned} &\text{trusted:} \quad \omega_A = \tau_A, \quad \operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k \Theta_k \ \ \forall k; \\ &\text{untrusted:} \quad \text{(no marginal condition)}, \quad \operatorname{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big] \bowtie_k c_k \ \ \forall k. \end{aligned} \tag{5.2}$$

The normalization $\operatorname{tr}\omega = 1$ is written *explicitly* in (5.1), and is not derived from anything: in trusted-source mode it also follows from $\omega_A = \tau_A$ (whence $\operatorname{tr}\omega = \operatorname{tr}\tau_A = 1$), but that derivation is unavailable in untrusted-source mode, where normalization is the *only* unconditional constraint the feasible set carries. Leaving it implied would empty the untrusted-mode program of its one defining condition. $\mathcal{W}$ is convex and compact in either mode and in either format (Definition "Feasible set" (protocol-compiler/def:feasible)). For $\alpha \in (1,\infty)$, $\mathbf{q}$ a distribution on $\widehat{\mathcal{C}}$ and $\omega \in \mathcal{W}$, define the *objective*

$$O_\alpha(\mathbf{q},\omega) \coloneqq \frac{\alpha}{\alpha-1} \sum_{\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}(\omega)} + \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}} \left[q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}(\omega)} + \frac{q(\hat{c})}{\alpha-1} \log_2 \frac{q(\hat{c})}{K_{\hat{c}}(\omega)} \right], \tag{5.3}$$

with the conventions of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) at vanishing arguments ($0 \log_2 \frac{0}{v} = 0$; $u \log_2 \frac{u}{0} = +\infty$ for $u > 0$), and

$$\underline{h}_\alpha \coloneqq \inf_{\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}} \ \inf_{\omega \in \mathcal{W}} \ O_\alpha(\mathbf{q},\omega). \tag{5.4}$$

The general format. For a declaration in the general format of Definition 1.1, $\mathcal{W}$ is the set of lists over the joint sectors of Definition "Feasible set" (protocol-compiler/def:feasible), $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$, with the s-block $\hat{O}_k^{\uparrow(s)}$ of the lifted operator of an entry scoped to f_k and, for a party-local entry scoped to f, the split $s = (s_{f^c}, s_f)$ and the product $Q_{f^c,s_{f^c}}$ of the other factors' blocks written there, with every block positive semidefinite; its conditions read, block by block,

$$\begin{aligned} &\textstyle\sum_{(t,s)} \operatorname{tr}\omega^{(t,s)} = 1 && \text{(normalization)}; \\ &\textstyle\sum_s \operatorname{tr}_{Q_s}\omega^{(t,s)} = \tau_A^{(t)} \quad \forall t && \text{(marginal, trusted-source mode)}; \\ &\textstyle\sum_s \operatorname{tr}_{Q_s}\big[(\mathbb{1}_{A_t} \otimes \hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\big] \bowtie_k \Theta_k^{(t)} \quad \forall t && \text{(trusted; } \Theta_k^{(t)} = c_k\tau_A^{(t)} \text{ if homogene} \\ &\textstyle\sum_{(t,s)} \operatorname{tr}\big[(\mathbb{1}_{A_t} \otimes \hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\big] \bowtie_k c_k && \text{(scalar entry, untrusted-source mod} \\ &\textstyle\sum_{s_f \in \mathbb{S}^f} \operatorname{tr}_{Q_{f,s_f}}\Big[\big(\mathbb{1} \otimes (\hat{O}_k^{(s_f)} - c_k\mathbb{1})\big)\,\omega^{(t,(s_{f^c},s_f))}\Big] \bowtie_k 0 \quad \forall (t,s_{f^c}) && \text{(party-local, either mode)}, \end{aligned} \tag{5.5}$$

the last as an operator on $\mathcal{H}(A_t) \otimes \mathcal{H}(Q_{f^c,s_{f^c}})$. The objective (5.3) and $\underline{h}_\alpha$ of (5.4) are defined on this $\mathcal{W}$ by the same formulas.

Proposition 5.2 (The two source modes coincide at a trivial Alice system)**.** Let $d_A = 1$. Then trusted-source mode admits exactly one marginal, $\tau_A = [1]$, and:

(i) the two feasible sets of (5.1) coincide;

(ii) the n-round condition $\omega_{A_1^n} = \tau_A^{\otimes n}$ of Lemma 2.9 (ii) holds for every normalized state; it is satisfied vacuously;

(iii) the licensed constraint classes of Definition 2.2 coincide: trusted-mode τ_A-homogeneity is exactly the scalar typing of untrusted-source mode;

(iv) A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) is vacuous: what it licenses is the condition $\operatorname{tr}\omega = 1$, satisfied by every normalized state, so it deletes no attack.

Theorem 5.4 (Reduction of the rate to the convex program). Let $\mathfrak{D}$ satisfy Definition 1.2 with $\mathfrak{L} = \varnothing$, in either source mode, and let $\alpha \in (1,\infty)$. Then, with $h_\alpha^{\uparrow,\mathrm{QKD}} = h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ the single-round rate of Definition 2.4 at the empty list,

$$h_\alpha^{\uparrow,\mathrm{QKD}} \;\geq\; \underline{h}_\alpha. \tag{5.6}$$

Proposition 5.5 (Value neutrality along Alice's sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, in either source mode and with an arbitrary list $\mathfrak{L}$, general and party-local entries included. Let $\mathcal{W}^\circ$ be the set defined as $\mathcal{W}$ is in Definition 5.1 but with block diagonality required along $\mathbb{S}$ only — the feasible set of the same declaration with the factor tagged A forgotten — and let $\mathcal{P}_{\mathbb{T}}(X) \coloneqq \sum_t \Pi_t^A X \Pi_t^A$. Then for every $\alpha \in (1,\infty)$, every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega \in \mathcal{W}^\circ$, the operator $\omega' \coloneqq (\mathcal{P}_{\mathbb{T}} \otimes \mathrm{id}_Q)(\omega)$ satisfies

$$\omega' \in \mathcal{W}, \qquad P_{\hat{c}}(\omega') = P_{\hat{c}}(\omega), \qquad K_{\hat{c}}(\omega') \geq K_{\hat{c}}(\omega) \quad \forall \hat{c}, \qquad O_\alpha(\mathbf{q},\omega') \leq O_\alpha(\mathbf{q},\omega), \tag{5.8}$$

the last in the extended-real sense. Consequently $\inf_{\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}} O_\alpha = \inf_{\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^\circ} O_\alpha$, and $\omega \mapsto \omega'$ maps minimisers over $\mathcal{W}^\circ$ to minimisers over $\mathcal{W}$.

Corollary 5.6 (The reduction for arbitrary constraint lists)**.** Let $\mathfrak{L}$ be arbitrary: Θ_k an arbitrary Hermitian operator on A in trusted-source mode, $\Theta_k = c_k$ an arbitrary real number in untrusted-source mode, and entries declared party-local in either mode (Definition 1.2 (vi)); no licensing restriction (Definition 2.2) is needed in either mode. Then

$$h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \;\geq\; \underline{h}_\alpha(\mathfrak{L}), \tag{5.9}$$

where $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ is the constrained single-round rate of Definition 2.4 and $\underline{h}_\alpha(\mathfrak{L})$ denotes (5.4) over the feasible set $\mathcal{W}$ of (5.1), which carries the same list ($\underline{h}_\alpha(\mathfrak{L})$ and the unadorned $\underline{h}_\alpha$ are the same quantity; the argument is written when the dependence on the declared list matters, as in $\underline{h}_\alpha(\mathfrak{L}) \geq \underline{h}_\alpha(\varnothing)$).

Proposition 5.8 (Range of the value and its finite entropy part)**.** Let $\alpha \in (1, \infty)$.

(i) *(lower bound)* $O_\alpha(\mathbf{q}, \omega) \geq 0$ for every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega \in \mathcal{W}$; hence $\underline{h}_\alpha \in [0, +\infty]$.

(ii) *(entropy-part ceiling)* For every $\omega \in \mathcal{W}$ and every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ with $P_{\hat{c}}(\omega) > 0$, the entropy term $e_{\hat{c}}(\omega) := \frac{1}{1-\alpha} \log_2 \frac{K_{\hat{c}}(\omega)}{P_{\hat{c}}(\omega)}$ satisfies $e_{\hat{c}}(\omega) \leq \log_2 |\mathcal{Z}_\perp| = \log_2(d_Z + 1)$.

(iii) *(no overall ceiling)* No finite a-priori upper bound on $\underline{h}_\alpha$ exists: the divergence penalty is unbounded already at fixed α.

(iv) *(infinite value)* $\underline{h}_\alpha = +\infty$ whenever no $(\mathbf{q}, \omega)$ with $\mathbf{q} \ll \mathbf{\Phi}[\omega]$ is feasible. If $\underline{h}_\alpha = +\infty$, then no $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ is absolutely continuous with respect to $\mathbf{\Phi}[\omega]$ for any $\omega \in \mathcal{W}$.

Theorem 5.10 (Joint convexity of the program)**.** For every $\alpha \in (1, \infty)$ and every declaration satisfying Definition 1.2 (any $\mathfrak{L}$), the extended-real objective O_α of (5.3) is jointly convex and lower semicontinuous on $\Delta(\widehat{\mathcal{C}}) \times \mathcal{W}$, where $\Delta(\widehat{\mathcal{C}})$ is the probability simplex. Consequently (5.4) is a convex minimization over the compact convex set $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$, and the infimum is attained whenever it is finite.

6 Sector classes and sector removal

Definition 6.1 (Sector classes and the removal checklist)**.** Let $\mathfrak{D}$ satisfy Definition 1.2. For $s \in \mathbb{S}$:

(a) s is *key-active* if $(\hat{c}, s)$ fails to be key-inert (Definition 4.5) for at least one $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$; these sectors carry the deficits $f_{\hat{c},s}$ in the objective.

(b) s is *constraint-active* if it is key-inert for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$; by Proposition 4.6 it contributes nothing to any $K_{\hat{c}}$ but remains an optimization variable (it enters $\mathbf{\Phi}$, every $P_{\hat{c}}$, the marginal constraint, and $\mathfrak{L}$).

(c) s is *state-blind* if $F_v^{(s)} = c_{v,s}\, \mathbb{1}_{Q_s}$ for scalars $c_{v,s} \geq 0$, for every $v \in \mathcal{V}$ (then $\sum_v c_{v,s} = 1$ by POVM completeness restricted to the sector); if $\mathfrak{L} \neq \varnothing$ we additionally require each $\hat{O}_k^{(s)} = o_{k,s}\, \mathbb{1}_{Q_s}$ for scalars $o_{k,s}$.

(d) s is *removable* if the hypotheses of Theorem 6.2 hold for it.

All four predicates are decided by finitely many linear-algebra checks on the declaration. Items (a) and (b) reduce to key-inertness (Definition 4.5), a finite combinatorial predicate on the declared function tables and on which of the M_x, $F_v^{(s)}$ are the zero operator, decided exactly, at no window. Items (c) and (d) are decided exactly as well: (c) is a condition on the declared values themselves (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)), so that a block that is only *nearly* scalar fails it and s is then not state-blind, and (d) is decided by the checklist below.

The removal checklist. A *candidate witness* for a sector s^* is a target list $s_1, \dots, s_L$ as index data, states χ_i as binary64 sector-block data, weights μ_i as binary64 data, and the evaluation order of (W6). It is a *witness* for s^* in the sense of (d) when $\mathfrak{L}$ carries no entry declared party-local and all six of the following hold:

(W1) s^* *is state-blind* in the sense of (c), the $\hat{O}_k^{(s^*)}$ half included whenever $\mathfrak{L} \neq \varnothing$;

(W2) *the witness does not name s^*, and names no sector twice*: $L \geq 1$, $s_i \in \mathbb{S} \setminus \{s^*\}$ for every i, and the s_i pairwise distinct, all compared as index data at no window. A candidate naming s^* fails (W2), and so does one that repeats a target;

(W3) *weights*: $\mu_i \geq 0$ for every i — a comparison of an exhibited binary64 datum against zero — and $\sum_i \mu_i = 1$, the sum formed in the evaluation order the witness exhibits and verified to round nowhere, exactly as (W6) requires of its residuals, so that the computed sum *is* the real sum;

(W4) *states*: $\chi_i \succeq 0$ and $\operatorname{tr} \chi_i = 1$ for every i, decided on the *exhibited form* and never on a computed spectrum: the witness presents each χ_i diagonal in the declared basis of Q_{s_i} — every off-diagonal entry the datum 0 in the sense of (c), every diagonal entry a datum ≥ 0, and their sum formed in the exhibited order, rounding nowhere, and equal to the datum 1 — so that both conditions hold of the declared numbers by construction. Any other presentation fails (W4), which can only *refuse* a removal;

(W5) *key-inert targets*, which is (RM2): every pair $(\hat{c}, s_i)$ with $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $i \in \{1, \dots, L\}$ is key-inert (Definition 4.5);

(W6) *both equality families close at residual exactly zero*: (RM1) for every $v \in \mathcal{V}$, and (RM3) for every k (vacuous when $\mathfrak{L} = \varnothing$).

The condition on $\mathfrak{L}$ and (W1)–(W6) are the hypotheses of Theorem 6.2 transcribed, all of them and in the theorem's own words — the condition on

$\mathfrak{L}$ from its first hypothesis, (W1) and (W2) from its opening sentence, (W3) and (W4) from its quantifiers over μ_i and χ_i, (W5) from (RM2), (W6) from (RM1) and (RM3) — which is what (d) means by "the hypotheses of Theorem 6.2 hold". Condition (W6) is read on the residual itself: each residual is formed in an evaluation order that the witness exhibits alongside its data, every elementary operation of which rounds nothing, so that the computed residual *is* the real residual and its comparison against zero decides the real identity.

Deleting a set of removable sectors needs one condition more, and it is not part of this definition. Removability is a property of one sector, and Theorem 6.2 licenses the deletion of one sector; that two sectors are each removable does not license deleting both. Proposition 6.5 states the further hypothesis on the set $\mathbb{S}_{\mathrm{del}}$ a run deletes. Nothing in (W1)–(W6) implies that hypothesis.

In the general format of Definition 1.1 the four classes remain predicates of the sectors $s \in \mathbb{S}$ of Q, decided as above, with $\hat{O}_k^{(s)}$ the block of the lifted operator of an entry. The finer notion along Alice's sectors — which joint pairs $(\hat{c}, (t, s))$ carry a deficit — is Definition 4.8, and a sector that is key-active here may have key-inert joint pairs. Theorem 6.2 is stated for a list carrying no entry declared party-local, so for a list carrying one, clause (d) is unsatisfiable and no sector carries a witness in its sense.

Theorem 6.2 (Sound removal of a state-blind sector)**.** Let $\mathfrak{L}$ carry no entry declared party-local, let $s^* \in \mathbb{S}$ be state-blind, and suppose there exist *pairwise distinct* sectors $s_1, \dots, s_L \in \mathbb{S} \setminus \{s^*\}$, states $\chi_i \in S_=(Q_{s_i})$, and weights $\mu_i \geq 0$ with $\sum_{i=1}^{L} \mu_i = 1$, such that

(RM1) *response match:* $c_{v,s^*} = \sum_i \mu_i \operatorname{tr}\left[F_v^{(s_i)} \chi_i\right]$ for every $v \in \mathcal{V}$;

(RM2) *key-inert targets:* every pair $(\hat{c}, s_i)$ with $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ is key-inert;

(RM3) *constraint match* (vacuous for $\mathfrak{L} = \varnothing$): $o_{k,s^*} = \sum_i \mu_i \operatorname{tr}\left[\hat{O}_k^{(s_i)} \chi_i\right]$ for every k.

Then restricting the infimum (5.4) to $\{\omega \in \mathcal{W} : \omega_{s^*} = 0\}$ changes neither its value nor its attainment: the sector s^* may be deleted from the variable list, and its rows deleted from the generated maps, without loss.

Proposition 6.3 (Unsoundness of dropping a constraint-active sector)**.** There exists a declaration satisfying Definition 1.2 and a constraint-active sector s_0 such that forcing $\omega_{s_0} = 0$ strictly *increases* $\underline{h}_\alpha$. Dropping a constraint-active sector from the variable list is therefore unsound: it can overstate the extractable key.

Proposition 6.4 (Unsoundness of averaging sectors)**.** Coarse-graining the sector decomposition of a declaration — replacing $Q = \bigoplus_s Q_s$ and $\{F_v^{(s)}\}$ by a single sector carrying averaged POVM elements — is unsound: there exists a declaration $\mathfrak{D}$ and an averaged declaration $\bar{\mathfrak{D}}$ reproducing the same outcome statistics on the averaged state space for which $\underline{h}_\alpha(\bar{\mathfrak{D}}) > \underline{h}_\alpha(\mathfrak{D})$.

Proposition 6.5 (Deleting a set of removable sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 and let $\mathbb{S}_{\mathrm{del}} \subseteq \mathbb{S}$ be the set of sectors a run deletes. Suppose *every* $s \in \mathbb{S}_{\mathrm{del}}$ *carries a witness in the sense of Definition 6.1 (d) all of whose targets lie outside* $\mathbb{S}_{\mathrm{del}}$. Then restricting the infimum (5.4) to $\{\omega \in \mathcal{W} : \omega_s = 0 \text{ for all } s \in \mathbb{S}_{\mathrm{del}}\}$ changes neither its value nor its attainment: the whole set $\mathbb{S}_{\mathrm{del}}$ may be deleted from the variable list, and its rows from the generated maps, without loss.

7 The Choi program

Lemma 7.1 (The admissible Alice subspace)**.** Let $\mathfrak{D}$ be a declaration, and write

$$\mathcal{S}_A \coloneqq \begin{cases} \operatorname{supp} \tau_A, & \text{trusted-source mode,} \\ \mathcal{H}(A), & \text{untrusted-source mode,} \end{cases} \tag{7.1}$$

the *admissible Alice subspace*; it is determined by the marginal slot alone and never by $\mathfrak{L}$ or by a particular feasible point. Then:

(i) every $\omega \in \mathcal{W}$ satisfies $\operatorname{supp} \omega \subseteq \mathcal{S}_A \otimes \mathcal{H}(Q)$;

(ii) the bound (i) is tight for $\mathfrak{L} = \varnothing$, and for a nonempty $\mathfrak{L}$ it remains a valid upper bound.

Definition 7.2 (Support frame and declared isometry)**.** Let the declaration be in trusted-source mode, with declared marginal τ_A. Let A_0 be a register of dimension d_{A_0} and let $\iota \in \mathcal{L}(A_0, A)$ satisfy

$$\iota^\dagger \iota = \mathbb{1}_{A_0}, \qquad \iota\, \iota^\dagger\, \tau_A = \tau_A. \tag{7.2}$$

Write $\Pi_\iota \coloneqq \iota\iota^\dagger$, which is the orthogonal projector onto $\operatorname{ran} \iota$ ($\Pi_\iota^\dagger = \Pi_\iota$ and $\Pi_\iota^2 = \iota(\iota^\dagger\iota)\iota^\dagger = \Pi_\iota$ by the first condition). The first condition says ι is an isometry; the second says its range *contains* the support of τ_A, since $\Pi_\iota \tau_A = \tau_A$ holds exactly when every column of τ_A lies in $\operatorname{ran} \iota$, i.e. when $\operatorname{ran} \tau_A \subseteq \operatorname{ran} \iota$. We call $\operatorname{ran} \iota$ the *support frame* of the run and ι the *declared isometry*; a support frame is given by any ι satisfying (7.2). The frame is *tight* when $\operatorname{ran} \iota = \operatorname{supp} \tau_A$, equivalently when $d_{A_0} = \operatorname{rank} \tau_A$; in general $\operatorname{rank} \tau_A \le d_{A_0} \le d_A$. The images under ι of a fixed orthonormal basis of A_0 are the *declared basis of* A_0.

Two constructions satisfy (7.2) exactly:

(a) *The* full frame. $A_0 = A$, $d_{A_0} = d_A$, $\iota = \mathbb{1}_A$. Then $\iota^\dagger \iota = \mathbb{1}_A$ and $\Pi_\iota \tau_A = \mathbb{1}_A \, \tau_A = \tau_A$: both conditions of (7.2) hold for *any* admissible τ_A, and both hold *exactly* as binary64 data, a product with the identity rounding nothing. This construction is always available and uses no property of τ_A beyond admission. It is tight exactly when τ_A is nonsingular.

(b) *The* diagonal frame. If the declared τ_A is diagonal in the declared basis of A, $\tau_A = \sum_i t_i \, |i\rangle\langle i|$, then $\operatorname{supp} \tau_A = \operatorname{span}\{|i\rangle : t_i \neq 0\}$, and one may take $d_{A_0} = \#\{i : t_i \neq 0\}$ and $\iota = \sum_j |i_j\rangle \, \langle j|$, the coordinate injection onto those indices in increasing order. Both conditions of (7.2) are then arithmetic in $\{0, 1\}$ and hold exactly, and the frame is tight. Whether $t_i \neq 0$ is a bit-level comparison of a declared binary64 datum against zero and not a threshold: τ_A is admitted only with $\tau_A \succeq 0$ and $\operatorname{tr} \tau_A = 1$ decided exactly (Definition 1.1, the exact-admissibility paragraph), so its diagonal entries *are* its eigenvalues and *are* the declared numbers. The construction is available only for a τ_A that is diagonal in the declared basis, every off-diagonal entry being the datum 0. Construction (a) is always available and exhibits no tightness at all.

Set

$$W_\tau \coloneqq \tau_A^{1/2} \, \iota \;\in\; \mathcal{L}(A_0, A). \tag{7.3}$$

Lemma 7.3 (Fixed-marginal parametrisation on a support frame)**.** Let the declaration be in trusted-source mode, let ι be the declared isometry of a support frame, satisfying (7.2), and let W_τ be (7.3) (Definition 7.2). Then:

(i) $\Pi_\iota \tau_A^{1/2} = \tau_A^{1/2}$.

(ii) For every support frame, tight or not,

$$W_\tau W_\tau^\dagger = \tau_A^{1/2} \, \Pi_\iota \, \tau_A^{1/2} = \tau_A^{1/2} \, \tau_A^{1/2} = \tau_A. \tag{7.4}$$

(iii) $\operatorname{ran} W_\tau = \operatorname{supp} \tau_A$, and the Moore–Penrose inverse W_τ^+ satisfies

$$W_\tau W_\tau^+ = \Pi_{\operatorname{supp} \tau_A}, \qquad W_\tau^+ W_\tau = P_\iota \coloneqq \Pi_{(\ker W_\tau)^\perp}, \tag{7.5}$$

where $\ker W_\tau = \{\xi \in \mathcal{H}(A_0) : \iota\xi \in (\operatorname{supp} \tau_A)^\perp\}$ has dimension $d_{A_0} - \operatorname{rank} \tau_A$. The frame is tight if and only if $\ker W_\tau = 0$, i.e. $P_\iota = \mathbb{1}_{A_0}$, and then W_τ has full column rank and $W_\tau^+ = (W_\tau^\dagger W_\tau)^{-1} W_\tau^\dagger$.

(iv) For any finite-dimensional system B', the map $\chi_{B'}(J) \coloneqq (W_\tau \otimes \mathbb{1}_{B'}) \, J \, (W_\tau^\dagger \otimes \mathbb{1}_{B'})$ maps $\{J \in \operatorname{Pos}(A_0 B') : \operatorname{tr}_{B'} J = \mathbb{1}_{A_0}\}$ *onto* $\{\omega \in \operatorname{Pos}(AB') : \operatorname{tr}_{B'} \omega = \tau_A\}$. It is *injective*, hence a bijection with inverse $\omega \mapsto (W_\tau^+ \otimes \mathbb{1}_{B'}) \, \omega \, (W_\tau^{+\dagger} \otimes \mathbb{1}_{B'})$, when the support frame is tight, and, if

$\dim \mathcal{H}(B') \geq 2$, only then; if $\dim \mathcal{H}(B') = 1$ its domain is the single point $\mathbb{1}_{A_0}$. In general $\chi_{B'}(J) = \chi_{B'}(J')$ if and only if $(P_\iota \otimes \mathbb{1}_{B'})J(P_\iota \otimes \mathbb{1}_{B'}) = (P_\iota \otimes \mathbb{1}_{B'})J'(P_\iota \otimes \mathbb{1}_{B'})$.

Lemma 7.4 (Fixed-marginal parametrisation with any factor)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in trusted-source mode, in the general format of Definition 1.1.

(i) *(any factor serves)* Let $\tau \in \mathrm{Pos}(A')$ on a finite-dimensional system A', of any trace, and let $W \in \mathcal{L}(A_0, A')$ satisfy $WW^\dagger = \tau$. Let P_W be the projector onto $(\ker W)^\perp$. For every finite-dimensional B' the map $\chi^W_{B'}(J) \coloneqq (W \otimes \mathbb{1}_{B'})J(W^\dagger \otimes \mathbb{1}_{B'})$ maps $\{J \in \mathrm{Pos}(A_0B') : \mathrm{tr}_{B'}\, J = \mathbb{1}_{A_0}\}$ onto $\{\omega \in \mathrm{Pos}(A'B') : \mathrm{tr}_{B'}\, \omega = \tau\}$; it is injective when $\ker W = 0$, and, if $\dim \mathcal{H}(B') \geq 2$, only then; and $\chi^W_{B'}(J) = \chi^W_{B'}(J')$ exactly when $(P_W \otimes \mathbb{1})J(P_W \otimes \mathbb{1}) = (P_W \otimes \mathbb{1})J'(P_W \otimes \mathbb{1})$. The operator W_τ of (7.3) is such a factor of τ_A for every admissible frame.

(ii) *(per Alice sector)* For every $t \in \mathbb{T}$ let $W_t \in \mathcal{L}(A_{0,t}, A_t)$ satisfy $W_t W_t^\dagger = \tau_A^{(t)}$, and put $A_0 \coloneqq \bigoplus_t A_{0,t}$, $W \coloneqq \bigoplus_t W_t$. Then $WW^\dagger = \tau_A$, and the assignment $(J_{t,s}) \mapsto \big(\omega^{(t,s)}\big)$, $\omega^{(t,s)} \coloneqq (W_t \otimes \mathbb{1}_{Q_s})J_{t,s}(W_t^\dagger \otimes \mathbb{1}_{Q_s})$, maps

$$\Big\{(J_{t,s})_{(t,s)} :\ J_{t,s} \in \mathrm{Pos}(A_{0,t}Q_s), \quad \sum_{s\in\mathbb{S}} \mathrm{tr}_{Q_s}\, J_{t,s} = \mathbb{1}_{A_{0,t}} \ \ \forall t \in \mathbb{T}\Big\} \tag{7.6}$$

onto the set of lists over the joint sectors with positive semidefinite blocks and $\sum_s \mathrm{tr}_{Q_s}\, \omega^{(t,s)} = \tau_A^{(t)}$ for every t, bijectively when $\ker W_t = 0$ for every t, and, if $\dim \mathcal{H}(Q) \geq 2$, only then. The set (7.6) is a Cartesian product over t. If every ι_t satisfies (7.2) for $\tau_A^{(t)}$ on A_t, then $\iota \coloneqq \bigoplus_t \iota_t$ satisfies (7.2) for τ_A and $W_\tau = \bigoplus_t (\tau_A^{(t)})^{1/2}\iota_t$.

(iii) *(the objective reads Alice's side through N_x)* For a factor $W = \bigoplus_t W_t$ as in (ii) put

$$N_x^W \coloneqq W^\dagger M_x W = \bigoplus_t W_t^\dagger M_x^{(t)} W_t \in \mathrm{Pos}(A_0), \tag{7.7}$$

let K_κ^N be (1.18) with $\sqrt{N_x^W}$ in place of $\sqrt{M_x}$, an operator $\mathcal{H}(A_0Q) \to \mathcal{H}(R\,XY\,C_AC_B\,A_0Q)$, and let $\mathcal{G}_{\hat{c}}^N$ be (1.19) with K_κ^N. Then for every $J \in \mathrm{Pos}(A_0Q)$, every (x, v) and every $\hat{c}$, with $\omega \coloneqq \chi_Q^W(J)$,

$$\mathrm{tr}[(M_x \otimes F_v)\,\omega] = \mathrm{tr}\big[(N_x^W \otimes F_v)\,J\big], \qquad \mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega) = \mathrm{tr}\,\mathcal{G}_{\hat{c}}^N(J), \qquad T\big(\mathcal{G}_{\hat{c}}(\omega)\big) = T\big(\mathcal{G}_{\hat{c}}^N(J)\big), \tag{7.8}$$

T the functional (4.7); hence $P_{\hat{c}}$, $K_{\hat{c}}$ and O_α at ω are functions of J, $\{N_x^W\}$, $\{F_v\}$ and the classical maps alone, and the same holds sector by sector for the joint-sector quantities of Definition 4.8.

(iv) *(ensemble form)* On the full frame, $A_{0,t} = A_t$ and $W_t = (\tau_A^{(t)})^{1/2}$, one has $N_x^W = \tau_A^{1/2} M_x \tau_A^{1/2} = N_x$, the operators of Definition 1.1 (DX4): for an ensemble-form declaration they are the *declared* numbers. So the program of an ensemble-form declaration is the program (7.6), (iii) in the declared $\{N_x\}$ and $\{F_v\}$, with every constraint row in the conjugated form of (7.9); it is defined, and equals $\underline{h}_\alpha$ of the denoted declaration, at every rank of every $\tau_A^{(t)}$, and no rank is computed.

Theorem 7.5 (The Choi program)**.** Let the declaration be in trusted-source mode, fix a support frame (Definition 7.2), apply Lemma 7.3 with $B' = Q$, and restrict to sector-block-diagonal operators. Then the assignment $J = \bigoplus_s J_s \mapsto \omega = \chi_Q(J)$, with $\omega_s = \chi_{Q_s}(J_s) \coloneqq (W_\tau \otimes \mathbb{1}_{Q_s}) J_s (W_\tau^\dagger \otimes \mathbb{1}_{Q_s})$, maps

$$\mathcal{W}^J \coloneqq \Big\{ (J_s)_{s\in\mathbb{S}} :\ J_s \in \mathrm{Pos}(A_0 Q_s),\ \sum_{s\in\mathbb{S}} \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0},\ W_\tau \Big(\textstyle\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_0} \otimes \hat{O}_k^{(s)}) J_s] \Big) W_\tau^\dagger \bowtie_k \Theta_k \ \forall k \Big\} \tag{7.9}$$

onto $\mathcal{W}$ of (5.1), bijectively when the support frame is tight (Lemma 7.3), and, if $\mathcal{W} \neq \varnothing$ and $\dim \mathcal{H}(Q) \geq 2$, only then. Writing $\mathbf{\Phi}_J[J] \coloneqq \mathbf{\Phi}[\chi_Q(J)]$, $\mathcal{G}_{\hat{c},s}^J \coloneqq \mathcal{G}_{\hat{c},s} \circ \chi_{Q_s}$, $f_{\hat{c},s}^J \coloneqq f_{\hat{c},s} \circ \chi_{Q_s}$, $P_{\hat{c}}^J(J) \coloneqq \mathbf{\Phi}_J[J](\hat{c})$ and $K_{\hat{c}}^J(J) \coloneqq P_{\hat{c}}^J(J) + \sum_{s\in\mathbb{S}_K(\hat{c})} f_{\hat{c},s}^J(J_s)$, the value (5.4) equals

$$\underline{h}_\alpha = \inf_{\mathbf{q},\ (J_s)_s} \ \frac{\alpha}{\alpha-1} \sum_{\hat{c}\notin\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}^J(J)} + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \left[q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}^J(J)} + \frac{q(\hat{c})}{\alpha-1} \log_2 \frac{q(\hat{c})}{K_{\hat{c}}^J(J)} \right]$$
$$\text{s.t.} \quad \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}, \qquad (J_s)_s \in \mathcal{W}^J, \tag{7.10}$$

an *exact* reformulation (not a relaxation), jointly convex in $(\mathbf{q}, (J_s)_s)$, with all constraints affine or conic.

In the general format of Definition 1.1, let the frame be a direct sum $\iota = \bigoplus_{t\in\mathbb{T}} \iota_t$ with every $\iota_t \in \mathcal{L}(A_{0,t}, A_t)$ satisfying (7.2) for $\tau_A^{(t)}$ — either construction of Definition 7.2 applied sector by sector — and put $W_t \coloneqq (\tau_A^{(t)})^{1/2} \iota_t$. Let $\mathcal{W}^J$ be the set of lists $(J_{t,s})$ over the joint sectors with $J_{t,s} \in \mathrm{Pos}(A_{0,t} Q_s)$, the coupling constraints $\sum_s \mathrm{tr}_{Q_s} J_{t,s} = \mathbb{1}_{A_{0,t}}$ for every t, and for every entry the row of (5.5) evaluated at $\omega^{(t,s)} = (W_t \otimes \mathbb{1}) J_{t,s} (W_t^\dagger \otimes \mathbb{1})$: for an entry read through its lifted operator, $W_t \big(\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_{0,t}} \otimes \hat{O}_k^{\dagger(s)}) J_{t,s}] \big) W_t^\dagger \bowtie_k \Theta_k^{(t)}$ for every t, and for a party-local entry the last line of (5.5) at J, conjugated by $W_t \otimes \mathbb{1}$. Then $(J_{t,s}) \mapsto (\omega^{(t,s)})$ maps $\mathcal{W}^J$ onto $\mathcal{W}$ of (5.5), bijectively when every sector's frame is tight, and, if $\mathcal{W} \neq \varnothing$ and $\dim \mathcal{H}(Q) \geq 2$, only then, and (7.10) holds with $K_{\hat{c}}^J(J) = P_{\hat{c}}^J(J) + \sum_{(t,s)\in\mathbb{J}_K(\hat{c})} f_{\hat{c},(t,s)}(\omega^{(t,s)})$. Every constraint of $\mathcal{W}^J$ involves the blocks of one Alice sector only, so $\mathcal{W}^J = \prod_t \mathcal{W}_t^J$, and a linear function of $(J_{t,s})$ is minimised over $\mathcal{W}^J$ by minimising, independently for each t, its part on the blocks of t over $\mathcal{W}_t^J$.

Proposition 7.6 (Frame-free constraint rows on a tight frame)**.** In (i)–(iii) let the declaration be in trusted-source mode, let W_τ be (7.3) for a support frame (Definition 7.2), and for an entry k of $\mathfrak{L}$ and $(J_s)_{s\in\mathbb{S}}$ with $J_s \in \mathrm{Pos}(A_0 Q_s)$ write $Y_k := \sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_0} \otimes \hat{O}_k^{(s)}) J_s]$. For a homogeneous entry, whose right-hand side is $\Theta_k = c_k \tau_A$ (Definition 2.2 (a)), call the k-th row of (7.9), $W_\tau Y_k W_\tau^\dagger \bowtie_k c_k \tau_A$, its *conjugated form*, and

$$Y_k \;\bowtie_k\; c_k\, \mathbb{1}_{A_0} \tag{7.11}$$

its *frame-free form*.

(i) If $\ker W_\tau = 0$, that is, if the frame is tight, the frame-free form of a homogeneous row holds for $(J_s)_s$ if and only if its conjugated form does; so imposing some homogeneous rows in the frame-free form leaves $\mathcal{W}^J$ unchanged.

(ii) On every support frame the frame-free form of a homogeneous row implies its conjugated form. So the set obtained from $\mathcal{W}^J$ by imposing some homogeneous rows in the frame-free form is contained in $\mathcal{W}^J$, and the infimum in (7.10) taken over that set in place of $\mathcal{W}^J$ is at least $\underline{h}_\alpha$.

(iii) In the general format, with $\iota = \bigoplus_t \iota_t$ and W_t as in Theorem 7.5:

  (a) for a homogeneous entry read through its lifted operator and every $t \in \mathbb{T}$, (i) and (ii) hold for the row of sector t, with W_t and $A_{0,t}$ in place of W_τ and A_0 and with the frame-free form $\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_{0,t}} \otimes \hat{O}_k^{\uparrow(s)}) J_{t,s}] \bowtie_k c_k\, \mathbb{1}_{A_{0,t}}$;

  (b) for a party-local entry, with Y_k the expression of the last line of (5.5) at J in place of ω, the unconjugated row $Y_k \bowtie_k 0$ is equivalent to the conjugated row $(W_t \otimes \mathbb{1})\, Y_k\, (W_t^\dagger \otimes \mathbb{1}) \bowtie_k 0$ if $\ker W_t = 0$, and implies it on every frame.

(iv) In untrusted-source mode there is no conjugation to remove: every row is a condition on the state variable as written.

Definition 7.10 (The program in untrusted-source mode)**.** In untrusted-source mode the Choi step is the identity. This is the degenerate case of Theorem 7.5, not a second program:

(a) *The parametrization.* $\chi_{B'} = \mathrm{id}$, $J = \omega$, $J_s = \omega_s$; no A_0, no ι, no W_τ.

(b) *The variables and constraints.* $\mathcal{W}$ of (5.1) is already the feasible set, and by (5.2) it reads

$$\mathcal{W} = \Big\{ (\omega_s)_{s\in\mathbb{S}} :\ \omega_s \in \mathrm{Pos}(A Q_s),\ \sum_{s\in\mathbb{S}} \mathrm{tr}\, \omega_s = 1,\ \sum_{s\in\mathbb{S}} \mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k^{(s)})\, \omega_s\right] \bowtie_k c_k\ \forall k \Big\}, \tag{7.13}$$

so the coupling constraint that binds the sector blocks is the single *scalar* equation $\sum_s \operatorname{tr} \omega_s = 1$, in place of the operator equation $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ of (7.9).

(c) *The objective.* (7.10) holds verbatim with J_s replaced by ω_s and $P^J_{\hat{c}}$, $f^J_{\hat{c},s}$, $K^J_{\hat{c}}$ by their state-form originals $P_{\hat{c}}$, $f_{\hat{c},s}$, $K_{\hat{c}}$ of (4.1): the substitution $\omega_s = \chi_{Q_s}(J_s)$ that Theorem 7.5 performs is the identity here. Hence the value is $\underline{h}_\alpha$ of (5.4) by definition rather than by a theorem, the program is jointly convex by Theorem 5.10 directly, and the feasible set is convex and compact by Definition 5.1.

(d) *Consistency at $d_A = 1$.* There trusted-source mode admits only $\tau_A = [1]$, the one state on a one-dimensional A, and its Choi parametrization degenerates: $d_{A_0} = \operatorname{rank} \tau_A = 1$, $\iota = [1]$, $W_\tau = [1]$, $\chi_{B'} = \mathrm{id}$ and $J_s = \omega_s$. So $\mathbb{1}_{A_0} = [1]$ and $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ *is* $\sum_s \operatorname{tr} \omega_s = 1$, the constraint rows agree because every 1×1 Hermitian Θ_k is $c_k \tau_A$ with $c_k = \Theta_k$ (Proposition 5.2 (iii)), and the two feasible sets of (5.1) coincide (Proposition 5.2 (i)): (7.9) and (7.13) are the same set, and the two modes give literally the same program.

Lemma 7.11 (Upper estimate of the value at any feasible point)**.** Let $\mathfrak{D}$ satisfy Definition 1.2, let $\mathcal{S}_{\mathrm{acc}}$ be any set of probability vectors on the acceptance alphabet, and let $\alpha \in (1, \infty)$.

(i) *(State form.)* For every $(\mathbf{q}, \omega) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$,

$$O_\alpha(\mathbf{q}, \omega) \;\geq\; \underline{h}_\alpha. \tag{7.14}$$

(ii) *(Choi form.)* Write $O^J_\alpha(\mathbf{q}, J) := O_\alpha(\mathbf{q}, \chi_Q(J))$ for the objective of the Choi program (7.10) of Theorem 7.5, with χ_Q the map of that theorem. For every $x = (\mathbf{q}, J) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$,

$$O^J_\alpha(x) \;\geq\; \underline{h}_\alpha. \tag{7.15}$$

8 Gradients and the limit of order one

Definition 8.1 (Maximal branch support)**.** For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}$ the *maximal branch support* is the subspace

$$W_{\hat{c},s} := \sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} K^{(s)}_\kappa\big(\mathcal{S}_A \otimes \mathcal{H}(Q_s)\big) \;\subseteq\; \mathcal{H}(R\,XY\,C_A C_B\,AQ_s), \tag{8.1}$$

the span of the images, under the Kraus operators of $\mathcal{G}_{\hat{c},s}$ (Definition 1.6), of the admissible Alice subspace $\mathcal{S}_A$ of (7.1) tensored with the sector, that is, of the subspace the declaration admits: $\operatorname{supp} \tau_A$ in trusted-source mode, all of $\mathcal{H}(A)$ in untrusted-source mode.

Listing the operators of (8.1) side by side, with W_τ of (7.3) on the support frame of Definition 7.2, put

$$\mathbf{K}_{\hat{c},s} \coloneqq \Big[\, K_\kappa^{(s)}\,(W_\tau \otimes \mathbb{1}_{Q_s}) \,\Big]_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} : \bigoplus_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} \mathcal{H}(A_0 Q_s) \longrightarrow \mathcal{H}(R\,XY\,C_A C_B\,AQ_s), \tag{8.2}$$

with $W_\tau \otimes \mathbb{1}_{Q_s}$ replaced by $\mathbb{1}_{AQ_s}$, and the domain by $\bigoplus_\kappa \mathcal{H}(AQ_s)$, in untrusted-source mode. The matrix $\mathbf{K}_{\hat{c},s}$ depends on the declaration and on W_τ alone and on no iterate.

Lemma 8.2 (Properties of the maximal branch support)**.** Let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}$, and let $W_{\hat{c},s}$ and $\mathbf{K}_{\hat{c},s}$ be as in Definition 8.1.

(i) In trusted-source mode every $\omega_s = \chi_{Q_s}(\Delta)$ with $\Delta \in \mathrm{Pos}(A_0 Q_s)$, in particular every feasible ω_s and the image of the positive and the negative part of every Hermitian direction, satisfies $\operatorname{supp}\omega_s = (W_\tau \otimes \mathbb{1})(\operatorname{supp}\Delta) \subseteq \operatorname{supp}\tau_A \otimes \mathcal{H}(Q_s) = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$; in untrusted-source mode $\mathcal{S}_A = \mathcal{H}(A)$ and the inclusion holds for every $\omega_s \in \mathrm{Pos}(AQ_s)$. In either mode, therefore,

$$\operatorname{supp}\mathcal{G}_{\hat{c},s}(\omega_s) \subseteq W_{\hat{c},s}, \tag{8.3}$$

with equality whenever ω_s attains that maximal support, $\operatorname{supp}\omega_s = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$. In trusted-source mode $J_s \succ 0$ for the Choi block suffices for it, and on a tight support frame is equivalent to it; in untrusted-source mode it is $\omega_s \succ 0$ on $\mathcal{H}(AQ_s)$, which is $J_s \succ 0$, as $J_s = \omega_s$ there (Definition 7.10 (a)).

(ii) $W_{\hat{c},s}$ is the range of one matrix, and its dimension is bounded:

$$W_{\hat{c},s} = \operatorname{ran}\mathbf{K}_{\hat{c},s}, \qquad \dim W_{\hat{c},s} = \operatorname{rank}\mathbf{K}_{\hat{c},s} \;\le\; \big|\mathcal{K}_{\mathrm{pass}}(\hat{c})\big| \cdot \dim\mathcal{S}_A \cdot d_s, \tag{8.4}$$

where $\dim\mathcal{S}_A = \operatorname{rank}\tau_A$ in trusted-source mode and d_A in untrusted-source mode.

(iii) $\dim W_{\hat{c},s}$ is *strictly* smaller than $\dim\mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$, so 0 always lies in the ambient spectrum of a branch operator $X_{\hat{c},s} = \mathcal{G}_{\hat{c},s}(\omega_s)$ of (i) and "maximal support" is never the statement that $X_{\hat{c},s} \succ 0$ on the ambient space.

(iv) $W_{\hat{c},s}$ is *not* in general $\mathcal{Z}$-invariant.

Theorem 8.4 (Gradient of the pinched trace power on a face)**.** Let $\alpha \in (1,\infty)$, $p = 1/\alpha$, let T be the pinched trace power (4.7) on the positive

semidefinite cone of a branch space (Lemma 8.3), and let $X \succ 0$. Then T is Fréchet differentiable at X with gradient

$$\nabla T(X) = \alpha\, DX^p\Big[\mathcal{Z}\big(Y^{\alpha-1}\big)\Big], \qquad Y \coloneqq \mathcal{Z}(X^p), \tag{8.7}$$

i.e. $\frac{\mathrm{d}}{\mathrm{d}t} T(X+tH)\big|_0 = \operatorname{tr}[\nabla T(X) H]$, where $DX^p[\cdot]$ is the self-adjoint map (8.5) with $f(x) = x^p$ and $Y^{\alpha-1}$ is defined by functional calculus ($0^{\alpha-1} \coloneqq 0$; Y may be singular). Explicitly, in the eigenbasis $X = \sum_i \lambda_i P_i$:

$$\nabla T(X) = \alpha \sum_{i,j} \frac{\lambda_i^p - \lambda_j^p}{\lambda_i - \lambda_j}\, P_i\, \mathcal{Z}\big(Y^{\alpha-1}\big)\, P_j \qquad \text{(diagonal kernel } p\lambda_i^{p-1}\text{)}. \tag{8.8}$$

Moreover:

(i) *(faces)* Let $X \succeq 0$, write $S \coloneqq \operatorname{supp} X$, and let H be Hermitian with $\operatorname{supp} H \subseteq S$. Then $t \mapsto T(X+tH)$ is differentiable at $t = 0$, and (8.7)–(8.8) hold in the following *split* form:

  (a) the divided-difference map $DX^p[\cdot]$ is compressed to S, equivalently, the sum in (8.8) runs over the strictly positive eigenvalues of X only, where X is positive definite and the kernel $f^{[1]}$ is finite;

  (b) $Y = \mathcal{Z}(X^p)$, $Y^{\alpha-1}$ and $\mathcal{Z}(Y^{\alpha-1})$ are formed on the *ambient* branch space and are *not* compressed to S.

  The resulting $\nabla T(X)$ determines the derivative in every direction supported in S; its components outside S are neither determined by this nor needed, support-enlarging directions being the subject of (ii). Compressing the *whole* computation to a subspace S' is licensed when $\sum_{j \in \mathcal{Z}_\perp} Z_j S \subseteq S'$, in particular when $S \subseteq S'$ and S' is $\mathcal{Z}$-invariant ($Z_j S' \subseteq S'$ for every $j \in \mathcal{Z}_\perp$): then the ambient complement contributes 0 to T, and the compressed and ambient computations agree. Compressing to S itself is licensed *only* when S is $\mathcal{Z}$-invariant; otherwise $\mathcal{Z}$ transports weight out of S and the compressed value of T is strictly smaller (the proof gives the argument and a rank-one witness). For the branch operators $X = \mathcal{G}_{\hat{c},s}(\omega_s)$, whose faces are in general not $\mathcal{Z}$-invariant, the relevant face is $S = W_{\hat{c},s}$ of (8.1) at maximal support, and (a)–(b) is what makes the gradient available there.

(ii) *(one-sided derivatives)* For $X \succeq 0$ and any Hermitian H with $X + tH \succeq 0$ for small $t > 0$, the one-sided directional derivative $\lim_{t \downarrow 0} \frac{T(X+tH) - T(X)}{t}$ exists in $(-\infty, +\infty]$, and can equal $+\infty$ only when H strictly enlarges the support of X.

Corollary 8.5 (Gradient of the objective along feasible directions)**.** Let $\alpha \in (1, \infty)$ and let $x = (\mathbf{q}, (J_s)_s)$ be a point with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $J_s \in \operatorname{Pos}(A_0 Q_s)$

for every $s \in \mathbb{S}$ — per-block positive semidefiniteness of the J factor and nothing else: no normalization, no coupling condition $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ and no declared row $\bowtie_k$ is assumed of it, so that every point of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ qualifies, at every $\varepsilon \geq 0$ and the exact $\mathcal{W}^J$ included (Definition 17.1: per-block positive semidefiniteness is the one condition that family does not relax) — and such that $q(\hat{c}) > 0$, $P^J_{\hat{c}} > 0$ and $K^J_{\hat{c}} > 0$ for all $\hat{c}$, and such that each branch operator $X_{\hat{c},s} \coloneqq \mathcal{G}^J_{\hat{c},s}(J_s)$ ($\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, $s \in \mathbb{S}_K(\hat{c})$) has *maximal support*, $\operatorname{supp} X_{\hat{c},s} = W_{\hat{c},s}$ of (8.1), for which $J_s \succ 0$ suffices, so that Theorem 8.4 (i) applies on that face. The displayed formulas are written for trusted-source mode; the untrusted-source-mode statement is at the end of this corollary.

Then the objective O^J_α of (7.10) is differentiable at that point *along every feasible direction*, and that is the whole of the conclusion: for every $y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ the one-variable map $t' \mapsto O^J_\alpha\big(x + t'(y - x)\big)$ is differentiable at $t' = 0$, with derivative the pairing of $y - x$ against the gradient

$$\frac{\partial O^J_\alpha}{\partial q(\hat{c})} = \begin{cases} \dfrac{\alpha}{\alpha - 1}\Big(\log_2 \dfrac{q(\hat{c})}{P^J_{\hat{c}}} + \log_2 \mathrm{e}\Big), & \hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}, \\ \log_2 \dfrac{q(\hat{c})}{P^J_{\hat{c}}} + \log_2 \mathrm{e} + \dfrac{1}{\alpha - 1}\Big(\log_2 \dfrac{q(\hat{c})}{K^J_{\hat{c}}} + \log_2 \mathrm{e}\Big), & \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}, \end{cases} \tag{8.10}$$

$$\nabla_{J_{s'}} O^J_\alpha = -\log_2 \mathrm{e} \sum_{\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}} \frac{\alpha}{\alpha - 1} \frac{q(\hat{c})}{P^J_{\hat{c}}} \Gamma^{(s')}_{\hat{c}} \; - \; \log_2 \mathrm{e} \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}} \left[\frac{q(\hat{c})}{P^J_{\hat{c}}} \Gamma^{(s')}_{\hat{c}} + \frac{1}{\alpha - 1} \frac{q(\hat{c})}{K^J_{\hat{c}}} \Big(\Gamma^{(s')}_{\hat{c}} + \Delta^{(s')}_{\hat{c}} \Big) \right], \tag{8.11}$$

where

$$\Gamma^{(s')}_{\hat{c}} \coloneqq \sum_{(x,v)\colon \phi(a(x,v)) = \hat{c}} \big(W^\dagger_\tau M_x W_\tau\big) \otimes F^{(s')}_v, \qquad \Delta^{(s')}_{\hat{c}} \coloneqq \mathbb{1}[s' \in \mathbb{S}_K(\hat{c})]\; \chi^\dagger_{Q_{s'}}\Big(\mathcal{G}^\dagger_{\hat{c},s'}\big[\nabla T(X_{\hat{c},s'}) - \mathbb{1}\big]\Big), \tag{8.12}$$

with $\mathcal{G}^\dagger_{\hat{c},s'}(Y) = \sum_\kappa K^{(s')\dagger}_\kappa Y K^{(s')}_\kappa$ the Kraus adjoint and $\chi^\dagger_{Q_{s'}}(A) = (W^\dagger_\tau \otimes \mathbb{1})A(W_\tau \otimes \mathbb{1})$. Each $\nabla T(X_{\hat{c},s'})$ is read in the split form of Theorem 8.4 (i): divided differences over the strictly positive spectrum of $X_{\hat{c},s'}$, the pinched operators $\mathcal{Z}(X^p_{\hat{c},s'})$ and $\mathcal{Z}(Y^{\alpha-1})$ on the ambient branch space.

Untrusted-source mode. There the *state form* is primary, the Choi form being its pullback rather than the other way round: the variables are the sector blocks ω_s themselves (Definition 7.10), $\chi_{Q_{s'}} = \mathrm{id}$, and (8.10)–(8.11) hold verbatim with $J_{s'} \to \omega_{s'}$, $P^J_{\hat{c}} \to P_{\hat{c}}$, $K^J_{\hat{c}} \to K_{\hat{c}}$, $X_{\hat{c},s} = \mathcal{G}_{\hat{c},s}(\omega_s)$ and the pieces (8.12) losing their W_τ conjugation:

$$\Gamma^{(s')}_{\hat{c}} = \sum_{(x,v)\colon \phi(a(x,v)) = \hat{c}} M_x \otimes F^{(s')}_v, \qquad \Delta^{(s')}_{\hat{c}} = \mathbb{1}[s' \in \mathbb{S}_K(\hat{c})]\; \mathcal{G}^\dagger_{\hat{c},s'}\big[\nabla T(X_{\hat{c},s'}) - \mathbb{1}\big], \tag{8.13}$$

Hermitian operators on $AQ_{s'}$. The derivation is the proof below with $\chi_{Q_{s'}}$ deleted; the positivity hypothesis on the point reads $\omega_s \in \mathrm{Pos}(AQ_s)$ per block, there being no A_0 in that mode; and the maximal-support hypothesis reads $\operatorname{supp} X_{\hat{c},s'} = W_{\hat{c},s'}$ with $\mathcal{S}_A$ now all of $\mathcal{H}(A)$, for which $\omega_{s'} \succ 0$ suffices. In trusted-source mode the same formulas conjugated by χ give the state-form gradients, and at $d_A = 1$ the two agree because $W_\tau = [1]$.

Lemma 8.6 (Kraus freedom in the branch maps)**.** Let $\alpha \in (1,\infty)$, $p \coloneqq 1/\alpha$ and $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, and let $\mathcal{H}_{\mathrm{in}}$ be a finite-dimensional Hilbert space. For every outcome pair (x,v) of a pass class of $\hat{c}$, that is $a(x,v) \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$, let S_{xv} and S'_{xv} be linear maps from $\mathcal{H}_{\mathrm{in}}$ into finite-dimensional Hilbert spaces $\mathcal{K}$ and $\mathcal{K}'$ with

$$S'^{\dagger}_{xv} S'_{xv} \;=\; S^{\dagger}_{xv} S_{xv}. \tag{8.14}$$

For $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ put

$$K_\kappa \coloneqq \sum_{(x,v)\in a^{-1}(\kappa)} |r(x,v)\rangle_R \, |x,v\rangle_{XY} \, |\kappa\rangle_{C_A C_B} \otimes S_{xv}, \tag{8.15}$$

a map into $\mathcal{B} \coloneqq \mathcal{H}(R\,XY\,C_A C_B) \otimes \mathcal{K}$, and K'_κ likewise with S'_{xv}, into $\mathcal{B}' \coloneqq \mathcal{H}(R\,XY\,C_A C_B) \otimes \mathcal{K}'$. For every positive semidefinite operator ω on $\mathcal{H}_{\mathrm{in}}$ put $X(\omega) \coloneqq \sum_\kappa K_\kappa \omega K_\kappa^\dagger$ and $X'(\omega) \coloneqq \sum_\kappa K'_\kappa \omega K'^{\dagger}_\kappa$, the sums over $\mathcal{K}_{\mathrm{pass}}(\hat{c})$. Let $\mathcal{Z}$ and $\mathcal{Z}'$ be the key pinchings on $\mathcal{B}$ and $\mathcal{B}'$, by the projectors $Z_j = |j\rangle\langle j|_R \otimes \mathbb{1}$, $j \in \mathcal{Z}_\perp$, and T and T' the functionals (4.7) with these pinchings. For $X \succeq 0$ write $\nabla T(X)$ for the operator of (8.8) with the sum over the strictly positive eigenvalues of X only, $Y = \mathcal{Z}(X^p)$ and $0^{\alpha-1} \coloneqq 0$, which is the split form of Theorem 8.4 (i), and $\nabla T'$ likewise.

(i) *(A partial isometry.)* Let $\mathcal{W} \subseteq \mathcal{B}$ be the span of the subspaces $|r(x,v)\rangle_R \, |x,v\rangle_{XY} \, |a(x,v)\rangle_{C_A C_B} \otimes \operatorname{ran} S_{xv}$. There is a linear map $\mathbf{U} : \mathcal{B} \to \mathcal{B}'$ that is isometric on $\mathcal{W}$ and vanishes on $\mathcal{W}^\perp$, with $\mathbf{U} K_\kappa = K'_\kappa$ for every κ and $(Z_j \otimes \mathbb{1}_{\mathcal{K}'})\,\mathbf{U} = \mathbf{U}\,(Z_j \otimes \mathbb{1}_{\mathcal{K}})$ for every j; and $\mathcal{W}$ is invariant under every $Z_j \otimes \mathbb{1}_{\mathcal{K}}$. Hence $X'(\omega) = \mathbf{U} X(\omega) \mathbf{U}^\dagger$, with $\operatorname{supp} X(\omega) \subseteq \mathcal{W}$, for every $\omega \succeq 0$.

(ii) *(Values.)* For every $\omega \succeq 0$: $T'(X'(\omega)) = T(X(\omega))$, $\operatorname{tr} X'(\omega) = \operatorname{tr} X(\omega)$, $\operatorname{supp} X'(\omega) = \mathbf{U} \operatorname{supp} X(\omega)$ and $\operatorname{rank} X'(\omega) = \operatorname{rank} X(\omega)$.

(iii) *(Gradients.)* For every $\omega \succeq 0$, with $X \coloneqq X(\omega)$ and $X' \coloneqq X'(\omega)$,

$$\nabla T'(X') = \mathbf{U}\, \nabla T(X)\, \mathbf{U}^\dagger, \qquad \sum_\kappa K'^{\dagger}_\kappa \big(\nabla T'(X') - \mathbb{1}\big) K'_\kappa = \sum_\kappa K^\dagger_\kappa \big(\nabla T(X) - \mathbb{1}\big) K_\kappa. \tag{8.16}$$

(iv) *(Compression to the support.)* Let $A \succeq 0$ be an operator on $\mathcal{B}$.

(a) *(Containment.)* For every $\omega \succeq 0$, $\operatorname{supp} X(\omega) \subseteq \sum_\kappa \operatorname{ran} K_\kappa$; and $\operatorname{supp}\big((Z_j \otimes \mathbb{1}_{\mathcal{K}})A(Z_j \otimes \mathbb{1}_{\mathcal{K}})\big) \subseteq (Z_j \otimes \mathbb{1}_{\mathcal{K}}) \operatorname{supp} A$ for every j; and $\operatorname{supp} A^q = \operatorname{supp} A$ for every $q > 0$.

(b) *(A floor proves equality.)* If $Q : \mathbb{C}^m \to \mathcal{B}$ is an isometry with $\operatorname{ran} Q \supseteq \operatorname{supp} A$ and $Q^\dagger A Q$ is positive definite, then $\operatorname{ran} Q = \operatorname{supp} A$.

(c) *(The compression.)* If $Q : \mathbb{C}^m \to \mathcal{B}$ is an isometry with $\operatorname{ran} Q = \operatorname{supp} A$ and $m \geq 1$, then $N \coloneqq Q^\dagger A Q$ is positive definite, $A = QNQ^\dagger$, $A^q = QN^qQ^\dagger$ for every $q > 0$, and for every operator H on $\mathcal{B}$ the divided-difference sum of (8.9) at A, over the strictly positive eigenvalues of A, equals $Q\,\mathrm{D}_N[Q^\dagger H Q]\,Q^\dagger$, with D_N the map of Lemma "Perturbation bounds for matrix functions" (foundations/lem:matrix-function-perturbation) (iii) for $f(x) = x^p$.

(v) *(The branch maps.)* With $\mathcal{H}_{\mathrm{in}} = \mathcal{H}(AQ_s)$ and $S_{xv} = \sqrt{M_x} \otimes \sqrt{F_v^{(s)}}$, the K_κ are the $K_\kappa^{(s)}$ of Definition 1.6 and $X(\omega) = \mathcal{G}_{\hat{c},s}(\omega)$. In trusted-source mode, with $\mathcal{H}_{\mathrm{in}} = \mathcal{H}(A_0 Q_s)$, a support frame and $S_{xv} = \sqrt{M_x}\,W_\tau \otimes \sqrt{F_v^{(s)}}$, the K_κ are the $K_\kappa^{(s)}(W_\tau \otimes \mathbb{1}_{Q_s})$, $X(J_s) = \mathcal{G}^J_{\hat{c},s}(J_s)$, and $\sum_\kappa K_\kappa^\dagger(\nabla T(X) - \mathbb{1})K_\kappa = \chi_{Q_s}^\dagger\big(\mathcal{G}^\dagger_{\hat{c},s}[\nabla T(X) - \mathbb{1}]\big)$, the operator of (8.12); the same holds on a joint sector (t,s) with $M_x^{(t)}$ and the frame of the Alice sector t (Definition 4.8). In each case $S_{xv}^\dagger S_{xv}$ is $M_x \otimes F_v^{(s)}$, respectively $(W_\tau^\dagger M_x W_\tau) \otimes F_v^{(s)}$. So for any S'_{xv} with that Gram matrix, the deficit $f_{\hat{c},s}$, the branch operator's rank and the operator $\Delta_{\hat{c}}^{(s)}$ of (8.12) are given, by (ii) and (iii), by X' in place of X.

Theorem 8.9 (Limit of the value as the Rényi order tends to one)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. Assume that there is $p_0 > 0$ with $q(\hat{c}) \geq p_0$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, and that the asymptotic feasible set $\mathcal{W}_1 \coloneqq \{\omega \in \mathcal{W} : \mathbf{\Phi}[\omega] \in \mathcal{S}_{\mathrm{acc}}\}$ is nonempty. Then

$$\lim_{\alpha \downarrow 1} \underline{h}_\alpha(\alpha) \;=\; h_1 \coloneqq \inf_{\omega \in \mathcal{W}_1} \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}} \sum_{s \in \mathbb{S}_K(\hat{c})} D\Big(\mathcal{G}_{\hat{c},s}(\omega_s) \,\Big\|\, \mathcal{Z}\big(\mathcal{G}_{\hat{c},s}(\omega_s)\big)\Big). \tag{8.20}$$

9 The dropped program

Lemma 9.1 (Linear deficit on a rank-one Alice sector)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in trusted-source mode and the general format, and let $t \in \mathbb{T}$ with $\tau_A^{(t)} = p_t\,|\hat{v}\rangle\langle\hat{v}|$, $p_t > 0$, $|\hat{v}\rangle \in \mathcal{H}(A_t)$ a unit vector.

(i) *(product blocks)* Every list $(\omega^{(t,s)})_{s \in \mathbb{S}}$ of positive semidefinite blocks with $\sum_s \operatorname{tr}_{Q_s} \omega^{(t,s)} = \tau_A^{(t)}$, in particular the sector-t part of every $\omega \in$

$\mathcal{W}$, has the form $\omega^{(t,s)} = |\hat{v}\rangle\langle\hat{v}| \otimes \sigma_s$ with $\sigma_s \in \mathrm{Pos}(Q_s)$ and $\sum_s \operatorname{tr}\sigma_s = p_t$.

(ii) *(pass classes are product sets)* If the readout maps are the default ones, $\pi_{\mathsf{A}}(x,v) = x$ and $\pi_{\mathsf{B}}(x,v) = v$, then for every transcript protocol and every $\kappa \in \mathcal{K}$ there are sets $\mathcal{X}_\kappa \subseteq \mathcal{X}$, $\mathcal{V}_\kappa \subseteq \mathcal{V}$ with $a^{-1}(\kappa) = \mathcal{X}_\kappa \times \mathcal{V}_\kappa$; if moreover $\mathsf{h} = \mathsf{A}$, then $r(x,v) = r_{\mathsf{A}}(x,\kappa)$ on $a^{-1}(\kappa)$.

(iii) *(linear deficit)* Under the hypotheses of (ii), let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, $s \in \mathbb{S}$ and $\alpha \in (1,\infty)$, put $\mu_x := \langle\hat{v}|\, M_x^{(t)}\, |\hat{v}\rangle$ and, for $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ and $\mathsf{z} \in \mathcal{Z}$,

$$m_\kappa := \sum_{x\in\mathcal{X}_\kappa} \mu_x, \qquad m_{\kappa,\mathsf{z}} := \sum_{x\in\mathcal{X}_\kappa:\, r_{\mathsf{A}}(x,\kappa)=\mathsf{z}} \mu_x, \qquad \eta_\kappa := \begin{cases} 1 - \sum_{\mathsf{z}} \left(m_{\kappa,\mathsf{z}}/m_\kappa\right)^\alpha, & m_\kappa > 0, \\ 0, & m_\kappa = 0. \end{cases} \tag{9.1}$$

Then for every $\sigma_s \in \mathrm{Pos}(Q_s)$

$$f_{\hat{c},(t,s)}\big(\,|\hat{v}\rangle\langle\hat{v}| \otimes \sigma_s\big) = -\sum_{\kappa\in\mathcal{K}_{\mathrm{pass}}(\hat{c})} \eta_\kappa\, m_\kappa \sum_{v\in\mathcal{V}_\kappa} \operatorname{tr}\left[F_v^{(s)}\sigma_s\right], \tag{9.2}$$

a *linear* function of σ_s, with $0 \le \eta_\kappa < 1$. In ensemble form $\mu_x = \operatorname{tr} N_x^{(t)}/p_t$ and $p_t = \operatorname{tr}\tau_A^{(t)}$, declared numbers.

(iv) *(two values)* If within every pass class of $\hat{c}$ the key takes d values of equal weight, $m_{\kappa,\mathsf{z}} = m_\kappa/d$, then $\eta_\kappa = 1 - d^{1-\alpha}$ and the sector contributes to $\frac{1}{1-\alpha}\log_2$ of the ratio K/P *as if the conditional entropy were* $\log_2 d$ *on the pass weight*: one bit per unit of pass weight for a uniform key bit. The qualifier matters: what the expression assigns to the pair is its fail weight plus $d^{1-\alpha}$ times its pass weight, so with a non-zero fail weight in the same joint pair the contribution is strictly less than that of a sector of conditional entropy $\log_2 d$, and equals it exactly when the pass weight is the whole of the pair's weight. If the key is constant on every $\mathcal{X}_\kappa \cap \{x : \mu_x > 0\}$ then $\eta_\kappa = 0$ and $f_{\hat{c},(t,s)}$ vanishes on these blocks.

Definition 9.2 (Dropped set and dropped program)**.** Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1. A *dropped set* is a set $\mathcal{D} \subseteq \widehat{\mathcal{C}}_{\mathrm{key}} \times (\mathbb{T}\times\mathbb{S})$ of pairs of a key-live acceptance value and a joint sector. For $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$ block diagonal along the joint sectors put

$$K_{\hat{c}}^{\mathcal{D}}(\omega) := P_{\hat{c}}(\omega) + \sum_{(t,s):\,(\hat{c},(t,s))\notin\mathcal{D}} f_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big) \qquad (\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}), \tag{9.3}$$

with $f_{\hat{c},(t,s)}$ of (4.8); let $O_\alpha^{\mathcal{D}}$ be (5.3) with $K_{\hat{c}}^{\mathcal{D}}$ in place of $K_{\hat{c}}$, with the same conventions at vanishing arguments; and let

$$\underline{h}_\alpha^{\mathcal{D}} := \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}} O_\alpha^{\mathcal{D}}(\mathbf{q},\omega). \tag{9.4}$$

The minimisation in (9.4) is the *dropped program*. A joint sector (t,s) is *fully dropped* when $\widehat{\mathcal{C}}_{\text{key}} \times \{(t,s)\} \subseteq \mathcal{D}$. For $\mathcal{D} = \varnothing$ these are $K_{\hat{c}}$ ((4.11)), O_α and $\underline{h}_\alpha$.

What is not changed. The feasible set is $\mathcal{W}$, unchanged: a dropped sector's block remains a variable, enters $\mathbf{\Phi}$ and every $P_{\hat{c}}$ with its full weight — the term $P_{\hat{c},(t,s)}(\omega^{(t,s)})$ stays inside $P_{\hat{c}}$ in (9.3) — and enters the marginal condition and every constraint row. Setting a dropped block to zero, omitting its statistics, or removing its weight from $P_{\hat{c}}$ in (9.3) are different operations, and each can overstate the value (Proposition 6.3 for the first). *The dropped set is a formulation choice of a run and not declared data.*

Proposition 9.3 (The dropped program bounds the value from below)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format, in either source mode and with an arbitrary list $\mathfrak{L}$; let $\alpha \in (1,\infty)$ and let $\mathcal{D} \subseteq \mathcal{D}'$ be dropped sets (Definition 9.2). For every ω block diagonal along the joint sectors, every $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ and every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$:

(i) $0 \le K_{\hat{c}}(\omega) \le K_{\hat{c}}^{\mathcal{D}}(\omega) \le K_{\hat{c}}^{\mathcal{D}'}(\omega) \le P_{\hat{c}}(\omega)$;

(ii) $O_\alpha^{\mathcal{D}'}(\mathbf{q},\omega) \le O_\alpha^{\mathcal{D}}(\mathbf{q},\omega) \le O_\alpha(\mathbf{q},\omega)$ in the extended-real sense, and $O_\alpha^{\mathcal{D}'}(\mathbf{q},\omega) \ge 0$ whenever $\operatorname{tr}\omega = 1$;

(iii) consequently $0 \le \underline{h}_\alpha^{\mathcal{D}'} \le \underline{h}_\alpha^{\mathcal{D}} \le \underline{h}_\alpha \le h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$;

(iv) if every pair of $\mathcal{D}$ is key-inert (Definition 4.8) then $K_{\hat{c}}^{\mathcal{D}} = K_{\hat{c}}$ and $O_\alpha^{\mathcal{D}} = O_\alpha$ identically.

Corollary 9.4 (Convexity, gradient and limit of the dropped program)**.** Under the hypotheses of Proposition 9.3, with $\mathcal{S}_{\text{acc}}$ convex and compact (Definition 2.5):

(i) $K_{\hat{c}}^{\mathcal{D}}$ is concave and continuous on the cone of operators block diagonal along the joint sectors;

(ii) $O_\alpha^{\mathcal{D}}$ is jointly convex and lower semicontinuous on $\Delta(\widehat{\mathcal{C}}) \times \mathcal{W}$ with values in $[0,+\infty]$, and the infimum (9.4) is attained whenever it is finite;

(iii) if (t,s) is fully dropped, then $O_\alpha^{\mathcal{D}}$ depends on the block $\omega^{(t,s)}$ only through the real numbers $\operatorname{tr}\left[(M_x^{(t)} \otimes F_v^{(s)})\,\omega^{(t,s)}\right]$, $(x,v) \in \mathcal{X} \times \mathcal{V}$; at a point where every $P_{\hat{c}}$ with $q(\hat{c}) > 0$ and every $K_{\hat{c}}^{\mathcal{D}}$ with $q(\hat{c}) > 0$, $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$, is positive, its derivative in the direction of a Hermitian Δ supported on that joint sector is

$$-\log_2 \mathrm{e} \sum_{\hat{c} \in \widehat{\mathcal{C}}} \beta_{\hat{c}}\, \operatorname{tr}\left[\Gamma_{\hat{c}}^{(t,s)} \Delta\right], \qquad \Gamma_{\hat{c}}^{(t,s)} := \sum_{(x,v):\, \phi(a(x,v)) = \hat{c}} M_x^{(t)} \otimes F_v^{(s)}, \qquad \beta_{\hat{c}} := \begin{cases} \frac{\alpha}{\alpha-1}\frac{q(\hat{c})}{P_{\hat{c}}}, & \hat{c} \notin \\ \frac{q(\hat{c})}{P_{\hat{c}}} + \frac{1}{\alpha-1}\frac{q(\hat{c})}{K_{\hat{c}}^{\mathcal{D}}}, & \hat{c} \in \end{cases} \tag{9.5}$$

with $\beta_{\hat{c}} := 0$ where $q(\hat{c}) = 0$;

(iv) under the additional hypotheses of Theorem 8.9,

$$\lim_{\alpha\downarrow 1} \underline{h}_\alpha^{\mathcal{D}} = \inf_{\omega\in\mathcal{W}_1} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \sum_{\substack{(t,s)\in\mathbb{J}_K(\hat{c}) \\ (\hat{c},(t,s))\notin\mathcal{D}}} D\Big(\mathcal{G}_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big) \,\Big\|\, \mathcal{Z}\big(\mathcal{G}_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big)\big)\Big). \tag{9.6}$$

Proposition 9.5 (The minimal divergence as the value with every pair dropped)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format, in either source mode and with an arbitrary list $\mathfrak{L}$, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let $\alpha \in (1,\infty)$. Let $\mathcal{D}_{\mathrm{all}} \coloneqq \widehat{\mathcal{C}}_{\mathrm{key}} \times (\mathbb{T}\times\mathbb{S})$ be the dropped set of every pair (Definition 9.2), and put

$$D_{\min} \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}} D\big(\mathbf{q} \,\big\|\, \mathbf{\Phi}[\omega]\big) \in [0,+\infty], \qquad m_{\mathrm{key}} \coloneqq \sup_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) \in [0,1], \tag{9.7}$$

with $\inf\varnothing = +\infty$ and $\sup\varnothing = 0$.

(i) *(The value with every pair dropped.)* For every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega\in\mathcal{W}$, $O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) = \frac{\alpha}{\alpha-1}\, D(\mathbf{q}\|\mathbf{\Phi}[\omega])$; hence $\underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}} = \frac{\alpha}{\alpha-1}\, D_{\min}$.

(ii) *(The value between its floor and its ceiling.)*

$$\frac{\alpha}{\alpha-1}\, D_{\min} \;\le\; \underline{h}_\alpha \;\le\; \frac{\alpha}{\alpha-1}\, D_{\min} + \log_2(d_Z+1)\, m_{\mathrm{key}}. \tag{9.8}$$

(iii) *(When the floor is positive.)* $D_{\min} = 0$ if and only if $\mathbf{q} = \mathbf{\Phi}[\omega]$ for some $\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}$ and some $\omega\in\mathcal{W}$. Equivalently, $D_{\min} > 0$ if and only if no distribution of $\mathcal{S}_{\mathrm{acc}}$ is the statistics $\mathbf{\Phi}[\omega]$ of a state $\omega\in\mathcal{W}$.

(iv) *(Bounds at one order serve every order.)* Let b be a real number with $b \le \underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}}$, and let $(\mathbf{q},\omega)\in\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}$. Then

$$\frac{\alpha-1}{\alpha}\, b \;\le\; D_{\min} \;\le\; \frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega), \tag{9.9}$$

and for every $\alpha'\in(1,\infty)$ the value $\underline{h}_{\alpha'}$ of (5.4) at the order α' satisfies $\underline{h}_{\alpha'} \ge \frac{\alpha'}{\alpha'-1}\cdot\frac{\alpha-1}{\alpha}\, b$. If $b>0$, then $D_{\min}>0$, and no distribution of $\mathcal{S}_{\mathrm{acc}}$ is the statistics of a state of $\mathcal{W}$. For every $\alpha'\ge\alpha$,

$$\frac{\alpha'}{\alpha'-1}\, D_{\min} \;\le\; \frac{\alpha'}{\alpha'-1}\cdot\frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) \;\le\; O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega), \tag{9.10}$$

so that $\underline{h}_{\alpha'} \le \frac{\alpha'}{\alpha'-1}\cdot\frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) + \log_2(d_Z+1)\, m_{\mathrm{key}}$.

10 Key length

Definition 10.1 (Fixed-length post-processing model)**.** Fix a declaration $\mathfrak{D}$ with key holder h (Definition 1.1), a round count n, an acceptance set $\mathcal{S}_{\mathrm{acc}}$, an output length ℓ, a transcript length λ_{EC} and error budgets $\varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}} \in (0,1]$. A hash family is a set of functions with a probability law on it [3, Def. 3.5]. The *fixed-length post-processing model* is the following classical processing, applied after the n round channels of Definition 2.3. The raw string Z_1^n is the content of the classical registers $Z_1, \dots, Z_n$ of the round channels, which the single-round protocol channel $\mathcal{M}$ retains (Theorem 1.5); the key holder computes it from its own outcome record and the announcements I_1^n through the function r_{h} of (1.1), and we call the other party the partner. The acceptance values $\widehat{C}_1^n$ are announced. The processing appends, in order:

(i) the *acceptance test*: the event $\Omega_{\mathrm{AT}} \coloneqq \{\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$, a function of $\widehat{C}_1^n$;

(ii) *error correction*, one-way from the holder: a public seed S_{pub}, drawn with the law the error-correction scheme prescribes, and the public transcript $L_{\mathrm{EC}} = g_{\mathrm{EC}}(Z_1^n, S_{\mathrm{pub}})$, which occupies a fixed-length register of λ_{EC} bits: a shorter transcript is padded to length λ_{EC}, so that $\dim(L_{\mathrm{EC}}) = 2^{\lambda_{\mathrm{EC}}}$ exactly; from L_{EC} and its own data the partner forms a guess $\hat{Z}_1^n$ of Z_1^n;

(iii) *error verification*: a public seed S_{EV} selects a function h_{EV}, with the law of the error-verification family, from a hash family of functions with $\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$-bit values; the holder announces $T_{\mathrm{EV}} \coloneqq h_{\mathrm{EV}}(Z_1^n)$, the partner compares it with $h_{\mathrm{EV}}(\hat{Z}_1^n)$ and announces the one-bit outcome $F_{\mathrm{acc}} \in \{\mathrm{acc}, \mathrm{rej}\}$, which is acc exactly when the two values agree;

(iv) *privacy amplification*: a function G is drawn, with the law of the privacy-amplification family, from a hash family of functions with ℓ-bit values, and announced; the holder outputs $K \coloneqq G(Z_1^n)$ and the partner $\hat{K} \coloneqq G(\hat{Z}_1^n)$.

We write K_A and K_B for Alice's and Bob's outputs, which are K and $\hat{K}$ in the order the key holder fixes, and

$$S \coloneqq (S_{\mathrm{pub}}, S_{\mathrm{EV}}), \qquad L \coloneqq (L_{\mathrm{EC}}, T_{\mathrm{EV}}, F_{\mathrm{acc}}), \qquad E_{\mathrm{tot}} \coloneqq \widehat{C}_1^n I_1^n \widehat{E}\, S\, L\, G, \tag{10.1}$$

where L has at most $\lambda \coloneqq \lambda_{\mathrm{EC}} + \lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil + 1$ bits and $\widehat{E}$ is the adversary's retained quantum memory of Definition 2.3. The protocol *accepts* on $\Omega_{\mathrm{acc}} \coloneqq \Omega_{\mathrm{AT}} \wedge \{F_{\mathrm{acc}} = \mathrm{acc}\}$; on its complement both parties abort and output the symbol $\perp$ in place of a key. The model is stated in the *honest-authentication setting*: every classical message of the protocol, the round announcements

included, reaches its recipient exactly as it was sent, in the order sent and at some time after it was sent, the adversary receives a copy of it, and no message fails authentication.

Definition 10.2 (Authenticated classical channel and authentication post-processing)**.** Label each classical message of a party by its position i in that party's sequence of sent messages, and the messages a party receives by their position in its sequence of received messages; times are global times, which the parties need not know.

(a) The *authenticated classical channel* between the two parties gives the adversary a copy of every message sent over it and lets her choose when each message is delivered, subject to: (A1) if the i-th message a party receives arrives before the other party sent its i-th message, the received symbol is auth-abort; (A2) otherwise the received symbol is either the i-th sent message, unaltered, or auth-abort.

(b) An *authentication scheme* is a pair of procedures, one per party, that use a pre-shared key and an unauthenticated classical channel, on which the adversary may read, alter, inject, delay and suppress messages. It *ε-realises* the authenticated classical channel if there is a simulator at the adversary's interface such that the distinguishing advantage between the scheme run on the key and the unauthenticated channel, and the authenticated classical channel with that simulator, is at most ε [2, Sec. II.E, Def. 1]. Its *key consumption* is the number of bits of the pre-shared key it uses in one run of the protocol.

(c) The *authentication post-processing* appended to the fixed-length post-processing model (Definition 10.1) is: a party that receives auth-abort for any message aborts, replaces its key by $\perp$ and sends abort in place of each later message; after privacy amplification the partner sends accept if it holds a key and abort otherwise; the key holder sends accept if it received accept and holds a key, and otherwise sends abort and replaces its key by $\perp$; the partner keeps its key if it receives accept and otherwise replaces it by $\perp$.

Lemma 10.4 (Correctness from error verification)**.** Let the protocol run the fixed-length post-processing model (Definition 10.1) in the honest-authentication setting, let Assumptions 13.2, 13.3 and 13.5 hold, and let $\varepsilon_{\mathrm{EV}} \in (0, 1]$. Then for every attack and every output length ℓ the protocol is $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, in the sense of [4, Sec. 2.3.1, Def. 2.2, Eq. (4)]:

$$\Pr\left[\Omega_{\mathrm{acc}} \wedge K_A \neq K_B\right] \leq \varepsilon_{\mathrm{EV}}. \tag{10.2}$$

Lemma 10.5 (Approximating an attack by all-at-once attacks with finite memory)**.** Read a run of the protocol and its attacks as follows.

(a) *(the run)* Its quantum phase is an n-round protocol of Definition "Measurement-factorizing n-round protocol" (reductions/def:protocol), whose measured system B_j of round j is separable and of any dimension, followed by the fixed-length post-processing model of Definition 10.1 in the honest-authentication setting. In trusted-source mode its source releases, for $i = 1, \dots, n$, a separable transmitted system A'_i of any dimension, prepared together with Alice's letter register X_i in a state ρ_i on $X_i A'_i$ that depends on nothing else, and the run reads X_i from round i on; in untrusted-source mode there are no X_i and no A'_i. Every system of the run other than the B_j and the A'_i is finite-dimensional, and its public messages are classical registers it keeps.

(b) *(attacks)* An *attack* consists of a release schedule $0 = k_0 \le k_1 \le \cdots \le k_n = n$ with $k_j \ge j$ for every j, a state ζ on the adversary's initial register E_0, and channels, in the sense of Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression),

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}\big(A'_{k_{j-1}+1} \cdots A'_{k_j} E_{j-1},\ B_j E'_j\big), & \text{trusted-source mode,} \\ \mathrm{CPTP}\big(E_{j-1},\ A_j B_j E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{10.3}$$

for $j = 1, \dots, n$, where E'_j is the adversary's memory after round j, E_j is E'_j together with a copy of the public messages of round j, and $\mathcal{H}(E_0)$ and every $\mathcal{H}(E'_j)$ are Hilbert spaces of any dimension. Under the attack the run starts from ζ and its own registers, prepares ρ_i in round j for $k_{j-1} < i \le k_j$ before $\mathcal{A}_j$ acts, takes B_j, and in untrusted-source mode Alice's system A_j, from the output of $\mathcal{A}_j$, and places a copy of the public messages of round j in E_j. Its *output state* is the state of the run's output registers, its public messages and E'_n. The attack is *all-at-once* if $k_1 = n$, and has *finite registers* if E_0 and every E'_j are finite-dimensional.

Then:

(i) *(approximation)* for every attack and every $\eta > 0$ there is an all-at-once attack with finite registers whose memory after round n is a subspace of $\mathcal{H}(E'_n)$, such that the output states ρ and ρ° of the two attacks satisfy $\|\iota(\rho^\circ) - \rho\|_1 \le \eta$, with ι the inclusion of that subspace into $\mathcal{H}(E'_n)$, acting as the identity on every other register;

(ii) *(trace-distance criteria)* let L_1 and L_2 be completely positive maps, not increasing the trace, from the run's output registers and public

messages to a finite-dimensional system, put $L \coloneqq L_1 - L_2$, and for a state σ of those registers and the adversary's memory after round n put $f(\sigma) \coloneqq \frac{1}{2}\|(L \otimes \mathrm{id})(\sigma)\|_1$; if $f(\rho^\circ) \leq \varepsilon$ at the output state ρ° of every all-at-once attack with finite registers, then $f(\rho) \leq \varepsilon$ at the output state ρ of every attack;

(iii) *(processing by the adversary)* let, in (ii), each L_t keep the public messages and act on the other registers given their value m, $L_t(|m\rangle\langle m| \otimes x) = |m\rangle\langle m| \otimes L_{t,m}(x)$, and let $\Phi = \sum_m |m\rangle\langle m| \otimes \Phi_m$ act on a state that is classical on the public messages by a channel Φ_m on the adversary's register, given their value m, keeping them. Then $f((\mathrm{id} \otimes \Phi)(\sigma)) \leq f(\sigma)$ for every state σ classical on the public messages.

Lemma 10.6 (Adversaries of the honest-authentication setting as round attacks)**.** Let the run be as in Lemma 10.5 (a), with A-ALICE-MEAS (Assumption "Alice's declared measurement where no source certificate is demanded" (protocol-compiler/ass:alice-meas)) (i) for Alice's measurement where no source-replacement certificate is demanded, and let A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), its clause (v) included, Assumption 13.8 (A-LAB) at each of the two parties and Assumption 13.5 (A-CONFORM) hold. Let an adversary act on the run in the honest-authentication setting (Definition 10.1) by any operations, at any times, each chosen given the classical outcomes of her earlier ones, on her own registers, of any dimension, on each transmitted system once the source has released it and on her copies of the public messages once she holds them, delivering the measured system of each round, and in untrusted-source mode Alice's system, for the round's measurements. Then there are an all-at-once attack in the sense of Lemma 10.5 (b), with output state ρ, and a channel $\Phi = \sum_m |m\rangle\langle m| \otimes \Phi_m$ that applies a channel Φ_m to her register E'_n given the value m of the run's public messages and keeps them, such that the state of the run's output registers, its public messages and the adversary's final register is $(\mathrm{id} \otimes \Phi)(\rho)$.

Theorem 10.7 (Secrecy of the key from the accumulated entropy). Let the protocol run the fixed-length post-processing model (Definition 10.1) in the honest-authentication setting, with the registers (10.1). Let one of the two accumulation branches hold: *(unconstrained)* the hypotheses of Theorem 2.10 with $\mathfrak{L} = \varnothing$; or *(constrained)* the hypotheses of Theorem 3.1 ($\mathfrak{L}$ licensed for the declared source mode, Definition 2.2; A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk))), in either branch save the condition $\Pr[\Omega_{\mathrm{AT}}] > 0$ that the accumulation theorem places on the attack, which the attack below need not meet. Let the run be as in Lemma 10.5 (a), as A-SPEC (Assumption "Specification faithful to the apparatus" (protocol-compiler/ass:spec)) (i) attests for its measurement, with A-

ALICE-MEAS (Assumption "Alice's declared measurement where no source certificate is demanded" (protocol-compiler/ass:alice-meas)) (i) for Alice's measurement where no source-replacement certificate is demanded, and A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), or the premise a source model declares in its place, for its source states; let it be attacked by an attack in the sense of Lemma 10.5 (b) that is all-at-once and has finite registers; and let the reduction chain from which the declaration was compiled carry it onto the round model, as A-LINK (Assumption "Soundness of every reduction link" (protocol-compiler/ass:link)) (i)(a), (i)(b), (ii)(a) and (iv) attest, the existence of each link's channel, the pull-back identity, the source identity and complete positivity, with A-ATK in the constrained branch:

(H-CHAIN) the marginal on $Z_1^n \widehat{C}_1^n I_1^n E_n$ of the run's state after the n rounds is the state the round channels of Definition 2.3 produce under an attack sequence of that definition, $\mathfrak{L}$-admissible in the constrained branch, from an input state as in the branch's accumulation theorem; the run's state is an extension of it, holding the parties' further records.

Let Assumptions 13.1, 13.2, 13.3, 13.4, 13.5 and 13.8, the last at each of the two parties, and A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)) hold, let $\alpha \in (1,2)$ and $\varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}} \in (0,1]$. If the output length satisfies

$$\ell \;\leq\; n\,\underline{h}_\alpha - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \tag{10.4}$$

then, under that attack, the protocol is $\varepsilon_{\mathrm{PA}}$-secret on the accepted state:

$$\frac{1}{2} \left\| \rho_{K\,E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} - \frac{\mathbb{1}_K}{2^\ell} \otimes \rho_{E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} \right\|_1 \leq \varepsilon_{\mathrm{PA}}.$$

Corollary 10.8 (Security of the protocol and its net key). Let the hypotheses of Theorem 10.7, the run as in Lemma 10.5 (a) and (H-CHAIN) among them, hold at every attack in the sense of Lemma 10.5 (b) that is all-at-once and has finite registers, with $\ell \geq 1$ satisfying (10.4), A-SEQ with its clause (v) among them; let those of Lemma 10.4 hold; and put $\tau_{K_A K_B} \coloneqq 2^{-\ell} \sum_{k \in \{0,1\}^\ell} |k\rangle\langle k|_{K_A} \otimes |k\rangle\langle k|_{K_B}$.

(i) In the honest-authentication setting, at every output state the setting admits, whatever the dimension of the adversary's registers and the order in which the source releases its signals, the protocol is $\varepsilon_{\mathrm{PA}}$-secret on the accepted state as Theorem 10.7 states it, and $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-secure in the sense of [4, Sec. 2.3, Def. 2.1, Eq. (1)]:

$$\frac{1}{2} \left\| \rho_{K_A K_B\, E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} - \tau_{K_A K_B} \otimes \rho_{E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} \right\|_1 \leq \varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}}, \tag{10.7}$$

and on the complement of Ω_{acc} both parties output $\perp$.

(ii) Run with the authentication post-processing (Definition 10.2 (c)) over the authenticated classical channel (Definition 10.2 (a)), the protocol is $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}})$-secure in the sense of [3, Sec. 3.3, Def. 3.8], which admits asymmetric aborts, for every output state that channel admits.

(iii) Let Assumptions 13.6, 13.4 and 13.5 hold, with $\varepsilon_{\mathrm{auth}}$ and k_{auth} the parameters of Assumption 13.6. Run with the authentication post-processing over the classical channel of Assumption 13.6, the protocol is $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}}+2\varepsilon_{\mathrm{auth}})$-secure in the sense of [3, Sec. 3.3, Def. 3.8]: for every attack on that run, with output state ρ' on the key registers and the adversary's final information,

$$\frac{1}{2}\left\|\rho' - \mathcal{R}_{\mathrm{ideal}}(\rho')\right\|_1 \leq \varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}}, \tag{10.8}$$

with $\mathcal{R}_{\mathrm{ideal}}$ the map of [3, Sec. 3.3, Eqs. (13)–(17)]. It consumes at most k_{auth} bits of the pre-shared key, and when both parties output an ℓ-bit key its *net key gain* is $\ell_{\mathrm{net}} := \ell - k_{\mathrm{auth}}$ bits; for $\ell_{\mathrm{net}} \leq 0$ the run yields no net key.

Corollary 10.9 (Security parameters of separate executions add)**.** Let executions $1, \ldots, N$ of the protocol run against one adversary, in sequence, concurrently or interleaved, the adversary carrying information from each to the others. For each i let the hypotheses of Corollary 10.8 (iii) hold for execution i, with its parameters $\varepsilon_{\mathrm{PA}}^{(i)}$, $\varepsilon_{\mathrm{EV}}^{(i)}$ and $\varepsilon_{\mathrm{auth}}^{(i)}$. Write $K^{(i)} := K_A^{(i)} K_B^{(i)}$ for its key registers, $\mathcal{R}^{(i)}$ for the map $\mathcal{R}_{\mathrm{ideal}}$ of [3, Sec. 3.3, Eqs. (13)–(17)] on $K^{(i)}$, and $\varepsilon_i := \varepsilon_{\mathrm{PA}}^{(i)} + \varepsilon_{\mathrm{EV}}^{(i)} + 2\varepsilon_{\mathrm{auth}}^{(i)}$. Let Assumption 13.7 hold. Then for every attack on the N executions under which each of them reaches its end of an execution (Definition 10.3) on every branch, with ρ the joint state of $K^{(1)} \cdots K^{(N)}$ and the adversary's information E at the end of the last of them,

$$\frac{1}{2}\left\|\rho - \left(\mathcal{R}^{(1)} \otimes \cdots \otimes \mathcal{R}^{(N)} \otimes \mathrm{id}_E\right)(\rho)\right\|_1 \leq \sum_{i=1}^{N} \varepsilon_i. \tag{10.9}$$

Proposition 10.12 (The pass probability of devices the declared model admits)**.** Let $\mathfrak{D}$ and $\mathcal{S}_{\mathrm{acc}}$ satisfy the hypotheses of Proposition 9.5, let $D_{\min}$ be as in (9.7) and $\mathcal{D}_{\mathrm{all}}$ the dropped set of every pair there, and let $n \geq 1$ be an integer. Let Pr be a probability distribution on $\widehat{\mathcal{C}}^n$, write $\widehat{C}_1, \ldots, \widehat{C}_n$ for the coordinates and $\widehat{C}_1^t := (\widehat{C}_1, \ldots, \widehat{C}_t)$, and suppose:

(H) for every $t \in \{1, \ldots, n\}$ and every $h \in \widehat{\mathcal{C}}^{t-1}$ with $\Pr\left[\widehat{C}_1^{t-1} = h\right] > 0$ there is a state $\omega_t(h) \in \mathcal{W}$ with

$$\Pr\left[\widehat{C}_t = \hat{c} \;\middle|\; \widehat{C}_1^{t-1} = h\right] = \mathbf{\Phi}[\omega_t(h)](\hat{c}) \qquad \text{for every } \hat{c} \in \widehat{\mathcal{C}}, \tag{10.11}$$

where $\mathbf{\Phi}[\omega](\hat{c})$ is the entry of $\mathbf{\Phi}[\omega]$ at $\hat{c}$ and, for $t = 1$, h is the empty sequence and the left-hand side is $\Pr\left[\widehat{C}_1 = \hat{c}\right]$.

The case of independent rounds with one state $\omega \in \mathcal{W}$ is $\omega_t(h) = \omega$ for all t and h. Let $\Omega_{\mathrm{AT}} = \{\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$ be the event of Definition 10.1 (i), where $\mathrm{freq}_{\hat{c}_1^n}(\hat{c}) := \frac{1}{n}\,|\{t : \hat{c}_t = \hat{c}\}|$.

(i) *(The bound.)* For every real number $\delta \le D_{\min}$,

$$\Pr[\Omega_{\mathrm{AT}}] \;\le\; 2^{-n\delta}. \tag{10.12}$$

In particular $\Pr[\Omega_{\mathrm{AT}}] = 0$ if $D_{\min} = +\infty$.

(ii) *(From a bound at one order.)* Let $\alpha_0 \in (1, \infty)$ and let b be a real number with $b \le \underline{h}_{\alpha_0}^{\mathcal{D}_{\mathrm{all}}}$. Then $\Pr[\Omega_{\mathrm{AT}}] \le 2^{-x}$ for every real number $x \le n\,\frac{\alpha_0 - 1}{\alpha_0}\,b$. For every $\alpha' \in (1, \infty)$ the number $f(\alpha') := \frac{\alpha'}{\alpha' - 1} \cdot \frac{\alpha_0 - 1}{\alpha_0}\,b$ satisfies $\frac{\alpha' - 1}{\alpha'}\,n\,f(\alpha') = n\,\frac{\alpha_0 - 1}{\alpha_0}\,b$.

Definition 10.13 (Maximal key length)**.** The *maximal key length* is

$$\ell_{\max} := \max\{0, \lfloor n\underline{h}_\alpha - \lambda_{\mathrm{EC}} - \lceil \log_2 \tfrac{1}{\varepsilon_{\mathrm{EV}}} \rceil - \tfrac{\alpha}{\alpha - 1} \log_2 \tfrac{1}{\varepsilon_{\mathrm{PA}}} + 1 \rfloor\},$$

the right-hand side of (10.4) under a floor and a clamp at 0; the floor and the clamp are both in the safe direction: an integer at most the floor of a number is at most that number, and a key of length 0 is trivially secret and correct and claims no key.

Proposition 10.17 (Monotonicity of the maximal key length)**.** The maximal key length $\ell_{\max}$ of Definition 10.13 is

(i) non-increasing in λ_{EC};

(ii) non-decreasing in $\varepsilon_{\mathrm{EV}}$;

(iii) non-decreasing in $\varepsilon_{\mathrm{PA}}$;

(iv) unchanged under any modification of the declaration that leaves $\underline{h}_\alpha$ unchanged.

Consequently increasing leakage or tightening either error budget can never increase $\ell_{\max}$.

Corollary 10.18 (Key length from any lower bound on the value). Let $\alpha \in (1, 2)$, let the hypotheses of Theorem 10.7 hold in its unconstrained or its constrained branch, the run as in Lemma 10.5 (a) and hypothesis (H-CHAIN) included, at every all-at-once attack with finite registers, A-SEQ

with its clause (v) among them, and let those of Lemma 10.4 hold. Let $h \in \mathbb{R} \cup \{-\infty\}$ satisfy

$$h \;\le\; \underline{h}_\alpha, \tag{10.17}$$

with $\underline{h}_\alpha$ the exact value (5.4) of the declaration (for the constrained branch: with $\mathcal{W}$ carrying the declared $\mathfrak{L}$). If the output length satisfies

$$\ell \;\le\; n\,\max\{0,h\} - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha-1}\log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \tag{10.18}$$

then the protocol is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, hence, for $\ell \ge 1$, $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}})$-secure in the honest-authentication setting; under the further hypotheses of Corollary 10.8 (iii) it is $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}})$-secure over the classical channel of that part, with net key gain $\ell - k_{\mathrm{auth}}$ when both parties output a key. The secrecy and the security statements hold at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals, as Corollary 10.8 states them, and the correctness statement at every attack. Moreover:

(a) if h depends on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$, the four conclusions of Proposition 10.17 hold verbatim for the maximal key length (Definition 10.13) built from (10.18), with $\max\{0,h\}$ in place of $\underline{h}_\alpha$;

(b) if $h \le 0$ and $\varepsilon_{\mathrm{EV}} < 1$ then no positive ℓ satisfies (10.18) and no key is claimed (at $\varepsilon_{\mathrm{EV}} = 1$ the correctness statement is empty and the corner carries no security claim).

Corollary 10.19 (Key length from the constrained value)**.** Under the hypotheses of Theorem 3.1, with Corollary 5.6,

$$\widetilde{H}^{\uparrow}_{\alpha}\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\ge\; n\,\underline{h}_\alpha(\mathfrak{L}) - \frac{\alpha}{\alpha-1}\log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]},$$

which is the form Theorem 10.7 consumes; under A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) and the remaining hypotheses of Theorem 10.7, the key-length statement of Theorem 10.7 therefore holds with the constrained $\underline{h}_\alpha$ (its constrained branch).

11 Perturbations of the declared measurement

Proposition 11.1 (Instability of the rate under measurement perturbations)**.** Let $\alpha \in (1,\infty)$ and $B > 0$, write $N := \lceil B \rceil$, and let $\epsilon \in (0, \frac{1}{2}]$ satisfy $2^N \le \epsilon\, 2^{53}$. Write $\mathfrak{D}^{(i)}$ for the declaration obtained from a given one by replacing its measurement $\{F_v\}$ on Q by $\{F_v^{(i)}\}$, and $\underline{h}_\alpha[\,\cdot\,]$ for the value (5.4) of a declaration.

(i) There is a declaration $\mathfrak{D}$ satisfying Definition 1.2 in full, with $\{F_v\}$ itself exactly admissible, and there are two exactly admissible families $\{F_v^{(1)}\}$, $\{F_v^{(2)}\}$ with $\max_v \|F_v^{(i)} - F_v\| \le \epsilon$ for $i = 1, 2$, such that

$$\underline{h}_\alpha\big[\mathfrak{D}^{(1)}\big] - \underline{h}_\alpha\big[\mathfrak{D}^{(2)}\big] \;\ge\; B. \tag{11.1}$$

The dimensions and alphabets of $\mathfrak{D}$ are $d_A = d_Q = 1$, one sector, $|\mathcal{X}| = 1$, $|\mathcal{V}| = |\widehat{\mathcal{C}}| = 2$ and $d_Z = 1$, the same for every B and every ϵ.

(ii) Let err be a function of the declared dimensions, alphabet sizes and α whose value on the data of (i) is less than $\frac{\alpha}{\alpha-1}N$. Then the bound $\big|\underline{h}_\alpha[\mathfrak{D}^{(1)}] - \underline{h}_\alpha[\mathfrak{D}^{(2)}]\big| \le \mathrm{err}$, with err evaluated on the data of the declaration, fails for some declaration and some pair of exactly admissible families within ϵ of each other.

(iii) For $\alpha \in (1, 2)$ and every number n of rounds, the maximal admissible key lengths $\ell_{\max}$ of $\mathfrak{D}^{(1)}$ and $\mathfrak{D}^{(2)}$ (Definition 10.13), computed with the same λ_{EC}, $\varepsilon_{\mathrm{EV}}$ and $\varepsilon_{\mathrm{PA}}$, differ by at least $nB - 1$ bits whenever the smaller of the two is positive.

12 The reference companion

Proposition 12.1 (The compiled reference companion is an admissible declaration)**.** Let $\mathfrak{D}$ be an admissible declaration (Definition 1.2) that the compilation map (Definition "The compilation map" (protocol-compiler/def:compile)) emits from a specification, and let $\mathbb{F}^{\mathrm{ref}}$ be a non-empty set of its measured factors (the one measured system in the basic format) such that, for each $f \in \mathbb{F}^{\mathrm{ref}}$, the last link of the receiver segment of f is the emission of a reduction instance $\mathcal{J}_f$ satisfying (H-REF) of Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion). Let $\mathfrak{D}^{\mathrm{ref}}$ be the tuple the compilation map gives from the same specification with, for every $f \in \mathbb{F}^{\mathrm{ref}}$, that link's target family replaced by the target $\{F_v[A_f^{\mathrm{ref}}]\}$ of the reference companion of $\mathcal{J}_f$ of that corollary, A_f^{ref} its datum, and that link's entries replaced by the relaxed entries (Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation)) of the companion's entries, each rounded operator Hermitian and with the sector blocks of the entry it rounds. The companion's entries of such a link are the instance's own entries at the companion's datum and the companion forms (Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (vi)) of those entries requested of the link whose companion forms carry their certificates at that datum; a requested entry whose companion form is not established to carry one has no entry in $\mathfrak{D}^{\mathrm{ref}}$. Then:

(i) *(what is shared)* $\mathfrak{D}^{\mathrm{ref}}$ has the systems, sectors and dimensions, the alphabets, Alice's marginal slot and measurement, the party partition, the announcement and sifting maps, the raw-key map and the index map of $\mathfrak{D}$, and in the general format its factor list and Alice's block lists; it differs from $\mathfrak{D}$ at most in the measured families of the factors of $\mathbb{F}^{\mathrm{ref}}$ and in the entries of their last links. In particular the two have the same joint sectors (Definition 4.8) with the same block dimensions and the same coupling condition;

(ii) *(admissible)* $\mathfrak{D}^{\mathrm{ref}}$ satisfies conditions (i)–(vi) and (viii) of Definition 1.2;

(iii) *(the relaxed entries enlarge the feasible set)* let $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$ be formed as $\mathfrak{D}^{\mathrm{ref}}$ but with the companions' entries themselves in place of their relaxed entries; then the feasible set of $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$ (Definition 5.1) is contained in that of $\mathfrak{D}^{\mathrm{ref}}$, and for every acceptance set $\underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}] \le \underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}]$.

Corollary 12.2 (Factor enclosures of the compiled reference companion)**.** Let $\mathfrak{D}$, $\mathbb{F}^{\mathrm{ref}}$, the instances $\mathcal{J}_f$ and $\mathfrak{D}^{\mathrm{ref}}$ be as in Proposition 12.1, and for $f \in \mathbb{F}^{\mathrm{ref}}$ let G_f be the family of Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) from which the datum A^{ref}_f of the reference companion of $\mathcal{J}_f$ is formed. Let $f \in \mathbb{F}^{\mathrm{ref}}$, and read the block on a data sector s of $\mathcal{J}_f$'s target of an element of the measured family of factor f of $\mathfrak{D}^{\mathrm{ref}}$ formed from the target element $F_v[A^{\mathrm{ref}}_f]$ with the triple that part (i) of Corollary "Factor enclosures carried to the reference companion" (reductions/cor:companion-factor-enclosure) forms for the element $A^{\mathrm{ref},(s)}_{f,v}$ of A^{ref}_f on s, where the rule of (I1) of $\mathcal{J}_f$ is a non-negative combination (that corollary) and that part forms one from factor enclosures (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) that blocks of G_f carry, its core multiplied by $P_f(\varphi \,|\, t(v))$ where the family is weighted after the chain by factor f's weighting P_f and the element is $P_f(\varphi \,|\, t(v))\, F_v[A^{\mathrm{ref}}_f]$ with $(v, \varphi) \in \mathcal{V}_{P_f}$ (Definition "Choice-weighted families" (protocol-compiler/def:weighted), with $P := P_f$); and read every other block of the measured families of $\mathfrak{D}^{\mathrm{ref}}$ with none. Then every block so read with a triple is a member of it, the triple being a factor enclosure of that block.

13 Assumptions

Assumption 13.1 [**A-KEYREG**] (Dedicated key register and one-way reconciliation)**.** (i) Error correction, error verification and privacy amplification are applied to Z_1^n, the string of raw-key values $r(x, v)$ that the rounds write to the registers $Z_1, \ldots, Z_n$ (Definition 10.1); (ii) the declared key holder h (Definition 1.1) computes Z_1^n from its own outcome record and the public announcements; (iii) reconciliation is one-way from h: its transcript L_{EC} is a

function of Z_1^n and public seeds only; (iv) L_{EC} occupies a fixed-length register of λ_{EC} bits, shorter transcripts being padded, so that $\dim(L_{\mathrm{EC}}) = 2^{\lambda_{\mathrm{EC}}}$.

Assumption 13.2 [**A-HASH**] (Ideal two-universal privacy amplification)**.** (i) The privacy-amplification family of Definition 10.1, applied to Z_1^n, is ideal two-universal: for any two distinct inputs, the probability under the family's law that their ℓ-bit values coincide is exactly $2^{-\ell}$; (ii) the error-verification family of Definition 10.1 is two-universal: for any two distinct inputs, the probability under the family's law that their $\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$-bit values coincide is at most $2^{-\lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil}$.

Assumption 13.3 [**A-SEED**] (Private, independent and fresh local randomness)**.** (i) The error-correction seed S_{pub}, the error-verification seed S_{EV} and the privacy-amplification hash G of Definition 10.1 are drawn with the laws that definition gives them, S_{EV} and G with the laws of their hash families; (ii) each is drawn privately: its value is available to the adversary only once the protocol announces it; (iii) each is independent of every other register of the protocol and of the adversary's information at the time it is drawn; (iv) each is drawn afresh in every run; (v) in trusted-source mode, the letters Alice's source draws in the n rounds, and the systems it prepares from them, are jointly independent of the adversary's registers at the start of the protocol: the joint state of the letters, the prepared systems and those registers is the product of its marginal on the letters and the prepared systems together and its marginal on those registers; in particular the adversary holds no purification of a prepared system, and no purification of a prepared system is made available to the adversary during or after the protocol.

Assumption 13.4 [**A-FIXED**] (Protocol parameters fixed before the run)**.** (i) The acceptance set $\mathcal{S}_{\mathrm{acc}}$, the round count n, the output length ℓ, the transcript length λ_{EC}, the error budgets $\varepsilon_{\mathrm{PA}}$ and $\varepsilon_{\mathrm{EV}}$ of Definition 10.1, and the authentication parameters $\varepsilon_{\mathrm{auth}}$ and k_{auth}, the declared authentication error bound and key consumption (Definition 10.2 (b)), are fixed before the first round of the run; (ii) the run applies them unchanged: none of them is chosen, adjusted or recomputed from data observed in the run; (iii) the Rényi order $\alpha \in (1, 2)$ at which ℓ is computed is fixed before the first round of the run, from data available before it, and is not chosen, adjusted or recomputed from data observed in the run.

Assumption 13.5 [**A-CONFORM**] (The implementation runs the analysed post-processing)**.** (i) The implementation evaluates the acceptance test of Definition 10.1 (i) and aborts unless Ω_{AT} holds; (ii) after the rounds, the only classical messages either party sends are those of Definition 10.1 and of the authentication post-processing (Definition 10.2 (c)); (iii) both parties

abort when the verification flag is rej; (iv) each party outputs the ℓ-bit key that Definition 10.1 (iv) assigns to it, or $\perp$; (v) neither party uses any part of its key before the end of an execution (Definition 10.3); (vi) a party that receives an authentication failure acts as the authentication post-processing prescribes.

Assumption 13.6 [**A-AUTH**] (Authentication of the classical channel by a keyed scheme)**.** (i) All classical messages of the protocol, the round announcements included, are sent through one authentication scheme using a pre-shared key and an unauthenticated classical channel, or all over a classical channel that is itself the authenticated classical channel of Definition 10.2 (a), in which case $\varepsilon_{\mathrm{auth}} = k_{\mathrm{auth}} = 0$; (ii) in the first case, the scheme's key consumption is at most k_{auth} bits; (iii) run on two identical copies of a uniformly distributed key independent of every other register of the protocol and of the adversary's information, the scheme ε_0-realises the authenticated classical channel (Definition 10.2 (b)), for some $\varepsilon_0 \geq 0$; (iv) before the run starts, the joint state of the two parties' copies of the pre-shared key the run consumes with every other register of the protocol and with the adversary's information is within trace distance $\delta \geq 0$ of the state in which those copies are replaced by two identical copies of a uniformly distributed key independent of them; (v) $\varepsilon_0 + \delta \leq \varepsilon_{\mathrm{auth}}$.

Assumption 13.7 [**A-SEPARATE**] (Executions that share no key and no device state)**.** For executions $1, \ldots, N$ of the protocol, run against one adversary: (i) N and, for every execution, the parameters fixed before its run (its acceptance set, round count, output length, transcript length, error budgets and authentication parameters) are fixed before the first of the N executions starts; (ii) no operation of the honest parties of one execution acts on a register that the honest parties of another execution use, hold or output: no execution uses any part of another execution's key, local randomness, seeds or pre-shared key, no device, memory or random source carries a state from one execution into another, and the executions interact only through systems and classical channels the adversary has access to; (iii) from the end of an execution (Definition 10.3) until the end of the last of the N executions, no part of that execution's key is used, read or otherwise acted on, and no information about any other register of its parties leaves their laboratories; (iv) for every execution, every assumption on which its key and its security parameter rest holds when every register of the honest parties of every other execution is counted as part of the adversary's information, and every operation of those parties as part of the adversary's actions.

Assumption 13.8 [**A-LAB**] (Each party's laboratory is sealed)**.** Let $\mathsf{P} \in \{\mathsf{A}, \mathsf{B}\}$ be a party of the protocol declaration (Definition 1.1). Its *records*

are its letters, where the party partition assigns Alice's system A to P; the private settings and the outcomes of the measurements made in its laboratory, the public choices it draws before they are announced, and its readout π_{P}; the inputs of its announcements; its raw or final key; and its local randomness. The *systems handed out of its laboratory* are, in trusted-source mode and where the party partition assigns A to P, the systems its source emits into the quantum channel, which the round model (Definition 2.3) gives to the adversary, A being no output of the attack in that mode (2.5) and staying with P; in every other case there are none. In particular, in untrusted-source mode, where the attack supplies A and Q (2.5), no laboratory hands out any system, and a source the declaration treats as untrusted lies outside every laboratory wherever it physically stands. The *systems delivered into its laboratory* are, in untrusted-source mode and where the party partition assigns A to P, the system A; and, in either mode and in each sector s, the tensor factors $Q_{s,i}$ of the declared factorisation of $\mathcal{H}(Q_s)$ that the party partition labels P (in the general format, every measured factor labelled P), physically the input modes those factors model.

(i) During the protocol, no information about P's records leaves its laboratory, other than its key as the protocol's output, except through the systems handed out of its laboratory and through the public messages of the protocol (the announcements of the round channels of Definition 2.3, the messages of Definition 10.1 and of the authentication post-processing (Definition 10.2 (c)), and, where an authentication scheme carries them, the scheme's own transmissions over the unauthenticated classical channel): no light emitted by its devices other than those systems, no electromagnetic, acoustic, power or timing emanation and no side channel of its classical processing carries such information, and neither the timing of its messages nor any auxiliary or synchronisation signal does;

(ii) the adversary acts on P's laboratory only through the systems delivered into its laboratory, as the declared measurements receive them, and through the public messages the protocol delivers to P and, where a scheme carries them, the transmissions it receives over the unauthenticated channel: no light or signal sent into the laboratory, through the systems handed out of it or otherwise, reads out any of its records, and nothing else influences the state, configuration or calibration of its devices (its source included, where it has one), its measurements or its classical processing.

Assumption 13.9 [**A-EVE-DIM**] (The adversary's registers are finite-dimensional)**.** (i) Every register of the adversary in the round model (Definition "Round model, attack sequences and attack sets" (foundations/def:round-

model)), her initial register E_0, her outgoing memories E'_j and her registers E_j for $j = 1, \ldots, n$, is finite-dimensional, of arbitrary finite dimension.

Engine requirements

14 Requirements: admission and the declared structure

Requirement 14.1 (Exact admission of declared data). The engine takes no tolerance on declared data and decides no condition on it more loosely than exactly (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)): it admits a declaration only if the declaration is exactly admissible in the sense of Definition 1.1, and refuses it otherwise. No perturbation of a declared family is licensed to carry the key length of the family it approximates. Proposition 11.1 is the reason: it shows, for the measurement on Q, that no tolerance could license a key-length claim. A compiled reference companion is not admitted under this requirement but under Requirement 14.3, which decides exactly what it shares with the declaration and decides none of its family; the binary64 numbers it is evaluated at are no declared family whose key length is claimed, since the bound evaluated there is that of the reference, at the unbudgeted arithmetic of Remark 22.4.

Requirement 14.2 (Admission of the declaration). The engine decides conditions (i)–(vi) and (viii) of Definition 1.2, in the general format as that definition reads them there, exactly on the declared data, and computes nothing from a declaration that fails one of them, which it refuses. It does not decide condition (vii) on a declaration whose constraint list is non-empty. A compiled reference companion is admitted instead under Requirement 14.3, which decides exactly the components it shares with the declaration it accompanies and none of its measured family.

Requirement 14.3 (Admission of the reference companion beside its declaration). Where the protocol compiler emits a compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ beside a declaration $\mathfrak{D}$ (Requirement "The reference companion compiled beside the declaration" (protocol-compiler/req:reference-companion-compiled)), the engine admits it only beside $\mathfrak{D}$, after admitting $\mathfrak{D}$ (Requirements 14.1 and 14.2), and only as follows.

(i) *Decided equal.* It decides exactly, on the declared data, that every component Proposition 12.1 (i) names as shared is equal to $\mathfrak{D}$'s, entry for entry and label for label: the shapes, the sector maps and the joint sectors, the alphabets and their orders, Alice's slots, the party partition,

the announcement, sifting and raw-key maps; that the companion's entries are those of $\mathfrak{D}$ save the entries of the links the companion replaced, one per measured factor whose family it replaces, each of which carries the relaxed mark of Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion) (iii) and a Hermitian binary64 operator with the blocks of its scope; and it reads the acceptance set of $\mathfrak{D}$, the acceptance set being one per run. A companion that fails a decision of this item is not admitted, the reason naming the component, and the run proceeds without it. The statistics map, the single-round protocol map, the branch maps, the objective and its gradient of the companion are generated from the companion's own slots, its measured family among them (Definition 1.4), as for any declaration; no map generated from $\mathfrak{D}$'s measured family is used for the companion. The announcement, sifting and raw-key maps decided equal above are declared data, not generated maps.

(ii) *Imported, never decided.* It decides none of the conditions of Definition 1.2 on the companion's measured family. That the family it denotes is a POVM, block diagonal for its sector map, is Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (i), which rests on the claims that establish the handed reference (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) on the blocks where one is handed, on the exact decisions of the declarer's snap on the blocks where the companion carries the snapped block, and on (H-REF) for each instance whose link it replaces; and with (i) above it gives the companion conditions (i)–(vi) and (viii) by Proposition 12.1 (ii). Condition (vii) is not decided on it. Nor does it decide that a block of the companion's measured family is a member of a factor enclosure the companion carries with it, which is Corollary 12.2, on the claims that establish the factor enclosures of the handed reference.

(iii) *Evaluated and measured.* It reads the companion's family at the binary64 numbers nearest to the centres of its enclosure, rounded once, entrywise to nearest, when the family is docked, and records, per block of the measured family, the completeness defect and the least eigenvalue of those numbers, measured and not decided, and the largest radius of the enclosure. A defect so measured refuses nothing and marks nothing inadmissible; what these numbers mean for the bound evaluated on them is Remark 22.4.

(iv) *Nothing else.* It computes from the companion only what Requirements 24.7 and 28.1 ask. The one program solved over it is the polish of Requirement 24.7 (i), a search continued from a point of the search on $\mathfrak{D}$, whose points serve only as points at which those requirements

form brackets; no estimate is reported from it, and no key length is reported from it save through those requirements.

Requirements 14.1 and 14.2 bind every declaration but a companion admitted under this requirement.

Requirement 14.4 (Admission of the acceptance set). The engine admits an acceptance set only if it decides, exactly and on the declared data, that the set is an admissible acceptance set (Definition 2.5): the set carries a geometry tag every well-formed datum of which denotes such a set, and its data are well formed for that tag. It refuses every other declared acceptance set (`acceptance_set_ill_formed` for data ill formed for the geometry they claim).

Requirement 14.6 (The sector report). The engine emits, for every sector, the four predicates of Definition 6.1 in `sector_report`, and justifies there every sector it deletes by the six conditions (W1)–(W6) checked on the exhibited witness, by that witness with its evaluation order of (W6), declared with the witness and recorded with it as `witness_evaluation_order`, and by the retention of the witness's targets.

Requirement 14.7 (Bit-level test of state-blindness). The engine decides whether a sector s is state-blind (Definition 6.1 (c)) by a bit-level identity of declared data. It reads $c_{v,s}$ off the declared block as its first diagonal entry, $c_{v,s} \coloneqq (F_v^{(s)})_{11}$, and requires $F_v^{(s)} - c_{v,s}\mathbb{1}_{Q_s} = 0$ entrywise and exactly: every off-diagonal entry of the block is the datum 0, and every diagonal entry is the same datum. Likewise, whenever $\mathfrak{L} \neq \varnothing$, it requires $\hat{O}_k^{(s)} - o_{k,s}\mathbb{1}_{Q_s} = 0$ with $o_{k,s} \coloneqq (\hat{O}_k^{(s)})_{11}$. No subtraction, no norm and no window enters: two binary64 data are equal or they are not, equality being equality of the values they denote and not of their bit patterns (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)). The engine decides the sign condition $c_{v,s} \geq 0$ by comparing that same datum with zero as a real number. A block that is only *nearly* scalar fails, and s is then not state-blind.

Requirement 14.8 (Setting the removable flag). The engine sets the flag `removable` for a sector s^* of a declaration with one measured factor when, and only when, a candidate witness exhibited by the implementation is a witness for s^* in the sense of Definition 6.1 (d): the list $\mathfrak{L}$ carries no entry declared party-local and the candidate meets all six of (W1)–(W6); it sets the flag on no other ground. The witness may come from any search, and it is never read as declared input: a declaration supplies no witness. Each condition is decided exactly, at no window: (W1) by the bit-level identity of

Requirement 14.7, (W2) as a comparison of sector indices, (W5) as the finite combinatorial predicate of key-inertness, (W3) and (W4) on the exhibited data as those conditions state, and (W6) on the residual formed in the exhibited evaluation order; a witness whose evaluation rounds anywhere, or whose residual is not the datum 0, sets no flag. Every other outcome — a nonzero residual, a rounding operation, an absent witness, a witness that names s^*, a target that is not key-inert, a block that is not exactly scalar — retains the sector. The flag is set for no sector of a declaration whose list carries an entry declared party-local, nor of a declaration with more than one measured factor. No tolerance licenses a removal: `sector_criterion` is decided at exactly zero and not at the declared tolerance τ_{num}.

Requirement 14.9 (Operations on the declared sectors). The engine treats the declared sector list as input it never edits; sectors leave the variable list only by the deletion of (ii). The operations it may perform on the declared sectors are exactly: (i) evaluate the objective over the key-active sectors only (Definition 4.7; an identity, by Proposition 4.6); (ii) delete sectors carrying the flag `removable` of Definition 6.1 (d), set as Requirement 14.8 states — and only those, and among those only a set meeting the hypothesis of Proposition 6.5, every deleted sector's witness naming targets outside the deleted set (an identity, by Theorem 6.2 and that proposition); (iii) nothing else. In particular no simplification step merges sectors, and the only structure-changing simplification is the deletion of (ii).

15 Requirements: the Choi program and gradients

Requirement 15.1 (Construction and exact check of the support frame). In trusted-source mode the engine constructs the declared isometry ι of every support frame it uses and never takes ι as declared input: by the full frame, which is always available, by the diagonal frame, or by another construction for which it exhibits (7.2) exactly. It decides both identities of (7.2) on the isometry it constructed, entrywise as binary64 data at exactly zero (`choi_isometry`), and refuses a frame that fails either of them with `choi_isometry_inexact`.

Requirement 15.2 (Exact diagonality test for the diagonal frame). The engine uses the diagonal frame of Definition 7.2 (b) only when every off-diagonal entry of the declared τ_A is the datum 0. It decides this by comparing each off-diagonal entry with 0 as a binary64 value, at no window, as Requirement 14.7 compares the entries of a sector block, and the comparison is a clause of `alice_marginal_exact`. A τ_A that is only *nearly* diagonal

is not diagonal: the diagonal frame is unavailable for it, and the full frame always is.

Requirement 15.3 (Record of the support frame). In trusted-source mode every run records, beside any Choi block it reports, the support frame it used: the construction that produced it, the dimension d_{A_0} and the declared isometry ι, per Alice sector in the general format, as `choi_input_basis` in `optimizer_certificate`. In untrusted-source mode no A_0 and no ι exist (Definition 7.10 (a)), and the field is absent.

Requirement 15.5 (Frame-free rows only where the construction exhibits tightness). In trusted-source mode the engine imposes a homogeneous row of (7.9) in its frame-free form (7.11), and in the general format the row of an Alice sector t of a homogeneous or a party-local entry in its unconjugated form, only where the construction that produced the frame (of sector t) exhibits $\ker W_\tau = 0$ ($\ker W_t = 0$) exactly, with no rank computed and no window declared. Of the two constructions of Definition 7.2 that is the diagonal frame; on a frame produced by the full frame construction the conjugated form is imposed, whatever τ_A is, and so it is on every frame whose construction does not exhibit $\ker W_\tau = 0$. The engine licenses the frame-free form by no numerical test on τ_A.

Requirement 15.6 (Record of the form each constraint row was imposed in). In trusted-source mode a run records, for every declared constraint row, which of the two forms of Proposition 7.6 it imposed, per Alice sector in the general format, as `constraint_row_form` in `optimizer_certificate`, beside the frame it records under Requirement 15.3.

Requirement 15.7 (Split-form evaluation of the trace-power gradient). The engine evaluates the gradient of the pinched trace power at a branch operator X in the split form of Theorem 8.4 (i): the divided-difference map $DX^p[\cdot]$ compressed to $\operatorname{supp} X$, its kernel summed over the strictly positive eigenvalues of X only, and $\mathcal{Z}(X^p)$, $Y^{\alpha-1}$ and $\mathcal{Z}(Y^{\alpha-1})$ formed on the ambient branch space. It compresses the whole computation to a subspace S' only where S' contains the maximal branch support $W_{\hat{c},s}$ of Definition 8.1 and $Z_j S' \subseteq S'$ for every j.

Requirement 15.8 (Gradient evaluation at maximal support). The engine evaluates a gradient of the pinched trace power only at an iterate whose support is (kept) maximal, maximality being that of (8.1) for the branch operators, for instance after an explicit perturbation.

Requirement 15.9 (The order-one limit as a validation reference only). The engine never reports the limit h_1 of (8.20), however computed, as a bound of any kind; in particular it never reports it as the first-order bound $\underline{h}_\alpha^{\mathrm{FO}}$ of Theorem 19.1.

Requirement 15.10 (Recording the dropped set). A run records its dropped set $\mathcal{D}$ (Definition 9.2) with the run, and reports the number it computes from (9.4) as $\underline{h}_\alpha^{\mathcal{D}}$ together with $\mathcal{D}$, never, when $\mathcal{D} \neq \varnothing$, under the name $\underline{h}_\alpha$.

16 Requirements: values and key lengths

Requirement 16.1 (Propagation of an infinite value). The engine propagates $\underline{h}_\alpha = +\infty$, the infeasibility of the declared acceptance set, rather than clamping it to any finite range; its output `h_low` has the extended-real domain $[0, +\infty]$. If $\underline{h}_\alpha = +\infty$ reaches Theorem 10.7, the engine emits no key-length claim and no numeric key length, and reports the mismatch between the declaration and the acceptance set (`infeasible_acceptance_set`).

Requirement 16.2 (The check of the acceptance set against the feasible statistics). A run is *checked* when its acceptance set is built from an honest law or an honest envelope that the caller gives as numbers, such as one measured on a device or read from a table, rather than one the engine computes from the declared devices, or when its declaration carries a constraint entry that the caller declares rather than one the compiler or a reduction derives, unless the run sets the option `statistics_check` to `off` (default `on`). On every run, checked or not, the engine records under `statistics_consistency` whether the run is checked and, where it is not, why (out of scope, or opted out), and the report says so beside every key length; a run that is not checked names the premise Assumption "An honest law produced by a feasible state" (statistics/ass:reach) as unchecked, the caller's to vouch for. On every checked run the engine does the following once, before it evaluates the run at any Rényi order and before it reports a key length.

(i) *The order and the solve.* It takes α_0 to be fixed before any order is evaluated: the run's fixed order, or the lowest order its order search evaluates, where the search fixes that order before it evaluates any and evaluates none below it, and otherwise the lower end of the domain the search declares, which every order the search evaluates is at least. The engine evaluates the program only at an order in $(1, \infty)$ (Requirement 16.3), so the solve runs only where $\alpha_0 > 1$, and at any other α_0 it is refused, as (ii) records; a key length is reported only at an order in $(1, 2)$ (Requirement 16.6), so a fixed order at which one

is reported lies there, and that the orders a search evaluates and the lower end of its declared domain lie in $(1, 2)$ is the bound the run tier places on the domain it accepts, which no claim of this contract states. On the declaration the run searches and with the run's acceptance set, it runs the first-order search of the dropped program for the dropped set $\mathcal{D}_{\text{all}}$ of every pair (Proposition 9.5) at α_0, with the method, the iteration cap and the gap target the run applies to one evaluation of its own, and with the share `statistics_check_budget_share` of the wall budget of one evaluation of the run, where it has one, as its wall budget. A smaller share can leave a mismatch unproven: b smaller, or not positive, so that (iv) may leave unrefused a run that a longer solve would have refused, and the cap of (iv) looser; no bound is made unsound by it.

(ii) *The bound.* From that solve it forms the first-order number of the dropped program and, unless the run sets `certified_evaluation` to `off`, the certified evaluation of that number, on the declaration the run searches and at that solve's points, by parts (ii)–(v) of Requirement 28.1, whatever declaration the run's key length is formed on. The number b is the certified number where a point is certified and the first-order number otherwise, with its tier. The solve is not refused on the sign of its estimate O_0 of (v), which is no estimate of the run and which Requirement 24.4 does not read: b is formed from the solve's points whatever that sign, and a negative O_0 is recorded as a diagnostic and refuses nothing. Where no bound is formed (the first-order number withdrawn, the solve not completed, or the solve refused at an order outside $(1, \infty)$), the run proceeds with the reason recorded.

(iii) *The proven part.* Where $b > 0$, for every order α' the run evaluates, the proven floor $f(\alpha') \coloneqq \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha_0-1}{\alpha_0}\, b$ satisfies $f(\alpha') \le \frac{\alpha'}{\alpha'-1} D_{\min} \le \underline{h}_{\alpha'}$ (Proposition 9.5 (ii), (iv)), with $D_{\min}$ and $\underline{h}_{\alpha'}$ those of the declaration the run searches: a part of that declaration's value at α' that the mismatch alone accounts for. Where $b \le 0$, $f(\alpha') \coloneqq 0$. Where no bound was formed, $f(\alpha')$ is not defined.

(iv) *The cap and the refusal.* Let $\tau \in (0, 1]$ be the run's option `min_model_pass_probability`, an exact rational number, default $\frac{1}{2}$. Where b was formed and $b > 0$, the engine forms the exponent $x \coloneqq n\, \frac{\alpha_0-1}{\alpha_0}\, b$ exactly from the run's round count n, α_0 and b, with the tier of b. By Proposition 10.12 (ii), devices the declaration the run searches admits, run so that each round's acceptance value, given the earlier ones, follows the statistics $\mathbf{\Phi}[\omega]$ of some ω in that declaration's $\mathcal{W}$, pass the acceptance test with probability at most 2^{-x}, the *cap*. Where $2^{-x} \le \tau$, that is $x \ge \log_2 \frac{1}{\tau}$, the run is refused as a whole with `acceptance_set_unreachable`: it is evaluated at no order and reports

no key length. The engine decides $x \geq \log_2 \frac{1}{\tau}$ exactly where $\tau = 2^{-k}$ for an integer $k \geq 0$, by comparing x with k; for any other τ it forms a rigorous enclosure $[\lambda_-, \lambda_+] \ni \log_2 \frac{1}{\tau}$ and refuses where $x \geq \lambda_+$, refuses nothing where $x < \lambda_-$, and refuses where $\lambda_- \leq x < \lambda_+$, the comparison being undecided there. The decision carries the tier of b: where b is a first-order number not certified, the refusal or its absence is of the first-order tier. Where $b \leq 0$ or no bound was formed there is no cap, and this item refuses nothing. With $\tau = 1$ every run with $b > 0$ is refused. Where a reference companion carries the key length, the cap is still that of the declaration the run searches, which no claim relates to the companion's feasible set; the refusal only withholds a key.

(v) *The record.* On a checked run the engine records under `statistics_consistency`: the declaration the solve ran on; α_0 and which rule of (i) fixed it; b with its tier and the lower bound $\frac{\alpha_0 - 1}{\alpha_0}\, b$ on $D_{\min}$, or the reason none was formed; τ; where $b > 0$, the exponent x of (iv) with the tier of b, rounded downward where the record cannot carry it exactly, so that 2^{-x} at the recorded x stays an upper bound on the pass probability, and whether the comparison of (iv) was decided exactly, by the enclosure, or left undecided and refused; where $b \leq 0$ or no bound was formed, that there is no cap; whether (iv) refused the run, as (iv) decided it on the exact x and never as a comparison of the recorded or displayed values with τ; where the solve ran, the share of the wall budget and the wall budget it was given, the wall time it took and the cause it stopped on, as the solver reports it (such as the gap target, the iteration cap or the budget); where the solve returned a point, the estimate $O_0 \coloneqq O_{\alpha_0}^{\mathcal{D}_{\text{all}}}$ at that point, whether it is negative, and whether its membership in the relaxed set was established; where b was formed, the gap $O_0 - b$; and, where a bound was formed, $n\, f(\alpha')$ in bits at each order α' the run evaluates, a diagnostic of the part of the value the mismatch accounts for, which satisfies $\frac{\alpha' - 1}{\alpha'}\, n\, f(\alpha') = n\, \frac{\alpha_0 - 1}{\alpha_0}\, b$ (Proposition 10.12 (ii)). The report shows, beside every key length it reports and, on a run refused under (iv), in place of one, the cap as a probability: 2^{-x} at the recorded x together with a decimal approximation of it rounded upward, with the tier of b, as the probability with which devices the declaration the run searches admits, run as (iv) says, pass the acceptance test, both labelled as rounded upward; and whether (iv) refused the run, with τ, as the record carries that decision from the exact comparison, so that a displayed cap a little above τ beside a refusal reads as the upward rounding it is. Where there is no cap it says so and why ($b \leq 0$, or no bound formed and the reason). It states the cap as a probability and not as a number of bits, and it does not show $n\, f(\alpha')$. No key length and no other number of the run reads these entries.

Requirement 16.4 (Reported key length). The engine reports as `key_length` the maximal key length built from the lower bound $h \le \underline{h}_\alpha$ it uses, that is $\ell_{\max}$ of Definition 10.13 with $\max\{0, h\}$ in place of $\underline{h}_\alpha$ (Corollary 10.18 (a)), with its floor and its clamp at 0: that is its conservative integer-output rule.

Requirement 16.5 (Key-length penalty for a merely two-universal family). For a privacy-amplification family that is only two-universal, collision probability at most $2^{-\ell}$, (10.4) must be tightened by one further bit, $\ell \to \ell - 1$.

Requirement 16.6 (Rényi order of a reported key length). The engine reports a key length only at a Rényi order $\alpha \in (1, 2)$; at any other order it reports no key length (`alpha_in_the_profile_domain`).

Requirement 16.7 (Key lengths only on declarations the compiler emitted). The engine reports a key length, whether as a key-length claim of Theorem 10.7, Corollary 10.18 or Corollary 19.5 or as a numeric key length, only on a declaration that the protocol compiler emitted; on any other declaration it reports none.

Requirement 16.8 (Refusal of budgets with an empty security statement). The engine refuses a run whose budgets satisfy $\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}} \ge 1$, deciding the sum exactly on the three declared numbers before any program is built, and reports `security_statement_vacuous`: the $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}})$-security statement of Corollary 10.8 (iii), at $\varepsilon_{\mathrm{auth}} = 0$ equal to $\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}}$, the parameter of (ii) (and numerically of (i)), is then empty, and Corollary 10.18 (b) does not claim the corner.

Requirement 16.9 (Declared choice of the source's secrecy convention). An instance that adds the one bit of (10.4) back adopts the secrecy definition of the $\wedge\Omega_{\mathrm{AT}}$ state, the convention of the derivation Remark 10.14 compares, and declares explicitly that it does so. A key length so obtained is not one of Theorem 10.7 and carries no claim of this module.

Requirement 16.10 (Two-part gate on a key length with declared constraints). A key-length output may be combined with $\mathfrak{L} \neq \varnothing$ *only if* (i) the list lies in the licensed class of the declared source mode (Definition 2.2), a condition that is mechanically checkable and degenerate in the cases of Remark 10.20, and (ii) the declaration carries the attack-set licence `attack_set_license` (Assumption A-ATK, Assumption "Attack-set licence" (protocol-compiler/ass:atk)), an attestation no program can verify. The engine refuses any request that couples $\mathfrak{L} \neq \varnothing$ with a key-length output unless *both* parts are present; without them the numerical machinery may be used on $\mathfrak{L}$ for exploration, but its output does not reach (10.4).

Requirement 16.11 (A degenerate licensed-class check reported as no evidence). Where part (i) of the two-part gate of Requirement 16.10 is degenerate, at $d_A = 1$ and in untrusted-source mode (Remark 10.20), the engine never reports a passing part (i) as evidence of anything, and it never reports a passing part (i) as evidence of anything about a party-local entry either.

Requirement 16.12 (Source licence for a declared marginal). The refusal discipline of the two-part gate of Requirement 16.10 governs the marginal as well: the engine refuses a declared τ_A without `source_replacement_licen se` (Assumption A-SRC, Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) before any key-length output, and at $d_A = 1$, where the licence is vacuous, it does not demand it (Proposition 5.2 (iv)).

Frank-Wolfe first-order bound

17 The relaxed feasible set

Definition 17.1 (The relaxed feasible set)**.** For $\varepsilon \geq 0$ the *relaxed feasible set* $\mathcal{W}(\varepsilon)$ is the set of sector-block-diagonal $\omega \in \mathrm{Pos}(AQ)$ satisfying, in trusted-source mode,

$$\big\|\omega_A - \tau_A\big\|_{\max} \leq \varepsilon, \quad \big|\operatorname{tr}\omega - 1\big| \leq d_A\varepsilon, \quad \begin{cases} \lambda_{\min}\big(\Theta_k - A_k(\omega)\big) \geq -\varepsilon, & \bowtie_k = \preceq, \\ \lambda_{\min}\big(A_k(\omega) - \Theta_k\big) \geq -\varepsilon, & \bowtie_k = \succeq, \\ \big\|A_k(\omega) - \Theta_k\big\|_{\max} \leq \varepsilon, & \bowtie_k = =, \end{cases} \tag{17.1}$$

where $A_k(\omega) := \operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega]$ is the left-hand side of the k-th declared relation of (5.2), and, in untrusted-source mode, the normalisation $\big|\sum_s \operatorname{tr}\omega_s - 1\big| \leq \varepsilon$ together with the scalar forms $a_k(\omega) \leq c_k + \varepsilon$, $a_k(\omega) \geq c_k - \varepsilon$, $|a_k(\omega) - c_k| \leq \varepsilon$, with $a_k(\omega) := \operatorname{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big]$ and no marginal condition. In the Choi form $\mathcal{W}^J(\varepsilon)$ is the set of $(J_s)_s$ with $J_s \in \mathrm{Pos}(A_0Q_s)$, $\big|\operatorname{tr}[\chi_Q(J)] - 1\big| \leq d_{A_0}\varepsilon$, $\big\|\sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0}\big\|_{\max} \leq \varepsilon$ and the same relaxation of the $\bowtie_k$ rows.

The normalisation clause of each form is implied by the clause written before it, by the marginal in trusted-source mode and by the coupling residual in the Choi form, and is written out only so that the set can be read without re-deriving it; Proposition 17.2 (c′) proves both implications and derives the two dimension factors. In untrusted-source mode there is no clause before it: there the normalisation *is* the relaxed coupling condition (Definition 7.10 (b)), which is why it carries no dimension factor.

Which predicate is which is part of the definition and is never converted: an inequality between operators is relaxed by a *spectral floor* $\lambda_{\min}(\cdot) \geq -\varepsilon$,

an operator equality by an *entrywise* ball $\|\cdot\|_{\max} \le \varepsilon$, a scalar equality by an interval, and these are different quantities against the same number (Lemma "The positivity floor against entrywise closeness to the cone" (foundations/lem:floor-stricter): the entrywise reading and the spectral floor). The factors d_A and d_{A_0} are not such a conversion: each is the dimension factor of an explicit trace bound derived in Proposition 17.2 (c′), written into the clause, and neither reads an entrywise defect as a spectral one.

At $\varepsilon = 0$ both sets reduce to (5.1) and (7.9) respectively: each normalisation clause is then the exact $\operatorname{tr}\omega = 1$, respectively $\operatorname{tr}[\chi_Q(J)] = 1$, which the coupling condition already forces, so re-imposing it changes nothing. We write $\mathcal{W}$ for $\mathcal{W}(0)$ and $\mathcal{W}^J$ for $\mathcal{W}^J(0)$ throughout.

In the general format of Definition 1.1 both sets are read on lists over the joint sectors, with every line of (5.5), respectively of the general-format Choi set of Theorem 7.5, relaxed by the predicate its relation calls for, as in (17.1), the party-local rows included. The marginal condition and the coupling condition are then read per Alice sector: $\|\sum_s \operatorname{tr}_{Q_s} \omega^{(\mathsf{t},s)} - \tau_A^{(\mathsf{t})}\|_{\max} \le \varepsilon$, respectively $\|R_{\mathsf{t}}\|_{\max} \le \varepsilon$ with $R_{\mathsf{t}} := \sum_s \operatorname{tr}_{Q_s} J_{(\mathsf{t},s)} - \mathbb{1}_{A_{0,\mathsf{t}}}$, for every $\mathsf{t} \in \mathbb{T}$. The residual of the single marginal clause of (17.1), respectively of the single coupling clause of the Choi form, is then the block-diagonal operator $\omega_A - \tau_A = \bigoplus_{\mathsf{t}}(\sum_s \operatorname{tr}_{Q_s} \omega^{(\mathsf{t},s)} - \tau_A^{(\mathsf{t})})$, respectively $\bigoplus_{\mathsf{t}} R_{\mathsf{t}}$ with $J_s := \bigoplus_{\mathsf{t}} J_{(\mathsf{t},s)}$, whose entrywise maximum is the largest over t of the blocks' ones; so the clauses per Alice sector together are that single clause. The normalisation clauses are those of the first paragraph of this definition. At $\varepsilon = 0$ the two sets are $\mathcal{W}$ of (5.5) and the general-format $\mathcal{W}^J$ of Theorem 7.5.

Proposition 17.2 (Properties of the relaxed feasible set)**.** Let $\mathfrak{D}$ satisfy Definition 1.2, let $\varepsilon \ge 0$, and let $\mathcal{W}(\varepsilon)$ and $\mathcal{W}^J(\varepsilon)$ be as in Definition 17.1.

(a) *(Monotone.)* For $0 \le \varepsilon \le \varepsilon'$, $\mathcal{W}(\varepsilon) \subseteq \mathcal{W}(\varepsilon')$ and $\mathcal{W}^J(\varepsilon) \subseteq \mathcal{W}^J(\varepsilon')$; in particular $\mathcal{W} \subseteq \mathcal{W}(\varepsilon)$ and $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$ for every $\varepsilon \ge 0$.

(b) *(Convex and compact.)* $\mathcal{W}(\varepsilon)$ and $\mathcal{W}^J(\varepsilon)$ are convex and compact. Every $\omega \in \mathcal{W}(\varepsilon)$ has $\operatorname{tr}\omega \le 1 + d_A\varepsilon$ in trusted-source mode and $\operatorname{tr}\omega \le 1 + \varepsilon$ in untrusted-source mode, and every $J \in \mathcal{W}^J(\varepsilon)$ has $\sum_s \operatorname{tr} J_s \le d_{A_0}(1 + \varepsilon)$, a bound that reads the coupling clause and never the normalisation clause.

(c) *(What positivity carries.)* $\mathcal{W}(\varepsilon) \subseteq \operatorname{Pos}(AQ)$ and $\chi_Q\big(\mathcal{W}^J(\varepsilon)\big) \subseteq \operatorname{Pos}(AQ)$. *First*, the bounds $0 \le K_{\hat{c}}(\omega) \le P_{\hat{c}}(\omega)$ hold on the whole cone. *Second*, $K_{\hat{c}}$ is concave on the sector-block-diagonal part of $\operatorname{Pos}(AQ)$. *Third*, Theorem 5.10 holds verbatim with $\mathcal{W}(\varepsilon)$ in place of $\mathcal{W}$, and O_α^J is convex and lower semicontinuous on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$. *Fourth*, none of this survives a relaxation of positivity: the objective factors through $X \mapsto X^p$, which is undefined off the cone. One fact of the exact set

does not survive: for $\varepsilon > 0$ and $\omega \in \mathcal{W}(\varepsilon)$, $\sum_{\hat{c}} P_{\hat{c}}(\omega) = \operatorname{tr}\omega$ for an exactly admissible $\{F_v\}$, which the relaxed normalisation does not pin to 1, so $\mathbf{\Phi}[\omega]$ need not be a probability vector and the range fact $\underline{h}_\alpha \geq 0$ of Proposition 5.8 (i) is not available on $\mathcal{W}(\varepsilon)$ or on $\mathcal{W}^J(\varepsilon)$; it holds on the exact sets $\mathcal{W}$ and $\mathcal{W}^J$.

(c′) *(Each normalisation clause is implied by the clause before it.)* Let $J = (J_s)_s$ be per-block positive semidefinite with $\|R\|_{\max} \leq \varepsilon$, where $R \coloneqq \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0}$, and put $G \coloneqq \iota^\dagger \tau_A \iota = W_\tau^\dagger W_\tau$. Then G is a state,

$$\operatorname{tr}[\chi_Q(J)] = \operatorname{tr}\Big[W_\tau^\dagger W_\tau \textstyle\sum_s \operatorname{tr}_{Q_s} J_s\Big] = \operatorname{tr}\big[G(\mathbb{1}_{A_0} + R)\big] = 1 + \operatorname{tr}[GR], \tag{17.2}$$

and $\big|\operatorname{tr}[\chi_Q(J)] - 1\big| \leq d_{A_0}\varepsilon$; in particular $\operatorname{tr}[\chi_Q(J)] = 1$ holds at a point of nonzero residual R exactly where $\operatorname{tr}[GR] = 0$. In the state form in trusted-source mode, $\|\omega_A - \tau_A\|_{\max} \leq \varepsilon$ implies $|\operatorname{tr}\omega - 1| \leq d_A\varepsilon$. In untrusted-source mode the normalisation clause is the relaxed coupling condition itself.

(c″) *(What relaxing the normalisation costs.)* At a point of $\mathcal{W}^J(\varepsilon)$ put $\varsigma \coloneqq \operatorname{tr}[\chi_Q(J)] - 1$, so that $|\varsigma| \leq d_{A_0}\varepsilon$ and $\sum_{\hat{c}} P^J_{\hat{c}}(J) = 1 + \varsigma$, and, where $1 + \varsigma > 0$, $\bar{\mathbf{P}} \coloneqq \mathbf{P}/(1+\varsigma)$, a probability vector. Then for every probability vector $\mathbf{q}$

$$\sum_{\hat{c}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}} = \sum_{\hat{c}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{(1+\varsigma)\bar{P}_{\hat{c}}} = D(\mathbf{q}\|\bar{\mathbf{P}}) - \log_2(1+\varsigma), \tag{17.3}$$

and, provided $d_{A_0}\varepsilon < 1$,

$$O^J_\alpha(\mathbf{q}, J) \;\geq\; -\frac{\alpha}{\alpha - 1} \log_2\big(1 + d_{A_0}\varepsilon\big) \qquad \text{on } \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon), \tag{17.4}$$

whose right-hand side is 0 at $\varepsilon = 0$ and negative otherwise. The state form carries the same statement with $d_A\varepsilon$ in trusted-source mode and ε in untrusted-source mode in place of $d_{A_0}\varepsilon$. At $d_{A_0}\varepsilon \geq 1$ the identity (17.3) still holds wherever $1 + \varsigma > 0$ but carries no bound.

(d) *(Reading the set at a tolerance is conservative.)* With $\underline{h}_\alpha[\mathcal{W}(\varepsilon)] \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}(\varepsilon)} O_\alpha(\mathbf{q}, \omega)$,

$$\varepsilon \leq \varepsilon' \implies \underline{h}_\alpha\big[\mathcal{W}(\varepsilon')\big] \;\leq\; \underline{h}_\alpha\big[\mathcal{W}(\varepsilon)\big] \;\leq\; \underline{h}_\alpha\big[\mathcal{W}(0)\big] = \underline{h}_\alpha \;\leq\; h_\alpha^{\uparrow,\mathrm{QKD}}. \tag{17.5}$$

(d′) *(The Choi-form quantity.)* With

$$\underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon)\big] \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{J\in\mathcal{W}^J(\varepsilon)} O^J_\alpha(\mathbf{q}, J), \tag{17.6}$$

the two relaxed quantities agree at $\varepsilon = 0$, and

$$\varepsilon \le \varepsilon' \implies \underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon')\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon)\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}^J(0)\big] \;=\; \underline{h}_\alpha \;\le\; h_\alpha^{\uparrow,\mathrm{QKD}}. \tag{17.7}$$

At $\varepsilon > 0$ no relation between $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ and $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ is asserted.

(e) *(The general format.)* For a declaration in the general format, with both sets read as Definition 17.1 reads them there and χ_Q the assignment $(J_{(\mathsf{t},s)}) \mapsto (\omega^{(\mathsf{t},s)})$ of Theorem 7.5, (a) to (d′) hold as stated. In (c′) the residual is $R = \bigoplus_{\mathsf{t}} R_{\mathsf{t}}$, $G = \bigoplus_{\mathsf{t}} G_{\mathsf{t}}$ with $G_{\mathsf{t}} \coloneqq W_{\mathsf{t}}^\dagger W_{\mathsf{t}}$, and $\mathrm{tr}[\chi_Q(J)] = 1 + \sum_{\mathsf{t}} \mathrm{tr}[G_{\mathsf{t}} R_{\mathsf{t}}]$.

18 Admissible directions

Definition 18.1 (Ambient program, pairings, acceptance support and linear minimisation)**.** From this definition on, the ambient program of this submodule is the Choi program (7.10) of Theorem 7.5: the variable is $x = (\mathbf{q}, J)$ with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $J = (J_s)_{s\in\mathbb{S}} \in \mathcal{W}^J$ of (7.9), and the objective is the function O_α^J displayed in (7.10), so that $\underline{h}_\alpha = \inf\{O_\alpha^J(x) : x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J\}$ by that theorem. In untrusted-source mode the state form is primary: there J_s, $\mathcal{W}^J$, χ_{Q_s} and O_α^J denote ω_s, $\mathcal{W}$, the identity and O_α (Definition 7.10), and every later statement of this submodule is read with these substitutions, as Corollary 8.5 reads the gradients.

The *acceptance support* of the acceptance set $\mathcal{S}_{\mathrm{acc}}$ (Definition 2.5) is

$$\widehat{\mathcal{C}}_{\mathrm{acc}} \coloneqq \big\{\hat{c} \in \widehat{\mathcal{C}} :\; q(\hat{c}) > 0 \text{ for some } \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}\big\}. \tag{18.1}$$

Pairings are the real inner products $\langle \mathbf{a}, \mathbf{b}\rangle = \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{acc}}} a(\hat{c})b(\hat{c})$ on $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$, the acceptance support being where the acceptance-side gradient lives (Lemma 18.2), and $\langle A, B\rangle = \mathrm{tr}[AB]$ on Hermitian operators, the latter real and, on a tuple $(A_s)_s$, summed over s.

For linear forms $g_t \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and $G_t = (G_{t,s})_{s\in\mathbb{S}}$, a tuple of Hermitian operators of the block shape of $\mathcal{W}^J$, a pair (v_t, S_t) of a vector in $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and such a tuple satisfies the *acceptance-side inequality* of (18.2) when the first inequality below holds, and the *feasible-side inequality* when the second holds:

$$\langle g_t, v_t\rangle \le \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}} \langle g_t, \mathbf{q}'\rangle \qquad \text{and} \qquad \langle G_t, S_t\rangle \le \min_{J'\in\mathcal{W}^J} \langle G_t, J'\rangle, \tag{18.2}$$

the second minimum being taken over the exact $\mathcal{W}^J$ of (7.9).

Lemma 18.2 (Positivity needed only on the acceptance support)**.** Let $\widehat{\mathcal{C}}_{\mathrm{acc}}$ be the acceptance support (18.1) of the acceptance set $\mathcal{S}_{\mathrm{acc}}$ of Definition 2.5.

(i) If $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$, $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is the support of a single element of $\mathcal{S}_{\mathrm{acc}}$; and, whether or not $\mathcal{S}_{\mathrm{acc}}$ is empty, every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ vanishes outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$.

(ii) On $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, for every $\varepsilon \geq 0$ and in particular on the exact $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, every term of O^J_α indexed by a $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$ vanishes identically, so O^J_α restricted to that set is a function of $(\mathbf{q}|_{\widehat{\mathcal{C}}_{\mathrm{acc}}}, J)$ alone; and for $\mathbf{q}, \mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}$ every feasible direction has $q'(\hat{c}) - q(\hat{c}) = 0$ at every $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$. The only property of the J factor the first clause reads is per-block positive semidefiniteness, so the first clause holds at every per-block positive semidefinite point whatever, of which $\mathcal{W}^J(\varepsilon)$ is one family.

(iii) Corollary 8.5 holds on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, for every $\varepsilon \geq 0$, with its hypothesis "$q(\hat{c}) > 0$, $P^J_{\hat{c}} > 0$ and $K^J_{\hat{c}} > 0$ for all $\hat{c}$" weakened to: $q(\hat{c}) > 0$ and $P^J_{\hat{c}} > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, and $K^J_{\hat{c}} > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$. Then (8.10) is read as a vector in $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and the sums of (8.11) are restricted to $\widehat{\mathcal{C}}_{\mathrm{acc}}$, respectively to $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$; both restrictions are identities on that set, not truncations.

Lemma 18.3 (Every direction is admissible at maximal support)**.** In the setting of Definition 18.1, let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}_K(\hat{c})$, let $J_s \in \mathrm{Pos}(A_0 Q_s)$, and put $X \coloneqq \mathcal{G}^J_{\hat{c},s}(J_s)$. Then for *every* Hermitian Δ on $\mathcal{H}(A_0 Q_s)$ the operator $H \coloneqq \mathcal{G}^J_{\hat{c},s}(\Delta)$ satisfies

$$\operatorname{supp} H \;\subseteq\; W_{\hat{c},s} \tag{18.3}$$

with $W_{\hat{c},s}$ the maximal branch support (8.1). Consequently, if X attains maximal support, $\operatorname{supp} X = W_{\hat{c},s}$, for which $J_s \succ 0$ suffices, then $\operatorname{supp} H \subseteq \operatorname{supp} X$ for every such Δ, so that:

(a) Theorem 8.4 (i) applies to the pair (X, H), and $t \mapsto T(X + tH)$ is differentiable at $t = 0$, two-sidedly, the hypothesis being symmetric in $H \mapsto -H$;

(b) the $+\infty$ case of Theorem 8.4 (ii) is *unreachable* from X along the image of any Hermitian direction Δ for which $X + tH \succeq 0$ for small $t > 0$. That is the class that theorem is stated on and the class along which T is defined at all, and it contains every direction $y - x$ that Theorem 19.1 pairs against, since $X + t\mathcal{G}^J_{\hat{c},s}(J'_s - J_s) = \mathcal{G}^J_{\hat{c},s}\big((1-t)J_s + tJ'_s\big) \succeq 0$ for $t \in [0,1]$ whenever J_s and J'_s are positive semidefinite. No such direction strictly enlarges the support of X, because by (18.3) none leaves $W_{\hat{c},s} = \operatorname{supp} X$.

19 The first-order lower bound

Theorem 19.1 (The estimate and the first-order lower bound). Let $\alpha \in (1,\infty)$, let $\mathfrak{D}$ satisfy Definition 1.2, in the basic format or in the general format of Definition 1.1, and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. In the general format $\mathcal{W}^J$ is the set of lists $(J_{(\mathsf{t},s)})$ over the joint sectors of Theorem 7.5, with $J_s := \bigoplus_{\mathsf{t}} J_{(\mathsf{t},s)}$ for every sector s of the measured system, and $\mathcal{W}^J(\varepsilon)$ is read as Definition 17.1 reads it there.

(i) *(The estimate over-estimates; Lemma 7.11 (ii) of the core, restated.)* For every $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$,

$$O^J_\alpha(x) \;\geq\; \underline{h}_\alpha. \tag{19.1}$$

(ii) *(The first-order lower bound.)* Let $\mathfrak{I}$ be a finite non-empty index set, and let $x_t = (\mathbf{q}_t, J_t) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, $t \in \mathfrak{I}$, satisfy for each t:

(H-POS) $q_t(\hat{c}) > 0$ and $P^J_{\hat{c}}(J_t) > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, and $K^J_{\hat{c}}(J_t) > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$, with $\widehat{\mathcal{C}}_{\mathrm{acc}}$ the acceptance support (18.1);

(H-SUPP) every key-active branch operator $X_{\hat{c},s} := \mathcal{G}^J_{\hat{c},s}(J_{t,s})$, $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, $s \in \mathbb{S}_K(\hat{c})$, attains maximal support: $\operatorname{supp} X_{\hat{c},s} = W_{\hat{c},s}$, the maximal branch support (8.1), equivalently $\operatorname{rank} X_{\hat{c},s} = \dim W_{\hat{c},s} = \operatorname{rank} \mathbf{K}_{\hat{c},s}$ by (8.4), the inclusion $\subseteq$ holding always by (8.3); $J_{t,s} \succ 0$ suffices for it (Lemma 8.2 (i));

(H-NOPERT) the pair $(g_t, G_t) := \nabla O^J_\alpha(x_t)$ is the split-form gradient of Theorem 8.4 (i)(a)–(b) assembled by Corollary 8.5 *at x_t itself*: no perturbation of x_t, of any $J_{t,s}$, or of any branch operator built from them is applied before the gradient is taken.

Let (v_t, S_t) satisfy both inequalities of (18.2) at (g_t, G_t) (Definition 18.1): the acceptance-side inequality and the feasible-side inequality. Then, *in exact real arithmetic*,

$$\underline{h}^{\mathrm{FO}}_\alpha \;:=\; \max_{t\in\mathfrak{I}} \Big[O^J_\alpha(x_t) + \langle g_t, v_t - \mathbf{q}_t\rangle + \langle G_t, S_t - J_t\rangle \Big] \;\leq\; \underline{h}_\alpha, \tag{19.2}$$

and $\max\{0, \underline{h}^{\mathrm{FO}}_\alpha\} \leq \underline{h}_\alpha$ as well.

(iii) *(The iterate read at a tolerance, the minima over the exact set.)* Let $\varepsilon \geq 0$ and let $\mathcal{W}^J(\varepsilon)$ be the relaxed feasible set of Definition 17.1.

(iiia) *The estimate.* For every $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, $O^J_\alpha(x) \geq \underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$, the Choi-form relaxed infimum (17.6); and $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)] \leq \underline{h}_\alpha \leq h^{\uparrow,\mathrm{QKD}}_\alpha$ by (17.7).

(iiib) *The bound.* Let $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, $t \in \mathfrak{I}$, satisfy (H-POS), (H-SUPP) and (H-NOPERT) as stated in (ii), and let (v_t, S_t) satisfy (18.2) *as written*, its acceptance-side inequality and its feasible-side inequality, with the two minima over $\mathcal{S}_{\mathrm{acc}}$ and over the *exact* $\mathcal{W}^J$. Then, in exact real arithmetic, $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ and $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} \le \underline{h}_\alpha$, with $\underline{h}_\alpha^{\mathrm{FO}}$ of (19.2).

(iv) *(Lower bounds in place of the pairings.)* The conclusions of (ii) and of (iiib) hold as well when, at some or all $t \in \mathfrak{I}$, the pairing $\langle g_t, v_t\rangle$ in (19.2) is replaced by a real number b_t^{A}, or the pairing $\langle G_t, S_t\rangle$ by a real number b_t^{F}, or both, with

$$b_t^{\mathrm{A}} \le \min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle g_t, \mathbf{q}'\rangle \qquad \text{and} \qquad b_t^{\mathrm{F}} \le \min_{J' \in \mathcal{W}^J} \langle G_t, J'\rangle, \tag{19.3}$$

each inequality asked only where its number is used, the second minimum taken over the *exact* $\mathcal{W}^J$, and the pairings $\langle g_t, \mathbf{q}_t\rangle$ and $\langle G_t, J_t\rangle$ being kept. At such a t the inequality of (18.2) on a replaced side is not asked of (v_t, S_t), and where both sides are replaced no pair (v_t, S_t) is asked at all: the bracket of (19.2) at that t is then

$$O_\alpha^J(x_t) + b_t^{\mathrm{A}} - \langle g_t, \mathbf{q}_t\rangle + b_t^{\mathrm{F}} - \langle G_t, J_t\rangle.$$

(v) *(An empty feasible set.)* In place of the hypothesis on $\mathfrak{D}$ above, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1 that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, and form $\mathcal{W}$, the objective O_α and $\underline{h}_\alpha$ from it by the formulas of Definition 5.1. Let $\mathcal{W} = \varnothing$. Then $\underline{h}_\alpha = +\infty$, so every real number is a lower bound on $\underline{h}_\alpha$, the value of (19.2) that a run forms at its iterates included. Consequently the conclusions $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ and $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} \le \underline{h}_\alpha$ of (ii), (iiib) and (iv), for a real $\underline{h}_\alpha^{\mathrm{FO}}$, hold of every tuple of the form of Definition 1.1 that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, whether or not it satisfies condition (vii): under the hypotheses of those parts where it does, and because $\underline{h}_\alpha = +\infty$ where it does not.

Corollary 19.5 (Key length from the first-order bound)**.** Let $\alpha \in (1, 2)$, let the hypotheses of Theorem 10.7 hold in its unconstrained or constrained branch, the run as in Lemma 10.5 (a) and hypothesis (H-CHAIN) included, at every all-at-once attack with finite registers, with $\mathcal{W}^J$ carrying the declared $\mathfrak{L}$ in the constrained branch, let those of Lemma 10.4 hold, A-SEQ with its clause (v) among the former, and let $\underline{h}_\alpha^{\mathrm{FO}}$ be as in Theorem 19.1 (ii) or (iiib), at some or all iterates with the lower bounds of its part (iv) in place of the pairings. If the output length satisfies

$$\ell \le n \max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha - 1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \tag{19.7}$$

then the protocol is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, hence, for $\ell \geq 1$, $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}})$-secure in the honest-authentication setting, and under the further hypotheses of Corollary 10.8 (iii) $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}}+2\varepsilon_{\mathrm{auth}})$-secure over the classical channel of that part, with net key gain $\ell - k_{\mathrm{auth}}$ when both parties output a key, the secrecy and the security statements holding at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals, as Corollary 10.8 states them, and the correctness statement at every attack; the four conclusions of Proposition 10.17 hold verbatim for the maximal key length built from (19.7); and if $\underline{h}_{\alpha}^{\mathrm{FO}} \leq 0$ and $\varepsilon_{\mathrm{EV}} < 1$ no key is claimed.

Corollary 19.6 (First-order bound for the dropped program). Let $\alpha \in (1,\infty)$, let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, in either source mode, in trusted-source mode with a frame per Alice sector as in Theorem 7.5, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let $\mathcal{D}$ be a dropped set (Definition 9.2). Write an Alice sector as $\mathtt{t} \in \mathbb{T}$ in this claim, the letter t indexing iterates as in Theorem 19.1. Put $O_{\alpha}^{J,\mathcal{D}}(\mathbf{q}, J) \coloneqq O_{\alpha}^{\mathcal{D}}(\mathbf{q}, \chi(J))$ and $K_{\hat{c}}^{J,\mathcal{D}} \coloneqq K_{\hat{c}}^{\mathcal{D}} \circ \chi$, with χ the assignment of that theorem in trusted-source mode. In untrusted-source mode the program is read as Definition 7.10 reads it, with no frame: χ is the identity, $J = \omega$ is the list of the blocks $J_{(\mathtt{t},s)} = \omega^{(\mathtt{t},s)}$ over the joint sectors, $A_{0,\mathtt{t}}$ and A_0 stand for $A_{\mathtt{t}}$ and A, and $\mathcal{W}^J$ is $\mathcal{W}$ of Definition 5.1 in the general format, whose condition coupling the blocks is the normalisation $\sum_{(\mathtt{t},s)} \operatorname{tr} \omega^{(\mathtt{t},s)} = 1$; so $O_{\alpha}^{J,\mathcal{D}}$ and $K_{\hat{c}}^{J,\mathcal{D}}$ are $O_{\alpha}^{\mathcal{D}}$ and $K_{\hat{c}}^{\mathcal{D}}$. Let $x_t = (\mathbf{q}_t, J_t) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, $t \in \mathfrak{I}$, satisfy

(H-POS$^{\mathcal{D}}$) (H-POS) of Theorem 19.1 with $K_{\hat{c}}^{J,\mathcal{D}}$ in place of $K_{\hat{c}}^{J}$;

(H-SUPP$^{\mathcal{D}}$) for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and every joint pair $(\mathtt{t}, s) \in \mathbb{J}_K(\hat{c})$ with $(\hat{c}, (\mathtt{t}, s)) \notin \mathcal{D}$, the branch operator $\mathcal{G}_{\hat{c},s}^{J}(J_{t,(\mathtt{t},s)})$ of the block $J_{t,(\mathtt{t},s)} \in \mathrm{Pos}(A_{0,\mathtt{t}} Q_s)$ attains the maximal support of that joint pair, the support it has at $J_{t,(\mathtt{t},s)} \succ 0$; nothing is asked of a pair that is key-inert or dropped;

(H-NOPERT$^{\mathcal{D}}$) (g_t, G_t) is the gradient of $O_{\alpha}^{J,\mathcal{D}}$ at x_t itself, assembled as in (H-NOPERT) with the kept summands only and with $K_{\hat{c}}^{J,\mathcal{D}}$ in the scalar prefactors, its blocks on fully dropped joint sectors being (9.5) in Choi coordinates;

and let (v_t, S_t) satisfy (18.2) for these linear forms, or let, at some or all t, lower bounds b_t^{A}, b_t^{F} on one or both of its minima, as in (19.3) for these linear forms, stand in for the pairings as Theorem 19.1 (iv) lets them. Then, in exact real arithmetic, with $\underline{h}_{\alpha}^{\mathrm{FO},\mathcal{D}}$ the expression (19.2) for $O_{\alpha}^{J,\mathcal{D}}$, with those replacements where they are made,

$$\max\{0, \underline{h}_{\alpha}^{\mathrm{FO},\mathcal{D}}\} \;\leq\; \underline{h}_{\alpha}^{\mathcal{D}} \;\leq\; \underline{h}_{\alpha} \;\leq\; h_{\alpha}^{\uparrow,\mathrm{QKD}}(\mathfrak{L}). \tag{19.8}$$

(H-POS$^{\mathcal{D}}$) is implied by (H-POS), and (H-SUPP$^{\mathcal{D}}$) by (H-SUPP) read on joint blocks. For $\mathcal{D} = \varnothing$ and $|\mathbb{T}| = 1$ the statement is Theorem 19.1 (ii), with its part (iv) where lower bounds stand in.

The relaxed reading. Let $\varepsilon \geq 0$ and let $\mathcal{W}^J(\varepsilon)$ be the relaxed feasible set of Definition 17.1, read in the general format, with one coupling clause per Alice sector in trusted-source mode and its normalisation clause in untrusted-source mode. Let $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, $t \in \mathfrak{I}$, satisfy (H-POS$^{\mathcal{D}}$), (H-SUPP$^{\mathcal{D}}$) and (H-NOPERT$^{\mathcal{D}}$) as stated, and let (v_t, S_t) satisfy (18.2) *as written*, or the lower bounds standing in for its pairings satisfy (19.3), with the two minima over $\mathcal{S}_{\mathrm{acc}}$ and over the *exact* $\mathcal{W}^J$. Then (19.8) holds in full, and the case $\varepsilon = 0$ is the statement above.

An empty feasible set. In place of the hypothesis on $\mathfrak{D}$ above, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1, in the general format and in either source mode, that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, and form $\mathcal{W}$, O_α, $\underline{h}_\alpha$ and $\underline{h}_\alpha^{\mathcal{D}}$ from it by the formulas of Definitions 5.1 and 9.2. Let $\mathcal{W} = \varnothing$. Then $\underline{h}_\alpha^{\mathcal{D}} = \underline{h}_\alpha = +\infty$, so every real number is a lower bound on both, the value of (19.2) for $O_\alpha^{J,\mathcal{D}}$ that a run forms at its iterates included. Consequently the first two inequalities of (19.8), for a real $\underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$, hold of every such tuple, whether or not it satisfies condition (vii): under the hypotheses above where it does, and by the preceding sentence where it does not.

Corollary 19.7 (The first-order bound on the reference companion). Let $\mathfrak{D}$ and $\mathfrak{D}^{\mathrm{ref}}$ be as in Proposition 12.1, let $\alpha \in (1, \infty)$, and let $\mathcal{S}_{\mathrm{acc}}$ be an admissible acceptance set (Definition 2.5), the one of the run, read for both. Let $\varepsilon \geq 0$ and let $x_t = (\mathbf{q}_t, J_t)$, $t \in \mathfrak{I}$, be a finite non-empty set of points, produced by any computation, a search on $\mathfrak{D}$ among them, such that every x_t lies in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ of $\mathfrak{D}^{\mathrm{ref}}$ (Definition 17.1) and satisfies (H-POS), (H-SUPP) and (H-NOPERT) of Theorem 19.1 for the objective, the branch maps and the gradient of $\mathfrak{D}^{\mathrm{ref}}$, and let (v_t, S_t) satisfy (18.2) at the gradient of $\mathfrak{D}^{\mathrm{ref}}$ at x_t, its feasible-side minimum over the exact $\mathcal{W}^J$ of $\mathfrak{D}^{\mathrm{ref}}$ (Definition 18.1), or let, at some or all t, lower bounds on one or both of those minima stand in for the pairings as Theorem 19.1 (iv) lets them, (19.3) read at that gradient and over those sets. Write $\underline{h}_\alpha^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$ for (19.2) so formed. Then:

(i) in exact real arithmetic, $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]\} \leq \underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}]$, and $\underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}] \leq \underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}]$ for the companion with its unrelaxed entries;

(ii) the Choi variables of $\mathfrak{D}$ and $\mathfrak{D}^{\mathrm{ref}}$ are lists over the same joint sectors with the same block dimensions, so every point of the one program is a point of the other; where no link the companion replaces carries entries, as on the encoded route, $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is the same set for both; and a point with $J_{t,s} \succ 0$ for every sector s satisfies (H-SUPP) for $\mathfrak{D}^{\mathrm{ref}}$, whatever its measured family;

(iii) where the hypotheses of Corollary 19.5 hold for $\mathfrak{D}^{\mathrm{ref}}$, its conclusions hold with $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$ in (19.7): the key length so formed is secret and correct for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

20 The verified feasible side

Lemma 20.1 (A lower bound on the feasible-side minimum from any multipliers). (i) *(Weak duality with a trace identity.)* Let $\mathcal{H}_{\mathrm{in}}$ be a Hilbert space of finite dimension $d_{\mathrm{in}} \geq 1$, let P be a finite non-empty set, and for every $p \in P$ let $\mathcal{H}_p$ be a finite-dimensional Hilbert space. Let $\mathcal{R}$ be a finite, possibly empty, set of *rows*; a row r consists of a finite-dimensional Hilbert space $\mathcal{K}_r$, a Hermitian operator Θ_r on it, a relation $\bowtie_r \in \{\preceq, \succeq, =\}$ and, for every $p \in P$, a linear map $L_{r,p}$ from operators on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ to operators on $\mathcal{K}_r$ that maps Hermitian operators to Hermitian ones. Write $L_{r,p}^{*}$ for its adjoint, the real-linear map from Hermitian operators on $\mathcal{K}_r$ to Hermitian operators on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ with $\operatorname{tr}[\Lambda\, L_{r,p}(X)] = \operatorname{tr}\big[L_{r,p}^{*}(\Lambda)\, X\big]$ for all Hermitian Λ and X. Let $\mathcal{J}$ be the set of lists $J = (J_p)_{p \in P}$ of positive semidefinite operators J_p on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ with

$$\sum_{p \in P} \operatorname{tr}_{\mathcal{H}_p} J_p = \mathbb{1}_{\mathcal{H}_{\mathrm{in}}} \qquad \text{and} \qquad L_r(J) \coloneqq \sum_{p \in P} L_{r,p}(J_p) \;\bowtie_r\; \Theta_r \quad \text{for every } r \in \mathcal{R}. \tag{20.1}$$

Let $G = (G_p)_{p \in P}$ be a list of Hermitian operators G_p on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$, let Y be a Hermitian operator on $\mathcal{H}_{\mathrm{in}}$, and for every row r let Λ_r be a Hermitian operator on $\mathcal{K}_r$, with $\Lambda_r \succeq 0$ when $\bowtie_r$ is $\preceq$ or $\succeq$. Put $\sigma_r \coloneqq -1$ when $\bowtie_r$ is $\preceq$ and $\sigma_r \coloneqq +1$ otherwise,

$$R_p \;\coloneqq\; G_p - Y \otimes \mathbb{1}_{\mathcal{H}_p} - \sum_{r \in \mathcal{R}} \sigma_r\, L_{r,p}^{*}(\Lambda_r) \qquad (p \in P), \tag{20.2}$$

and let λ be a real number with $\lambda \leq \lambda_{\min}(R_p)$ for every $p \in P$. Then for every $J \in \mathcal{J}$

$$\langle G, J \rangle \coloneqq \sum_{p \in P} \operatorname{tr}[G_p J_p] \;\geq\; b(G; Y, \Lambda, \lambda) \coloneqq d_{\mathrm{in}}\, \lambda + \operatorname{tr} Y + \sum_{r \in \mathcal{R}} \sigma_r \operatorname{tr}[\Lambda_r \Theta_r], \tag{20.3}$$

so that $b(G; Y, \Lambda, \lambda) \leq \inf_{J \in \mathcal{J}} \langle G, J \rangle$. No other condition is placed on Y, on the Λ_r or on λ: Y and the Λ_r need not be feasible for any dual program, λ may be negative, and $Y = 0$ with every $\Lambda_r = 0$ is admitted.

(ii) *(The Choi feasible set is of that form.)* Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1, the basic format included, with its Alice sectors $\mathsf{t} \in \mathbb{T}$.

(a) In trusted-source mode, with a frame and the factors W_{t} as in Theorem 7.5, the factor $\mathcal{W}_{\mathsf{t}}^{J}$ of $\mathcal{W}^{J} = \prod_{\mathsf{t}} \mathcal{W}_{\mathsf{t}}^{J}$ (Theorem 7.5, the

general format) is the set $\mathcal{J}$ of (i) with $\mathcal{H}_{\text{in}} = \mathcal{H}(A_{0,\mathsf{t}})$, $P = \mathbb{S}$, $\mathcal{H}_s = \mathcal{H}(Q_s)$ and $J_s = J_{(\mathsf{t},s)}$, and with the following rows. For every entry k read through its lifted operator, one row with $\mathcal{K} = \mathcal{H}(A_{\mathsf{t}})$, $\Theta = \Theta_k^{(\mathsf{t})}$, the relation $\bowtie_k$ and

$$L_{k,s}(X) = W_{\mathsf{t}}\ \mathrm{tr}_{Q_s}\big[(\mathbb{1}_{A_{0,\mathsf{t}}} \otimes \hat{O}_k^{\uparrow(s)})X\big]\, W_{\mathsf{t}}^\dagger, \qquad L_{k,s}^*(\Lambda) = (W_{\mathsf{t}}^\dagger \Lambda W_{\mathsf{t}}) \otimes \hat{O}_k^{\uparrow(s)}.$$

For every party-local entry k, scoped to the measured factor f, and every s_{f^c}, one row with $\mathcal{K} = \mathcal{H}(A_{\mathsf{t}}) \otimes \mathcal{H}(Q_{f^c,s_{f^c}})$, $\Theta = 0$, the relation $\bowtie_k$ and, at $s = (s_{f^c}, s_f)$,

$$L_{(k,s_{f^c}),s}(X) = (W_{\mathsf{t}} \otimes \mathbb{1})\ \mathrm{tr}_{Q_{f,s_f}}\big[(\mathbb{1} \otimes D_k^{(s_f)})X\big]\, (W_{\mathsf{t}}^\dagger \otimes \mathbb{1}),$$
$$L_{(k,s_{f^c}),s}^*(\Lambda) = \big((W_{\mathsf{t}}^\dagger \otimes \mathbb{1})\Lambda(W_{\mathsf{t}} \otimes \mathbb{1})\big) \otimes D_k^{(s_f)},$$

where $D_k^{(s_f)} \coloneqq \hat{O}_k^{(s_f)} - c_k\mathbb{1}$ acts on $\mathcal{H}(Q_{f,s_f})$ at the place of that factor in $\mathcal{H}(Q_s)$, while $L_{(k,s_{f^c}),s'} = 0$ and $L_{(k,s_{f^c}),s'}^* = 0$ at every sector s' whose component on the other factors is not s_{f^c}. The two kinds of map, and the formulas for their adjoints, are defined, and the formulas hold, for any linear map W from $\mathcal{H}(A_{0,\mathsf{t}})$ to $\mathcal{H}(A_{\mathsf{t}})$ in place of W_{t}; we write $L_{r,s}^W$ for a row's map so built. A homogeneous or party-local row in the frame-free form of Proposition 7.6, on a tight frame, is a row of the same kind with $\mathbb{1}_{A_{0,\mathsf{t}}}$ in place of W_{t}, hence with $\mathcal{H}(A_{0,\mathsf{t}})$ in place of $\mathcal{H}(A_{\mathsf{t}})$ in its $\mathcal{K}$, and with $\Theta = c_k\mathbb{1}_{A_{0,\mathsf{t}}}$, respectively $\Theta = 0$; and $\mathcal{W}_{\mathsf{t}}^J$ is unchanged when it replaces the conjugated row. Hence, for Y_{t}, $\Lambda_{\mathsf{t}} = (\Lambda_r)_r$ and λ_{t} as in (i) for each t, and with $G_{\mathsf{t}} \coloneqq (G_{(\mathsf{t},s)})_{s\in\mathbb{S}}$,

$$b_{\mathrm{F}} \;\coloneqq\; \sum_{\mathsf{t}\in\mathbb{T}} b(G_{\mathsf{t}}; Y_{\mathsf{t}}, \Lambda_{\mathsf{t}}, \lambda_{\mathsf{t}}) \;\le\; \langle G, J'\rangle \qquad \text{for every } J' \in \mathcal{W}^J. \tag{20.4}$$

(b) In untrusted-source mode, $\mathcal{W}^J$, which is $\mathcal{W}$ (Definition 18.1), is the set $\mathcal{J}$ of (i) with $\mathcal{H}_{\text{in}}$ one-dimensional, so that $Y = y$ is a real number, $Y \otimes \mathbb{1} = y\,\mathbb{1}$, $\mathrm{tr}\, Y = y$ and $d_{\text{in}} = 1$; with P the set of joint sectors, $\mathcal{H}_{(\mathsf{t},s)} = \mathcal{H}(A_{\mathsf{t}}Q_s)$ and $J_{(\mathsf{t},s)} = \omega^{(\mathsf{t},s)}$; and with these rows. For every scalar entry k, one row with $\mathcal{K} = \mathbb{C}$, $\Theta = c_k$, the relation $\bowtie_k$, $L_{k,(\mathsf{t},s)}(X) = \mathrm{tr}\Big[(\mathbb{1}_{A_{\mathsf{t}}} \otimes \hat{O}_k^{\uparrow(s)})X\Big]$ and $L_{k,(\mathsf{t},s)}^*(\mu) = \mu\,\mathbb{1}_{A_{\mathsf{t}}} \otimes \hat{O}_k^{\uparrow(s)}$; for every party-local entry k and every pair (t, s_{f^c}), one row as in (a) with $\mathbb{1}_{A_{\mathsf{t}}}$ in place of W_{t}, its maps vanishing outside the joint sectors $(\mathsf{t}, (s_{f^c}, s_f))$. Hence $b(G; y, \Lambda, \lambda) \le \langle G, J'\rangle$ for every $J' \in \mathcal{W}^J$, the whole list being one set of the form of (i).

(iii) *(A scalar coupling and no rows.)* If $d_{\text{in}} = 1$ and $\mathcal{R} = \varnothing$ in (i), then, with $Y = 0$ and $\lambda = \min_{p\in P} \lambda_{\min}(G_p)$, the infimum in (i) is attained and $b(G; 0, \Lambda, \lambda) = \min_{J\in\mathcal{J}} \langle G, J\rangle$.

(iv) *(Rows built from an inexact frame.)* In trusted-source mode fix an Alice sector $\mathtt{t}$, a linear map $\widetilde{W}_{\mathtt{t}}$ from $\mathcal{H}(A_{0,\mathtt{t}})$ to $\mathcal{H}(A_{\mathtt{t}})$ and a real number $\varepsilon_{\mathtt{t}} \geq \left\|\widetilde{W}_{\mathtt{t}} - W_{\mathtt{t}}\right\|$. For every row r of $\mathcal{W}_{\mathtt{t}}^{J}$ in (ii)(a) let $\widetilde{L}_{r,s} \coloneqq L_{r,s}^{\widetilde{W}_{\mathtt{t}}}$, a row in frame-free form keeping its map, every other datum of the row $(\Theta_r,\ \hat{O}_k^{\uparrow(s)},\ D_k^{(s_f)},\ c_k)$ unchanged. Let $G_{\mathtt{t}}$, Y and the Λ_r be as in (i), form $\widetilde{R}_s$ by (20.2) with $\widetilde{L}_{r,s}^{*}$ in place of $L_{r,s}^{*}$, and let $\widetilde{\lambda} \leq \lambda_{\min}(\widetilde{R}_s)$ for every $s \in \mathbb{S}$. Then for every $J \in \mathcal{W}_{\mathtt{t}}^{J}$, the factor cut out by the rows built from the exact $W_{\mathtt{t}}$,

$$\begin{aligned} \langle G_{\mathtt{t}}, J\rangle \ &\geq\ b(G_{\mathtt{t}}; Y, \Lambda, \widetilde{\lambda}) - \sum_{r} \kappa_r, \\ \kappa_r \coloneqq d_{A_{0,\mathtt{t}}}\, \varepsilon_{\mathtt{t}} \big(2\left\|\widetilde{W}_{\mathtt{t}}\right\| + \varepsilon_{\mathtt{t}}\big)\ \|\Lambda_r\|\ &\max_{s\in\mathbb{S}} \|O_{r,s}\|\,, \end{aligned} \tag{20.5}$$

where $O_{r,s}$ is $\hat{O}_k^{\uparrow(s)}$ for a lifted-operator row, $D_k^{(s_f)}$ for a party-local row at $s = (s_{f^c}, s_f)$ and 0 at the other sectors, and $\kappa_r \coloneqq 0$ for a row in frame-free form. Summed over $\mathtt{t}$ as in (ii)(a), the right-hand sides bound $\langle G, J'\rangle$ from below for every $J' \in \mathcal{W}^{J}$.

21 The certified bracket

Lemma 21.1 (The certified bracket from enclosures of the gradient)**.** Let $\alpha \in (1,\infty)$, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1, in the basic format or the general format, that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. Let $x = (\mathbf{q}, J)$ with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and every block of J positive definite, on every joint sector in the general format, satisfy (H-POS) of Theorem 19.1, and let (g, G) be the gradient of O_α^J at x assembled by Corollary 8.5, read with Lemma 18.2 (iii) and, in the general format, on the joint sectors as Theorem 19.1 reads it.

(i) *(The Euler form.)*

$$\langle g, \mathbf{q}\rangle + \langle G, J\rangle = O_\alpha^J(x). \tag{21.1}$$

Consequently, for real numbers b^{A} and b^{F} with $b^{\mathrm{A}} \leq \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}}\langle g, \mathbf{q}'\rangle$ and $b^{\mathrm{F}} \leq \min_{J'\in\mathcal{W}^{J}}\langle G, J'\rangle$, the minimum over the exact $\mathcal{W}^{J}$, the bracket of Theorem 19.1 (iv) at x with both pairings replaced is $b^{\mathrm{A}}+b^{\mathrm{F}}$, and

$$\max\{0,\ b^{\mathrm{A}} + b^{\mathrm{F}}\} \ \leq\ \underline{h}_\alpha, \tag{21.2}$$

in exact real arithmetic.

(ii) *(The acceptance side from an enclosure.)* If $\hat{g} \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and $\delta \geq 0$ satisfy $|g(\hat{c}) - \hat{g}(\hat{c})| \leq \delta$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, then $\min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}}\langle g, \mathbf{q}'\rangle \geq \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}}\langle \hat{g}, \mathbf{q}'\rangle - \delta$.

(iii) *(The feasible side from enclosed residuals.)* For each group of Lemma 20.1 (ii), an Alice sector $\mathtt{t}$ in trusted-source mode, its rows built from the exact frame $W_{\mathtt{t}}$ of Theorem 7.5, or the one group in untrusted-source mode, let Y and the Λ_r be as in Lemma 20.1 (i), let R_p be the residual (20.2) formed with G, and for every block p of the group let M_p be a Hermitian matrix, $\rho_p \geq 0$ with $\|R_p - M_p\|_{\mathrm{F}} \leq \rho_p$, the Frobenius norm of Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure), and ℓ_p a lower bound on $\lambda_{\min}(M_p)$. Then the group's $\lambda := \min_p(\ell_p - \rho_p)$ satisfies $\lambda \leq \lambda_{\min}(R_p)$ for every p, and the sum over the groups of $b(G; Y, \Lambda, \lambda)$ of (20.3) is a number $b^{\mathrm{F}} \leq \min_{J' \in \mathcal{W}^J} \langle G, J' \rangle$.

(iv) *(The dropped program.)* In the setting of Corollary 19.6, with a dropped set $\mathcal{D}$, (i)–(iii) hold with $O_\alpha^{J,\mathcal{D}}$ in place of O_α^J, (H-POS$^{\mathcal{D}}$) in place of (H-POS), its gradient of (H-NOPERT$^{\mathcal{D}}$) in place of (g, G), the kept pairs in place of the pairs that are not key-inert, and $\underline{h}_\alpha^{\mathcal{D}}$ in place of $\underline{h}_\alpha$; and $\underline{h}_\alpha^{\mathcal{D}} \leq \underline{h}_\alpha$.

Lemma 21.2 (The certified bracket over a measured family known through enclosures)**.** Let $\mathfrak{D}^{\mathrm{ref}}$ be a compiled reference companion of Proposition 12.1, with $\mathbb{F}^{\mathrm{ref}}$ the set of the measured factors it replaces (the one measured system in the basic format), let α, $\mathcal{S}_{\mathrm{acc}}$ and $x = (\mathbf{q}, J)$ be as in Lemma 21.1 with $\mathfrak{D} := \mathfrak{D}^{\mathrm{ref}}$, save that (H-POS) of Theorem 19.1 is not assumed of x, and, where (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$ at x, let (g, G) be the gradient that lemma reads there, the blocks of the measured families of the factors of $\mathbb{F}^{\mathrm{ref}}$ each known through an entrywise enclosure (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)), some of them read, besides, with a triple (L_c, t, W) of a matrix box and a core as in Definition "Factor enclosure of a block" (foundations/def:factor-enclosure), the block's *factor enclosure*, and every other datum exactly. Call a tuple $F = (F^f)_{f \in \mathbb{F}^{\mathrm{ref}}}$ of families, each F^f indexed as the measured family of factor f is, a *family of the replaced factors*, its elements $F_v^{(s)}$ being the products (1.5) of its blocks and of the blocks of the other factors of $\mathfrak{D}^{\mathrm{ref}}$; call it an *entrywise member* when each of its blocks is a Hermitian matrix that the entrywise enclosure of that block contains, and a *member* when, moreover, each of its blocks that is read with a factor enclosure is a member of it. Suppose

(H-FAC) each block of the companion's own family of the replaced factors that is read with a factor enclosure is a member of it, the triple being a factor enclosure of that block, so that the companion's family of the replaced factors is a member.

Call a block *decided* when every entry of its entrywise enclosure has centre and radius 0; or some entry's centre C_{il} and radius r_{il} satisfy $|C_{il}| > r_{il}$; or

its factor enclosure has no column; or its factor enclosure has $k \geq 1$ columns and there are G, ρ and ℓ as in Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii) with $\ell - \rho > 0$. Call a block *of fixed rank* when it is of one of three kinds: of the first kind when it is zero for every member; of the second kind when there are a matrix E of full row rank and, for every member, a positive definite $\widetilde{X}$ with the member's block equal to $E^{\dagger}\widetilde{X}E$, E the same for every member (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iv) gives such an E); and of the third kind when there are an integer $k \geq 1$, a matrix box of complex $k \times n$ matrices, n the dimension of the block, and a positive definite $k \times k$ matrix $\widetilde{X}$ and, for every member, a matrix E_* of rank k that the box contains, with the member's block equal to $E_*^{\dagger}\widetilde{X}E_*$ (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii) gives such a box, that of the adjoints of the factors, with $\widetilde{X}$ the core). Suppose that every block of a factor of $\mathbb{F}^{\mathrm{ref}}$ that is a factor of an element $F_v^{(s)}$ with (x', v) an outcome pair of a pass class of some $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $M_{x'}^{(\mathsf{t})} \neq 0$, no other block of which is zero for every member, is decided, and that every such block is of fixed rank at the pairs $(\hat{c}, (\mathsf{t}, s))$ that are not key-inert. Then:

(a) every member gives the same effective outcome sets on every joint sector, the same key-inert pairs and the same joint key-active index sets, those read off the enclosures;

(b) fix, for every block that the hypothesis asks to be of fixed rank, one kind it is of, with the data of that kind, and, where that kind is the third, at every member one matrix E_* as that kind asks, one for the block and not one for each element it is a factor of; below, the kind of a block is the one so fixed for it. At every pair $(\hat{c}, (\mathsf{t}, s))$ that is not key-inert, every element $F_v^{(s)}$ of an outcome pair (x', v) of a pass class of $\hat{c}$ with $M_{x'}^{(\mathsf{t})} \neq 0$ is, for every member, $E_v^{(s)\dagger}\widetilde{F}_v^{(s)}E_v^{(s)}$ with $\widetilde{F}_v^{(s)}$ positive definite and $E_v^{(s)}$ of full row rank, the case $F_v^{(s)} = 0$ included as $E_v^{(s)}$ with no rows, where $E_v^{(s)}$ is the same for every member if no block of $F_v^{(s)}$ is of the third kind or some block of it is zero for every member, and otherwise its rank is the same for every member and it is the tensor product, in the order of $\mathbb{F}$, of one matrix for each block of $F_v^{(s)}$: the matrix E_* fixed for the block at that member where the block is of the third kind, and a matrix the same for every member otherwise. For an element of the latter case, call its *tuple* $\beta(v)$ the list, in the order of $\mathbb{F}$, of its blocks of the third kind, each block $F_{v_f}^{f,(s_f)}$ named by the triple (f, s_f, v_f), so that two entries of tuples agree exactly when they name the same block, whatever their boxes; and for a tuple β let E_{β}^{box} be the tensor product, in that order, of the matrices E_* fixed for its

blocks, which at each member is one matrix for every element whose tuple is β. Moreover, for every pass class of $\hat{c}$ and any factor forms $M_x^{(\mathsf{t})} = E_x^{(\mathsf{t})\dagger} \widetilde{M}_x E_x^{(\mathsf{t})}$ of Alice's elements with $E_x^{(\mathsf{t})}$ of full row rank and $\widetilde{M}_x$ positive definite, the matrix $\widetilde{E}$ that stacks the matrices $E_x^{(\mathsf{t})} \otimes E_v^{(s)}$ over the class's outcome pairs (x, v) with $M_x^{(\mathsf{t})} \neq 0$, in a fixed order, is, for every member, $\widetilde{E} = \widetilde{B}\, C$ with $\widetilde{B}$ of full column rank and the same for every member, and C a stack of a matrix C_0 and, one for each tuple β of an element of the latter case in the class, of a matrix $(C_\beta \otimes E_\beta^{\mathrm{box}}) U_\beta$, with C_0, every C_β and every permutation matrix U_β the same for every member; where the $E_x^{(\mathsf{t})}$, the $E_v^{(s)}$ that are the same for every member and the matrices the same for every member of which the other $E_v^{(s)}$ are tensor products have entries in $\mathbb{Q}(i)$, $\widetilde{B}$, C_0 and every C_β can be taken with entries in $\mathbb{Q}(i)$;

(c) call a pair (F', F'') of families of the replaced factors, the first read by the statistics and the second by the deficits, *admissible* when every block of F'' that is a factor of an element $F_v^{(s)}$ of an outcome pair (x', v) of a pass class of $\hat{c}$ with $M_{x'}^{(\mathsf{t})} \neq 0$, at a pair $(\hat{c}, (\mathsf{t}, s))$ that is not key-inert, no other block of which is zero for every member, is a positive semidefinite matrix. For such a pair form, at x, $P_{\hat{c}}^J$ and every $\Gamma_{\hat{c}}^{(s')}$ with F', every deficit $f_{\hat{c},s}^J$ and every $\Delta_{\hat{c}}^{(s')}$ with F'', an outcome pair whose element of Alice's is zero, or whose element $F_v^{(s)}$ has a block that is zero for every member, contributing nothing to a branch map or to the statistics, over the key-inert pairs and the joint key-active index sets of (a), and $K_{\hat{c}}^J$ as $P_{\hat{c}}^J$ plus those deficits; where $q(\hat{c})$, $P_{\hat{c}}^J$ and $K_{\hat{c}}^J$ so formed are positive as (H-POS) of Theorem 19.1 asks, let $(g, G)[F', F'']$ be the pair given by (8.10) and (8.11) with these pieces, read as Lemma 21.1 reads them. Let $\mathcal{P}$ be a set of admissible pairs that contains (F, F) for every member F, such as the set of the admissible pairs of an entrywise member F' and a family F'' of the replaced factors each of whose blocks lies in its entrywise enclosure or is a member of its factor enclosure, the one or the other chosen block by block. Then a set that contains $P_{\hat{c}}^J$, or $K_{\hat{c}}^J$, formed with every pair of $\mathcal{P}$ contains that of $\mathfrak{D}^{\mathrm{ref}}$ at x; and where (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$, enclosures $(\hat{g}, \delta)$ and (M_p, ρ_p) that hold, as in parts (ii) and (iii) of Lemma 21.1, at $(g, G)[F', F'']$ for every pair of $\mathcal{P}$ at which it is formed hold at the gradient of $\mathfrak{D}^{\mathrm{ref}}$, and parts (i)–(iii) of that lemma then give $\max\{0, b^{\mathrm{A}} + b^{\mathrm{F}}\} \leq \underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}]$;

(d) in the setting of Corollary 19.6, with a dropped set $\mathcal{D}$, (a)–(c) hold, the hypotheses read the same way, with $O_\alpha^{J,\mathcal{D}}$ in place of O_α^J, (H-POS$^{\mathcal{D}}$) in place of (H-POS), its gradient of (H-NOPERT$^{\mathcal{D}}$) in place of (g, G) and $K_{\hat{c}}^{J,\mathcal{D}}$ in place of $K_{\hat{c}}^J$, the kept pairs in place of the pairs that are not key-inert, part (iv) of Lemma 21.1 in place of its parts (i)–(iii), and $\underline{h}_\alpha^{\mathcal{D}}$

in place of $\underline{h}_\alpha$.

Corollary 21.3 (Key length from the certified bracket). Let α, $\mathfrak{D}$ and $\mathcal{S}_{\mathrm{acc}}$ be as in Lemma 21.1, let $\mathfrak{I}$ be a finite non-empty set of points $x_t = (\mathbf{q}_t, J_t)$ each satisfying the hypotheses of that lemma, and let b_t^{A} and b_t^{F} be real numbers with (19.3) at the gradient at x_t, as parts (ii) and (iii) of that lemma form them. Put

$$\underline{h}_\alpha^{\mathrm{cert}} := \max_{t \in \mathfrak{I}} \left(b_t^{\mathrm{A}} + b_t^{\mathrm{F}} \right). \tag{21.3}$$

(i) $\underline{h}_\alpha^{\mathrm{cert}}$ is the value of (19.2) at the points of $\mathfrak{I}$ with both pairings replaced as Theorem 19.1 (iv) replaces them, and $\max\{0, \underline{h}_\alpha^{\mathrm{cert}}\} \le \underline{h}_\alpha$ in exact real arithmetic.

(ii) Where $\mathfrak{D}$ satisfies Definition 1.2 and the hypotheses of Corollary 19.5 hold, its conclusions hold with $\underline{h}_\alpha^{\mathrm{cert}}$ in place of $\underline{h}_\alpha^{\mathrm{FO}}$ in (19.7): the key length so formed is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct, with the composed-security and monotonicity conclusions of that corollary, the secrecy and security statements at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals.

(iii) Under a dropped set, with the lemma's part (iv), $\max\{0, \underline{h}_\alpha^{\mathrm{cert}}\} \le \underline{h}_\alpha^{\mathcal{D}} \le \underline{h}_\alpha$; and where $\mathfrak{D}$ satisfies Definition 1.2 and the hypotheses of Corollary 10.18 other than (10.17) hold, the conclusions of that corollary hold at $h = \underline{h}_\alpha^{\mathrm{cert}}$.

Corollary 21.4 (Key length on a reference companion from the certified bracket)**.** Let $\mathfrak{D}^{\mathrm{ref}}$ be a compiled reference companion of Proposition 12.1, and let α, $\mathcal{S}_{\mathrm{acc}}$ and $\mathfrak{I}$, with its points x_t, be as in Corollary 21.3 with $\mathfrak{D} := \mathfrak{D}^{\mathrm{ref}}$. At each $t \in \mathfrak{I}$ let b_t^{A} and b_t^{F} be real numbers formed in one of two ways: as in that corollary, with (19.3) at the gradient of $\mathfrak{D}^{\mathrm{ref}}$ at x_t; or, where blocks of the measured family of $\mathfrak{D}^{\mathrm{ref}}$ are known through enclosures, entrywise or by factor enclosures, as parts (ii) and (iii) of Lemma 21.1 form them from enclosures $(\hat{g}, \delta)$ and (M_p, ρ_p) that hold at every pair of a set $\mathcal{P}$ as in Lemma 21.2 (c), at a point x_t at which the hypotheses of that lemma hold and (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$, the blocks it reads with factor enclosures being read with those of Corollary 12.2, under that corollary's hypotheses (for each replaced factor whose blocks are so read, the rule of (I1) of the reduction instance of its last link a non-negative combination, and the blocks of its handed reference carrying those factor enclosures), so that its conclusion is (H-FAC). Put $\underline{h}_\alpha^{\mathrm{cert}} := \max_{t \in \mathfrak{I}} (b_t^{\mathrm{A}} + b_t^{\mathrm{F}})$, as in (21.3).

(i) Where $\mathfrak{D}^{\mathrm{ref}}$ satisfies Definition 1.2 and the hypotheses of Corollary 19.5 hold, the conclusions of Corollary 19.7 (iii) hold with $\underline{h}_\alpha^{\mathrm{cert}}$ in place of $\underline{h}_\alpha^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$, for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

(ii) In the setting of Corollary 19.6, with a dropped set $\mathcal{D}$ and with the numbers formed at the gradient of the dropped objective and, in the second way, with part (d) of Lemma 21.2 in place of its part (c): $\max\{0, \underline{h}_\alpha^{\mathrm{cert}}\} \le \underline{h}_\alpha^{\mathcal{D}}[\mathfrak{D}^{\mathrm{ref}}] \le \underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}]$, and where $\mathfrak{D}^{\mathrm{ref}}$ satisfies Definition 1.2 and the hypotheses of Corollary 10.18 other than (10.17) hold, the conclusions of Corollary 10.18 hold at $h = \underline{h}_\alpha^{\mathrm{cert}}$, for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

22 The two minimisations and the arithmetic

Assumption 22.1 **[A-LMO]** (Exact linear minimisation over the feasible set)**.** (i) At each iterate x_t at which the engine accumulates a bracket of the first-order bound, the block list S_t it consumes, which is the backend's return after any positivity repair the adapter applies to it, satisfies the feasible-side inequality of (18.2) (Definition 18.1), its minimum taken over the *exact* $\mathcal{W}^J$ and not over the relaxed feasible set, at G_t, the block part of the gradient (g_t, G_t) the bracket consumes at x_t, of the objective, or of that of a dropped program.

24 Requirements: reported numbers and withdrawal

Requirement 24.1 (Withdrawal of the first-order bound). An iterate at which any of the following holds is not a member of $\mathfrak{I}$: it contributes no bracket to the maximum in (19.2), and the engine records the machine-readable reason.

(i) (H-POS) fails (`gradient_hypothesis_positivity`);

(ii) (H-SUPP) fails (`gradient_hypothesis_support`);

(iii) a gradient perturbation was applied, that is, (H-NOPERT) fails (`gradient_perturbation_applied`);

(iv) the point consumed at that iterate fails one of the three checks of Requirement 25.1 and is not repairable under Requirement 25.4, or its acceptance-side component $\mathbf{q}_t$ was not formed on the exact island under (α) and (β) of Remark 22.2, so that its membership in the product $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is not established (`iterate_not_feasible` in either case).

Neither the point the oracle returned at that iterate nor its status removes the iterate from $\mathfrak{I}$. If no iterate is left, $\mathfrak{I}$ is empty: the engine withdraws $\underline{h}_\alpha^{\mathrm{FO}}$ and the tier falls to the estimate, with a reason from the same machine-readable vocabulary.

Requirement 24.2 (Measurement of the returned point). Every run applies the three checks of Requirement 25.1 once more, at the end of the run, to the point it returned: per-block positivity at exactly 0, the coupling residual at `iterate_coupling` and each $\bowtie_k$ residual at `constraint_rows`. It records whether that point passes all three, together with the three quantities measured, in `optimizer_certificate`. The measurement enters neither the first-order bound, nor the tier, nor the key length.

Requirement 24.3 (Clamping of the first-order bound at zero). The engine passes $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\}$ to the key length of Corollary 19.5, and records the clamp: it emits the boolean `first_order_clamped` beside `h_lb`, and `h_lb` carries the unclamped value.

Requirement 24.4 (Refusal of a negative estimate). The engine refuses a run whose computed estimate, the objective at the point the run's search on the declaration it searches returned at an order the run evaluates, is negative, with the machine-readable reason `negative_objective` naming the value and the point, and never clamps the estimate to zero. A bare exception is not a conforming implementation of this rule; the reason must be readable by the caller. The rule binds that search alone: the polish of Requirement 24.7 (i) and the solve of Requirement 16.2 (i) form no estimate of the run, and a negative objective value of theirs refuses nothing.

Requirement 24.5 (The recorded difference between the bound and the estimate). The engine records the difference $\underline{h}_\alpha^{\mathrm{FO}}$ minus the reported estimate, with its sign, as a diagnostic, and asserts nothing of it.

Requirement 24.6 (The program's own objective, with no imported penalty). The engine imports no relative-entropy penalty onto O_α^J from any source: it subtracts from it no penalty term taken from a von-Neumann-entropy key-rate formulation.

Requirement 24.7 (The first-order bound evaluated on the reference companion). Where the engine has admitted a compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ beside the declaration $\mathfrak{D}$ it searched on (Requirement 14.3), it forms the run's first-order number as follows, after the search on $\mathfrak{D}$ has ended.

(i) *The points.* It evaluates the points of two sets. The first holds at most five points of the search on $\mathfrak{D}$: the at most four members of the index set of $\mathfrak{D}$'s bound (Requirement 24.1) with the largest brackets on $\mathfrak{D}$, ties broken by the order of the run, and the last point the search assessed; where $\mathfrak{D}$'s index set is empty, the last point alone. The second holds the points kept by the *polish*, a search on the Choi

program of $\mathfrak{D}^{\mathrm{ref}}$ that continues the search on $\mathfrak{D}$ after it has ended. The polish reads the objective, the gradient, the maps and the entries of $\mathfrak{D}^{\mathrm{ref}}$ (Requirement 14.3 (i)) and no map generated from $\mathfrak{D}$'s measured family; it starts at the point of the first set with the largest bracket on $\mathfrak{D}$, at the last point where $\mathfrak{D}$'s index set is empty, and forms its acceptance-side iterate as Requirement 26.2 requires, from that point's exact acceptance-side component. Where that component was not formed on the exact island (Requirement 26.2), the polish does not run, the second set is empty and the record names that reason. It assesses each point it consumes by the gate of (ii) and keeps the at most four of them that contribute with the largest brackets on $\mathfrak{D}^{\mathrm{ref}}$, ties broken by its own order, and the last point it assessed; a polish that assesses no point keeps none. It runs within the share `reference_polish_share` of the wall budget of one evaluation of the run, where the run has one, set aside for it before the search on $\mathfrak{D}$ begins, and ends where that share is spent, where an iteration cap it records is reached or where its own stopping rule holds, whichever comes first; a polish that fails or is cut short keeps, as above, the points it assessed until then. The share so set aside shortens the search on $\mathfrak{D}$, which bears on the run's estimate, on $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, on a number certified on $\mathfrak{D}$ in the fallback of Requirement 28.1 (i), and on the soundness of none of them. Its objective values are not the run's estimate: none is reported, and none refuses the run (Requirement 24.4 reads the estimate of the search on $\mathfrak{D}$ alone). Each point $x_t = (\mathbf{q}_t, J_t)$ of either set is taken as the search or the polish consumed it, $\mathbf{q}_t$ included, and is a point of the Choi program of $\mathfrak{D}^{\mathrm{ref}}$ (Corollary 19.7 (ii)).

(ii) *The gate.* At each such point it forms the bracket of (19.2) for $\mathfrak{D}^{\mathrm{ref}}$: the objective of $\mathfrak{D}^{\mathrm{ref}}$ at x_t, its gradient at x_t with no perturbation, the acceptance-side minimisation of Requirements 26.1 and 26.2, and the feasible side of Requirement 27.1, from a fresh feasible-side oracle call on the linear form of that gradient over $\mathfrak{D}^{\mathrm{ref}}$'s entries. It decides (H-POS) and (H-SUPP) on $\mathfrak{D}^{\mathrm{ref}}$'s maps (Requirement 25.2) and makes every check of Requirement 25.1 against $\mathfrak{D}^{\mathrm{ref}}$'s entries, on the point; a point at which Requirement 24.1 withdraws it on these terms contributes no bracket, with that requirement's reason.

(iii) *The number.* Where some evaluated point contributes and the bracket is not withdrawn by (iv), the run's first-order number is $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$, the largest bracket so formed, its clamp and key length are those of Requirement 24.3 and Corollary 19.5 at it, with the parameters of $\mathfrak{D}$'s run, and its tier is *first-order*. The bound on $\mathfrak{D}$, $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, is recorded beside it with its brackets as a diagnostic, and no key length is formed from it. The difference $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}] - \underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$ is recorded with its sign, as

a diagnostic, beside the difference of Requirement 24.5, which is taken against the estimate of $\mathfrak{D}$. A point of the second set that contributes counts as any evaluated point does: where no point of the first set contributes, it makes this item apply in place of the fallback of (iv), so that the run's first-order number moves from $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$ to the companion.

(iv) *The bracket withdrawn.* The bracket on the companion is withdrawn for one of two causes: no evaluated point contributes; or the certified evaluation finds the two enclosures of a block the companion carries inconsistent, the cause `factor_enclosure_inconsistent` (Requirement 28.1 (iv)). An inconsistency shows that a module carrying the enclosures is defective, and the first-order number on the companion reads its entrywise enclosure, which may be the defective one; the second cause is decided after that number was formed, and the run then forms its first-order number again as this item says, before any key length is reported. Where the bracket is withdrawn, the run's first-order number is $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, formed as without a companion, with the reason `reference_bracket_withdrawn`, its cause and the reasons of every evaluated point; no block is then evaluated off its snap. The record names in the group *reference bracket withdrawn*, with the cause (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)), every moved block on which a declaring instance whose link the companion replaces handed the reference and every moved class of each snapped emission of such a link whose normalizer the companion bypasses; it names as it does without a companion every other moved block or class that rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), so that a moved block on which a declaring instance whose link the companion does not replace handed a reference rests in the group *reference bracket not evaluated* (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(c)). A run that declares the strict option `strict_snaps` refuses there instead and reports no key length (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)).

(v) *The blocks.* Where (iii) applies and the key length the run reports is not formed in the fallback of Requirement 28.1 (i), the record of the run names as evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)) every moved block of the snap of each declaring instance whose link the companion replaces, on which the companion carries the handed reference, and every data sector of each snapped emission of such a link that holds a moved class and whose normalizer the compan-

ion bypasses; a kept block or sector is named as kept and is not counted among them. It names every moved block on which the companion carries the snapped block as not so evaluated, so that it rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)), and a run that declares `strict_snaps` refuses there (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). Where (iii) applies but the key length the run reports is formed from a number certified on $\mathfrak{D}$ in the fallback of Requirement 28.1 (i), no number formed on the companion reaches that key length, and the number it is formed from reads the snapped blocks: the moved blocks on which the companion carries the handed reference and the data sectors whose normalizer it bypasses are then not evaluated off their snap, condition (ii) of Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) failing for them. The record names them as resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) in the group *reference bracket withdrawn* with the cause `companion_not_certified` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)), beside the moved blocks on which the companion carries the snapped block, which rest as above. The run decides this after the certified evaluation and before any key length is reported, and no block is then named as evaluated off its snap. A run that declares `strict_snaps` forms no such number (Requirement 28.1 (i)).

(vi) *The record.* The record carries, per evaluated point, the set of (i) it belongs to, its bracket on $\mathfrak{D}^{\mathrm{ref}}$ and the withdrawal reasons there, and for a point of the first set its bracket on $\mathfrak{D}$; for the polish, whether it ran and, where it did not, why, the point it started from, the share of the wall budget set aside for it and the wall time it took, its iteration cap, the number of points it assessed and the cause it stopped on; the measurements of Requirement 14.3 (iii), the largest radius of the companion's enclosure among them, which bear on the arithmetic of Remark 22.4 and on nothing else; and, per block, whether the companion carries the handed reference or the snapped block.

A run with no admitted companion forms its first-order number on $\mathfrak{D}$ alone, as the other requirements of this submodule state; a moved block on which a reference was handed, and a moved class of an emission beside which a companion that bypasses its normalizer was formed, then rest in the group *reference bracket not evaluated* (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)).

25 Requirements: checks and repairs of the iterates

Requirement 25.1 (Feasibility checks at each iterate). On the point it consumes at each iterate, after any repair, the engine decides every condition defining $\mathcal{W}^J$ in (7.9), as the relaxed feasible set relaxes it, all three and none of them optional:

(i) per-block positive semidefiniteness, at exactly 0 and at no window (`iterate_positivity`);

(ii) the coupling condition, its residual $\| \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0} \|_{\max}$, or, in untrusted-source mode, where the coupling condition is the normalisation $\sum_s \operatorname{tr} \omega_s = 1$ (Definition 7.10 (b)), its residual $| \sum_s \operatorname{tr} \omega_s - 1 |$, at the declared window `iterate_coupling`;

(iii) the declared constraint rows $\bowtie_k$, each residual at the declared window `constraint_rows`.

No fourth check is required, and none is required of the point the oracle returned at that iterate. The ε at which a run reads the relaxed feasible set is the larger of the two declared windows of (ii) and (iii). No declared tolerance (Definition "The declared tolerance and the defects it bounds" (foundations/def:tolerance)) is that number: the windows govern conditions checked on an iterate and never on a declaration.

Requirement 25.2 (Rank test for maximal support). The engine decides (H-SUPP) as the rank equality (23.1), both sides computed and both read with the one declared threshold `support_cutoff`:

(i) the left side is the count of eigenvalues of $X_{\hat{c},s}$ above `support_cutoff`, read relative to that block's own spectral radius;

(ii) the right side is $\operatorname{rank} \mathbf{K}_{\hat{c},s}$ of (8.2), read at the square root of `support_cutoff` relative to that matrix's largest singular value, a window derived from the same threshold and not declared beside it; equivalently the count of eigenvalues of $\mathbf{K}_{\hat{c},s} \mathbf{K}_{\hat{c},s}^{\dagger}$ above `support_cutoff` relative to its spectral radius. It is computed once per $(\hat{c}, s)$ from the declaration and W_τ, and never from an iterate.

Requirement 25.3 (Repair of an oracle return). The engine may reject outright an oracle return a block of which has a negative spectrum, or project that block onto the positive semidefinite cone within the declared window `oracle_repair_window`, a number of its own, declared beside τ_{num} and never as an instance of it. It records every such repair, with the block, the iterate

and the measured depth, in `optimizer_certificate`, kept apart there from the iterate repairs of Requirement 25.4, and uses the repaired return only as a step of the search and in the diagnostic pairing of Requirement 27.1 (vi), never in a bracket and never as a bound on anything.

Requirement 25.4 (Repair of a formed iterate). When a block of the point consumed at iterate t fails the positivity check of Requirement 25.1, the engine either rejects it, in which case t contributes no bracket and is not a member of $\mathfrak{I}$, or projects it onto the positive semidefinite cone, provided its measured depth is within the declared window `iterate_repair_window`, a number of its own, declared beside τ_{num} and never as an instance of it. No other repair is made on an iterate. A projection is governed by three rules:

(i) the repaired point is x_t, the point at which the objective, the gradient and the pairings with x_t of (19.2) are evaluated;

(ii) the coupling and $\bowtie_k$ residuals of Requirement 25.1 are measured again on the repaired point and never carried over from the point before it, and a residual that then fails its window refuses the iterate;

(iii) every repair is recorded, with the block, the iterate and the measured depth, in `optimizer_certificate`, beside the oracle-return repairs of Requirement 25.3 and distinguished from them.

Requirement 25.5 (Coupling residual and record per Alice sector). On a declaration in the general format the engine decides the coupling condition of the relaxed feasible set of Definition 17.1 per Alice sector, one coupling residual per Alice sector, each at the declared window `iterate_coupling`; and the record names the frame and the row form per Alice sector (Requirements 15.3 and 15.6).

26 Requirements: the acceptance side

Requirement 26.1 (The admitted acceptance geometries). The engine solves the acceptance-side minimisation exactly for the two geometries of Remark 22.2, a singleton (a) and a polytope (b), and admits an acceptance set in no other geometry: a declaration whose $\mathcal{S}_{\mathrm{acc}}$ fits neither is refused at admission, before any solve, and no first-order bound is formed for it.

Requirement 26.2 (Exact formation of the acceptance-side iterate). The engine forms the acceptance-side iterate on the exact island, under conditions (α) and (β) of Remark 22.2: it initialises it at an exact rational point of $\mathcal{S}_{\mathrm{acc}}$, and updates it only by convex combinations, with exact rational weights in $[0, 1]$, of points already established to lie in $\mathcal{S}_{\mathrm{acc}}$. An iterate whose

acceptance-side component was not so formed has not met the membership hypothesis and contributes no bracket to $\underline{h}_{\alpha}^{\mathrm{FO}}$, with the reason of Requirement 24.1 (iv). The run's record states in `optimizer_certificate`, of the point consumed at each iterate, whether its acceptance-side component was so formed.

27 Requirements: the feasible side

Requirement 27.1 (The verified feasible side of every bracket). At every iterate t at which the engine forms a bracket of (19.2), on the declaration it searched and on an admitted reference companion alike, it uses in place of the pairing $\langle G_t, S_t \rangle$ the number b_t^{F} of Theorem 19.1 (iv), formed as follows, and never the pairing with a point the backend returned, a dual objective the backend reported, or the backend's status.

(i) *The groups and the data.* b_t^{F} is a sum over the groups of the program, one per Alice sector in trusted-source mode and one in untrusted-source mode, read as Lemma 20.1 (ii) reads them. Its G is the exact Hermitian part of the gradient blocks as the engine computed them at x_t, the same gradient whose acceptance-side part the bracket uses; its coupling is the exact coupling condition of the group; its rows are the maps of Lemma 20.1 (ii), in the conjugated or frame-free form the engine imposes, built from the declared data taken exactly: $\Theta_k^{(\mathsf{t})}$ as the declared block or the exact product $c_k \tau_A^{(\mathsf{t})}$, the lifted operators, and $D_k^{(s_f)} = \hat{O}_k^{(s_f)} - c_k \mathbb{1}$ formed exactly, all Hermitian by exact admissibility; and in trusted-source mode the frame in those maps is the binary64 $\widetilde{W}_{\mathsf{t}}$ the engine constructed for W_{t}, read as the dyadic matrix it is. Every binary64 number is read as the dyadic rational it is.

(ii) *The multipliers.* Y and every Λ_r are read off the dual the backend returned for that same solve, as the operators that the multipliers of the coupling condition and of row r represent in the encoding the oracle used, each replaced by its exact Hermitian part. On a group with a one-dimensional coupling and no rows, and on any group whose solve returned no finite dual, whatever its status, $Y = 0$ and every $\Lambda_r = 0$; on an Alice sector for which (iv) forms no ε_{t}, every Λ_r of its rows is 0 and Y is kept. The multiplier of an inequality row is replaced by $\Lambda_r + \mu_r \mathbb{1}$, with μ_r a dyadic rational at least $\max\{0, -\ell_r\}$ and ℓ_r a lower bound on $\lambda_{\min}(\Lambda_r)$ given by Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), so that $\Lambda_r \succeq 0$ holds exactly.

(iii) *The eigenvalue.* The group's λ is the least, over the blocks of the group, of the lower bounds on $\lambda_{\min}(R_p)$ that Lemma "Least-eigenvalue lower

bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) gives at a matrix V of the engine's choosing, with R_p formed from the rows of (i). The bound holds at every V that lemma admits, the identity ($\theta = 0$) always among them, and the choice bears only on how close λ lies to $\min_p \lambda_{\min}(R_p)$.

(iv) *The frame.* For each Alice sector the engine forms a dyadic $\varepsilon_{\mathsf{t}} \geq \left\|\widetilde{W}_{\mathsf{t}} - W_{\mathsf{t}}\right\|$, established in exact rational arithmetic, where it can: 0 where $\widetilde{W}_{\mathsf{t}} = W_{\mathsf{t}}$ is decided exactly; and where $\tau_A^{(\mathsf{t})}$ is diagonal, the frame is the full frame or the diagonal frame (Definition 7.2), and $\widetilde{W}_{\mathsf{t}}$ vanishes exactly wherever W_{t} does, so that both have at most one non-zero entry in each row and each column and $\left\|\widetilde{W}_{\mathsf{t}} - W_{\mathsf{t}}\right\|$ is the largest modulus of their entrywise difference: a dyadic at least $|\widetilde{w} - \sqrt{\tau_{ii}}|$ at every entry where W_{t} carries $\sqrt{\tau_{ii}}$, $\widetilde{w}$ being the entry of $\widetilde{W}_{\mathsf{t}}$ there (for a real $\widetilde{w} > 0$, $|\widetilde{w}^2 - \tau_{ii}|/\widetilde{w}$ is one, since $|\widetilde{w} - \sqrt{\tau_{ii}}| = |\widetilde{w}^2 - \tau_{ii}|/(\widetilde{w} + \sqrt{\tau_{ii}})$). Where every row of the sector is imposed in frame-free form, no deduction is formed: those rows read no frame, every κ_r is 0, and the sector's bound is that of Lemma 20.1 (i). In every other case no ε_{t} is formed. Where it is formed, the sector's bound is that of Lemma 20.1 (iv), $b(G_{\mathsf{t}}; Y, \Lambda, \widetilde{\lambda}) - \sum_r \kappa_r$, with each κ_r formed from dyadic upper bounds on $\left\|\widetilde{W}_{\mathsf{t}}\right\|$, $\|\Lambda_r\|$ and $\|O_{r,s}\|$ (for instance on their Frobenius norms, by an integer square root rounded up). Where it is not, the row multipliers are zero by (ii) and the sector's bound is that of Lemma 20.1 (i), which then reads no row. In untrusted-source mode there is no frame and no deduction.

(v) *The arithmetic.* The residuals, the shifted multipliers, every eigenvalue and norm bound, every deduction and b_t^{F} are computed in exact rational arithmetic on those dyadic data, as a certification of the exact island (Definition "The exact island" (foundations/def:exact-island) (iii)), the backend's dual entering it only as a candidate (Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island)). b_t^{F} enters the binary64 bracket rounded toward $-\infty$, the direction that weakens a lower bound, as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted one.

(vi) *The record.* Per bracket the engine records b_t^{F} (`feasible_side_bound`); the sum of the deductions κ_r (`frame_deduction`) and the sum of the bounds on $\|\Lambda_r\|$ it used (`row_multiplier_norm`); where the backend returned a point, the pairing $\langle G_t, S_t\rangle$ with it (`oracle_pairing`) and the difference $\langle G_t, S_t\rangle - b_t^{\mathrm{F}}$ with its sign (`dual_gap`); and, per group, whether its multipliers were the backend's, zero, or the

backend's with the row multipliers set to zero (`feasible_side_multipliers`). None of these but b_t^{F} enters any number.

Requirement 27.2 (The primal pairing in the first-order bracket). In the bracket of (19.2) the engine uses the primal pairing $\langle G_t, S_t \rangle$ at the returned point S_t and nothing else; in particular it never uses a dual objective reported by the backend in its place.

28 Requirements: the certified evaluation

Requirement 28.1 (The certified evaluation of the first-order bound). On every run that forms a first-order number, unless it sets the option `certified_evaluation` to `off`, its default being `on`, the engine forms a certified number as follows, after the first-order number is formed and before a key length is reported.

(i) *The declaration and the points.* It works on the declaration the run's first-order number is formed on, at the points that number is formed from: on an admitted reference companion where Requirement 24.7 (iii) applies, at the at most five of the points part (i) of that requirement evaluates, of the search and of the polish together, that contribute there with the largest brackets on the companion, a point that is in both sets counted once and ties broken by the sequence in which the search and then the polish consumed them, each with the solve part (ii) of it makes there; otherwise on the declaration the run searched, at the at most four members of the index set of its bound (Requirement 24.1) with the largest brackets, ties broken by the order of the run, and the last point the search assessed, each with the search's solve there. *The fallback*: where it has worked on a companion by (ii)–(v) and certified no point there, it works once more, by (ii)–(v), on the declaration the run searched, at the points and with the solves of the second case, unless the run declares the strict option `strict_snaps` of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv); the companion's results are kept for the record (ix). The fallback runs nowhere else: not where a point of the companion is certified, not where no companion's bracket counts, not where the companion's bracket is withdrawn by (iv), and not where the arithmetic the evaluation needs is not available. Where the companion's bracket is withdrawn by (iv), the run's first-order number is formed on the declaration the run searched (Requirement 24.7 (iv)), and the evaluation works once more there, by the second case above, as on any run whose first-order number is formed there. Each point is taken exactly: its acceptance-side component

as the exact rational iterate formed on the exact island (Requirement 26.2), a point whose component was not so formed being not certified (`acceptance_iterate_not_exact`), and each Choi block as the exact Hermitian part of the dyadic matrix its binary64 entries are.

(ii) *The data.* Every quantity is formed from the data of the declaration it works on, taken exactly, every binary64 number as the dyadic rational it is and the acceptance set as its exact rational data, and, for the blocks a reference companion replaces, from the exact centres and radii of the entrywise enclosures its compilation carries, recentred as Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure) recentres them, and from the exact centres and radii of the matrix box and the exact core of every factor enclosure its compilation carries (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)). No map, frame, gradient, support, rank, eigenvalue or check value the engine computed in binary64 enters it. In trusted-source mode the frame of each Alice sector is W_{t} of Theorem 7.5 for the support frame the engine constructed (Definition 7.2), held as an enclosure formed by a rigorously enclosed computation; where the engine forms none, every point is not certified (`frame_not_enclosed`).

(iii) *The maps in factor form.* At every joint pair that is not key-inert, every kept one under a dropped set, the branch operators and their pinched blocks are formed through factors S'_{xv} with the Gram matrices of Lemma 8.6 (v), built from factor forms $E^\dagger \widetilde{X} E$ of the elements of Alice's measurement or ensemble and of the measured family that enter them, $\widetilde{X}$ positive definite and E either an exact matrix of full row rank (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iii)) or, on a block whose rank the factor rule of (iv) fixes, the adjoint $L^\dagger$ of a factor L of its factor enclosure, held through the box of the adjoints, of full row rank at every factor (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii)), with $\widetilde{X}$ the exact core: on exact data from an exact factorisation, on a block a companion carries with a positive radius by the rule of (iv). In the general format an element of the measured family is the tensor product of its blocks, one per measured factor, and its factor form is the tensor product of theirs (Lemma 21.2 (b)), E and $\widetilde{X}$ the tensor products of the blocks' E and $\widetilde{X}$, however many of its blocks a companion replaces and however many of them are held through boxes: E is then held through an enclosure of the Kronecker product of their enclosures, which contains it at every member, and has full row rank at every member, $EE^\dagger$ being the tensor product of the blocks' $EE^\dagger$, each positive definite. No rank

is read off a computed spectrum, and a matrix function applied to an operator A, a branch operator X or a pinched block $(Z_j \otimes \mathbb{1})X^p(Z_j \otimes \mathbb{1})$ of one, acts on its compression $Q^\dagger AQ$ by an isometry Q, held through an enclosure, as follows.

(a) *Containment.* The range of Q is established to contain $\operatorname{supp} A$, exactly or by an argument on the factors and the enclosures that holds at every value they enclose. The range of the side-by-side listing of the pair's pass-class factors contains the support of every branch operator they form, and the pinching projector applied to a subspace containing $\operatorname{supp} X$ contains the support of the pinched block (Lemma 8.6 (iv)(a)). Where the listing factors as $T_{\mathrm{out}}\widetilde{B}\,T_{\mathrm{in}}$, with $\widetilde{B}$ an exact matrix of full column rank, formed exactly, T_{out} the maps applied after it, certified injective by a Gram matrix certified positive definite, and T_{in} the maps applied before it, an orthonormal basis of $\operatorname{ran}(T_{\mathrm{out}}\widetilde{B})$ is established from $\widetilde{B}$. On exact data $\widetilde{B}$ comes from an exact rank factorisation of the exact matrix of the factors of the pass classes, and T_{in} holds the rest of that factorisation and the frame $W_{\mathsf{t}} \otimes \mathbb{1}$ in the Choi form. Where an element is held through the box of the adjoints of the factors of one or more of its blocks, $\widetilde{B}$ is, for each pass class, the exact factor of Lemma 21.2 (b), formed by exact rank factorisations of the rows whose factors are exact and, per tuple of the blocks held through boxes, of the stack of Alice's factors tensored with the exact factors of the elements whose blocks held through boxes are that tuple, the kind that lemma fixes for a block being the third for a block that the factor rule of (iv)(b) decides not zero, fixing its rank, the first for a block decided zero, and the second otherwise, an exact block included. The rows of such elements are grouped by their tuples, each block named by its factor, its sector and its letter, and never by the equality of their boxes: elements whose tuples name the same blocks share a group, whatever their boxes, and two distinct blocks with equal boxes are distinct entries of the tuples. Where the form the engine holds of such an element is not of this shape (its parts and the blocks naming them differing in number, a split into exact and boxed parts held where no block of the element is held through a box or missing where one is, or one tuple of blocks at two positions or of two dimensions), no claim admits it, and every point is not certified (`boxed_form_inconsistent`), a defect of the engine and not of the declaration. T_{in} holds that lemma's matrix C, the stack of C_0 and of each tuple's $(C_\beta \otimes E_\beta^{\mathrm{box}})U_\beta$, the columns of each such block permuted by its own U_β and by no other permutation, which lies in an enclosure containing it at every member, and the frame.

That range contains the range of the listing at every member, and equals it at every member where the product of the maps in T_{in} is established to have full row rank at every member; otherwise the basis spans a subspace containing the range, which (a) admits and the floor of (b) decides. T_{out} is block diagonal over the outcome pairs, so the pinching projectors commute with it, and $(Z_j \otimes \mathbb{1})\operatorname{ran}(T_{\text{out}}\widetilde{B}) = \operatorname{ran}\big(T_{\text{out}}(Z_j \otimes \mathbb{1})\widetilde{B}\big)$, whose basis comes in the same way from an exact rank factorisation of $(Z_j \otimes \mathbb{1})\widetilde{B}$. A point at which containment is not established is not certified (`support_not_established`).

(b) *The floor.* A positive lower bound on $\lambda_{\min}(Q^\dagger A Q)$ is then established, over the enclosures at every value they contain, which with (a) proves $\operatorname{ran} Q = \operatorname{supp} A$ (Lemma 8.6 (iv)(b)) at every member and makes the compression that of part (iv)(c) of that lemma. A point at which it is not is not certified (`floor_not_certified`).

(iv) *The rank rule over an enclosure.* A block a companion carries with a positive radius is decided, and its rank fixed, by the first of two rules that does so.

(a) *The entrywise rule.* The block is decided when every entry of its entrywise enclosure has centre and radius 0, then zero, or some entry's centre exceeds its radius in modulus, then not zero. Its factor form is that of Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iv): the rows and columns whose every entry has centre and radius 0 are dropped, and the rest must have a positive lower bound on its least eigenvalue over the enclosure.

(b) *The factor rule*, where the compilation carries a factor enclosure of the block and the entrywise rule leaves the block undecided or fixes no rank for it. With no column the block is decided zero (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (i)). With $k \geq 1$ columns and a floor of part (ii) of that corollary established, G and ρ formed as its part (iii) forms them or by a rigorously enclosed computation of $L^\dagger L$ over the box, the block is decided not zero and its factor form is $L^\dagger$, held through the box of the adjoints, with the core as $\widetilde{X}$.

(c) *The cross-check*, wherever the compilation carries a factor enclosure of the block, whichever rule fixes its rank. The engine encloses $LWL^\dagger$ entry by entry over the box of the factor enclosure, in exact rational arithmetic or by a rigorously enclosed computation, so that the enclosure contains $LWL^\dagger$ for every factor L, every member of the factor enclosure; and it decides, as (b) decides them, whether the factor enclosure has no column and, where

it has columns, whether a floor of part (ii) of Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) is established.

A block that neither rule decides is undecided, and a block whose rank neither fixes has no certified rank. The two enclosures of a block are *inconsistent* where the enclosure of $LWL^{\dagger}$ of (c) and the block's entrywise enclosure are disjoint at some entry, or where one of the two decides the block zero and the other decides it not zero, as (a) and (c) decide them. The companion's own block lies in its entrywise enclosure and, by (H-FAC) of Lemma 21.2, is a member of its factor enclosure, so it lies in both enclosures and both decisions hold of it: an inconsistency shows that one of the two was not formed or carried as the claims that establish it say. Where the two enclosures of a block are inconsistent, the factor rule is set aside for the block, every point of the companion is not certified (`factor_enclosure_inconsistent`), whatever joint pairs the block is a factor at, key-inert or not, and whichever rule would fix its rank, and the companion's bracket is withdrawn with that cause (Requirement 24.7 (iv)), its first-order number reading the entrywise enclosure, which may be the defective one. On a recentred enclosure, the entrywise rule leaves a block undecided exactly when its centre vanishes and some radius does not, its enclosure then deciding neither that it vanishes nor that it does not. Where an undecided block is a factor of an element of an outcome pair of a pass class of a key-live acceptance value whose element of Alice's is not zero, or a block with no certified rank is such a factor at a joint pair that is not key-inert, every point is not certified (`rank_not_certified`), save where another block of that element is decided zero, an exact block that is zero or a block a rule decides zero: the element is then zero for every member, whatever its other blocks. What the rule establishes is the hypothesis of Lemma 21.2 that such blocks are decided and of fixed rank, in the sense of that lemma; its hypothesis (H-FAC), that the companion's block is a member of the factor enclosure carried with it, is established by the claims of the modules that form and carry that enclosure, and decided by no rule here.

(v) *What is established at a point.* At each point the engine establishes, each step in exact rational arithmetic or in a rigorously enclosed computation at a working precision of at least 128 bits:

(a) every Choi block positive definite, by a positive lower bound of Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) on its exact Hermitian part (`block_not_positive_definite`), which gives (H-SUPP) by Lemma 8.2 (i);

(b) (H-POS), with $q_t(\hat{c}) > 0$ decided exactly and the lower ends of the enclosures of $P^J_{\hat{c}}(J_t)$ and $K^J_{\hat{c}}(J_t)$ positive (`positivity_not_certified`), on the exact acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of (18.1), which is also where $\hat{g}$ of (c) lives: for a singleton the support of its one point, and for a polytope the union of the supports of its exact vertices (Remark 22.2);

(c) an enclosure $(\hat{g}, \delta)$ of the acceptance-side gradient and, for every block p of every group, an exact Hermitian M_p and a rational ρ_p with $\|R_p - M_p\|_{\mathrm{F}} \le \rho_p$ for the residual of Lemma 21.1 (iii), both formed by a rigorously enclosed computation of the gradient of Corollary 8.5 at the point (of the dropped objective under a dropped set) and of the frame: the enclosure of R_p is made Hermitian-centred with symmetric radii by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure) (iv), M_p is its centre rounded entrywise to dyadic rationals and kept Hermitian, and ρ_p is a rational at least the radius that Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (ii) gives for that centre plus a rational bound on the Frobenius norm of the rounding, so that the triangle inequality gives $\|R_p - M_p\|_{\mathrm{F}} \le \rho_p$. On a companion, the computation reads each block of the measured family through its entrywise enclosure where it forms $P^J_{\hat{c}}$, the term $P^J_{\hat{c}}$ of $K^J_{\hat{c}}$ and the $\Gamma^{(s')}_{\hat{c}}$, and, where it forms the deficits and the $\Delta^{(s')}_{\hat{c}}$, through the enclosure that holds the block's factor form of (iii), its entrywise enclosure where the entrywise rule of (iv) fixed the rank and its factor enclosure where the factor rule did, so that the enclosures of (b) and (c) hold at every pair of a set as in Lemma 21.2 (c). The multipliers Y and Λ_r are read off the dual of that point's solve as Requirement 27.1 (ii) reads them in these respects only: the operators the dual's multipliers represent, each replaced by its exact Hermitian part; zero where the solve returned no finite dual; and the exact shift that makes the multiplier of an inequality row positive semidefinite. The rule of that item that zeroes the row multipliers of an Alice sector without a bound on its frame's error is not read, the rows here being built from the enclosed exact frame;

(d) b^{A}_t as the exact minimum of $\langle \hat{g}, \cdot \rangle$ over $\mathcal{S}_{\mathrm{acc}}$ for a geometry of Requirement 26.1 that it solves exactly, less δ, a point of any other geometry being not certified (`acceptance_geometry_not_certified`); and b^{F}_t by Lemma 21.1 (iii), each ℓ_p from Lemma "Least-eigenvalue lower bound from an approximate eigenbasis"

(foundations/lem:least-eigenvalue-bound), in exact rational arithmetic;

and the point's certified bracket $b_t^{\mathrm{A}} + b_t^{\mathrm{F}}$. A point at which any of these is not established is not certified, with the reason named, and contributes nothing.

(vi) *The number, its key length and its tier.* Where some point is certified, the run's certified number $\underline{h}_\alpha^{\mathrm{cert}}$ (`certified_bound`) is the largest certified bracket, (21.3), an exact rational, entering every binary64 use rounded toward $-\infty$ as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted bound; the run reports it at the certified tier (Requirement "Reporting of the tiers" (foundations/req:tiers)). The key length it reports is that of Corollary 21.3 (on a reference companion, for the protocol it declares, Corollary 21.4) at $h := \max\{0, \underline{h}_\alpha^{\mathrm{cert}}\}$, with the parameters of the run, and it is the maximal key length of Definition 10.13 with h in place of $\underline{h}_\alpha$, computed exactly: $c := n\,h - \lambda_{\mathrm{EC}} - \lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil + 1$ is formed exactly, a rational, $\lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil$ being the least integer k with $2^k \geq 1/\varepsilon_{\mathrm{EV}}$, decided by comparing the rational $1/\varepsilon_{\mathrm{EV}}$ with powers of two; the order α is taken exactly, as the dyadic rational its binary64 value is, so that $\frac{\alpha}{\alpha-1}$ is a non-zero rational; the term $\tau := \frac{\alpha}{\alpha-1} \log_2(1/\varepsilon_{\mathrm{PA}})$ exactly where $1/\varepsilon_{\mathrm{PA}}$ is a power of two, τ being then rational, and otherwise through an enclosure $[L, U] \ni \tau$ formed in exact arithmetic or by a rigorously enclosed computation and refined until $\lfloor c - U \rfloor = \lfloor c - L \rfloor$, which is then $\lfloor c - \tau \rfloor$; the refinement ends, τ being irrational there, as a non-zero rational multiple of $\log_2(1/\varepsilon_{\mathrm{PA}})$, which is irrational for a dyadic rational $\varepsilon_{\mathrm{PA}}$ whose reciprocal is not a power of two, so that $c - \tau$ is not an integer. The key length is $\max\{0, \lfloor c - \tau \rfloor\}$, and the rule of Requirement 16.4 is met as written. The clamp is recorded as Requirement 24.3 records it. The first-order number and the key length formed from it, by Requirement 24.7 (iii) or without a companion, are recorded beside it (`first_order_key_length`). In the fallback of (i), the certified number is the largest certified bracket on the declaration the run searched, a lower bound on that declaration's value and not on the companion's, and its key length is that of Corollary 21.3 on that declaration, never that of Corollary 21.4; it is reported at the certified tier, and the blocks that Requirement 24.7 (v) names for the fallback rest on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement). The companion's first-order number stays the run's first-order number, recorded beside it with its key length, and no number is formed from the two together: the key length is that of the fallback's certified number alone, whether the companion's first-order number gives a larger key length or a smaller one. Where the companion's bracket

is withdrawn by (iv), the run's first-order number is that of Requirement 24.7 (iv) on the declaration the run searched, a number certified there is a lower bound on that declaration's value with the key length of Corollary 21.3, and the blocks the companion carried rest on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) with the cause `factor_enclosure_inconsistent`; a run that declares `strict_snaps` refuses there (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). Where no point is certified, or the arithmetic the evaluation needs is not available to the engine (`certifier_unavailable`), the run's number, key length and tier are those it has without this requirement, recorded with the reason `not_certified` and the reasons of every point.

(vii) *Premises and evaluations upstream of the declaration.* A run is not reported at the certified tier where (a) an assumption of the category numerical-trust is invoked on it, reason `numerical_trust_premise`: an assumption of its assumption closure counts as invoked unless the module that owns it records that it was invoked nowhere on the run, as the receiver records its eigenvalue-slack assumption beside the settings at which it was invoked (Requirement "Invoking settings on the eigenvalue-slack row of the ledger" (receiver/req:eig-ledger-row)); or (b) the declaration it is formed on carries a weight routed from a receiver's high-photon weights (Definition "The weight route" (reductions/wpfss/def:weight-route)) into which a per-setting weight enters that the receiver's artifact records as evaluated in floating point (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)), the uniform weight of a slot entry counting the weight of every setting as entering it, reason `receiver_weight_floating_point`. Its certified number is then recorded with the reason, and the run's number, key length and tier are those it has without this requirement.

(viii) *The strict option.* A run that sets `certified_evaluation` to `strict` and is not reported at the certified tier by (vi) and (vii) refuses (`certification_refused`), naming the reasons, and reports no key length. A run reported at the certified tier through the fallback of (i) is so reported under this option too and does not refuse: the premise it rests on there, Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), is named in its record, and the option that refuses a run resting on it is `strict_snaps`, under which the fallback does not run.

(ix) *The record.* The run records the option; per point, whether it was certified, b_t^{A}, b_t^{F}, its certified bracket, the largest radius of its enclosure of the gradient and its reasons; and per run $\underline{h}_\alpha^{\mathrm{cert}}$, the largest such

radius over the certified points (`max_gradient_radius`), the working precision (`working_precision_bits`), the library of the rigorous arithmetic with its version (`arithmetic_library`), and, as a diagnostic, $\underline{h}_{\alpha}^{\mathrm{cert}}$ less the first-order number on the same declaration, with its sign (`certified_minus_first_order`), which in the fallback is the bound on the searched declaration that Requirement 24.7 (iii) records and is absent where that declaration has none. It records further the declaration the per-point entries were evaluated on (`declaration`: `reference_companion` or `searched`); the premise at the snap the certified number rests on (`premises`): the short id of Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) wherever a moved block or class of the declaration the number is formed on rests on that assumption (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)), among them, on the declaration the run searched, every moved block or class it has, in the fallback, without a companion and where the companion's bracket is withdrawn alike, and, on a companion, every moved block on which it carries the snapped block and every moved block of another snap it shares with the declaration the run searched, such as that of another measured factor; and none otherwise; with it, the blocks and classes that rest on it (`resting_blocks`), with their groups and causes as the record of the run names them where the key length is formed from a number on that declaration, those of Requirement 24.7 (v) among them in the fallback, with the cause `companion_not_certified`; and, wherever the fallback ran or the companion's bracket was withdrawn by (iv), the companion's points, each with whether it was certified and its reasons (`companion_points`), which say why. The fields `premises` and `resting_blocks` describe the declaration that `declaration` names, whatever the status: they name what a number certified there rests on, also where none is reported at the certified tier. In a fallback not reported at the certified tier, the status being `withheld` or `not_certified`, they therefore name the blocks of Requirement 24.7 (v) with the cause `companion_not_certified`, under which that item names them where the fallback's number forms the key length, while the run's key length is formed from the companion's first-order number and the record of the run names those blocks as evaluated off their snap. None of these but $\underline{h}_{\alpha}^{\mathrm{cert}}$ enters a number.

References

[1] Amir Arqand and Ernest Y.-Z. Tan. Marginal-constrained entropy accumulation theorem, 2025.

[2] Christopher Portmann and Renato Renner. Security in quantum cryptography. *Rev. Mod. Phys. 94, 025008 (2022)*, 2022.

[3] Devashish Tupkary, Shlok Nahar, Amir Arqand, Ernest Y.-Z. Tan, and Norbert Lütkenhaus. A rigorous and complete security proof of decoy-state BB84 quantum key distribution, 2026.

[4] Devashish Tupkary, Ernest Y.-Z. Tan, Shlok Nahar, Lars Kamin, and Norbert Lütkenhaus. Security proofs for practical QKD: variations, techniques, gaps, and limitations, 2025.